

River Water Quality in Hong Kong in 2015



Environmental Protection Department
The Government of the Hong Kong Special Administrative Region

Mission

To conduct an effective and scientifically sound water monitoring programme that helps protect the health of Hong Kong's rivers and streams and ensure the water quality objectives can be achieved and maintained.



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Acknowledgement

We acknowledge the Government Laboratory for undertaking chemical analyses of the water samples.

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1. An overview of Hong Kong's rivers in 2015



Birds resting on streams in Lau Fau Shan.

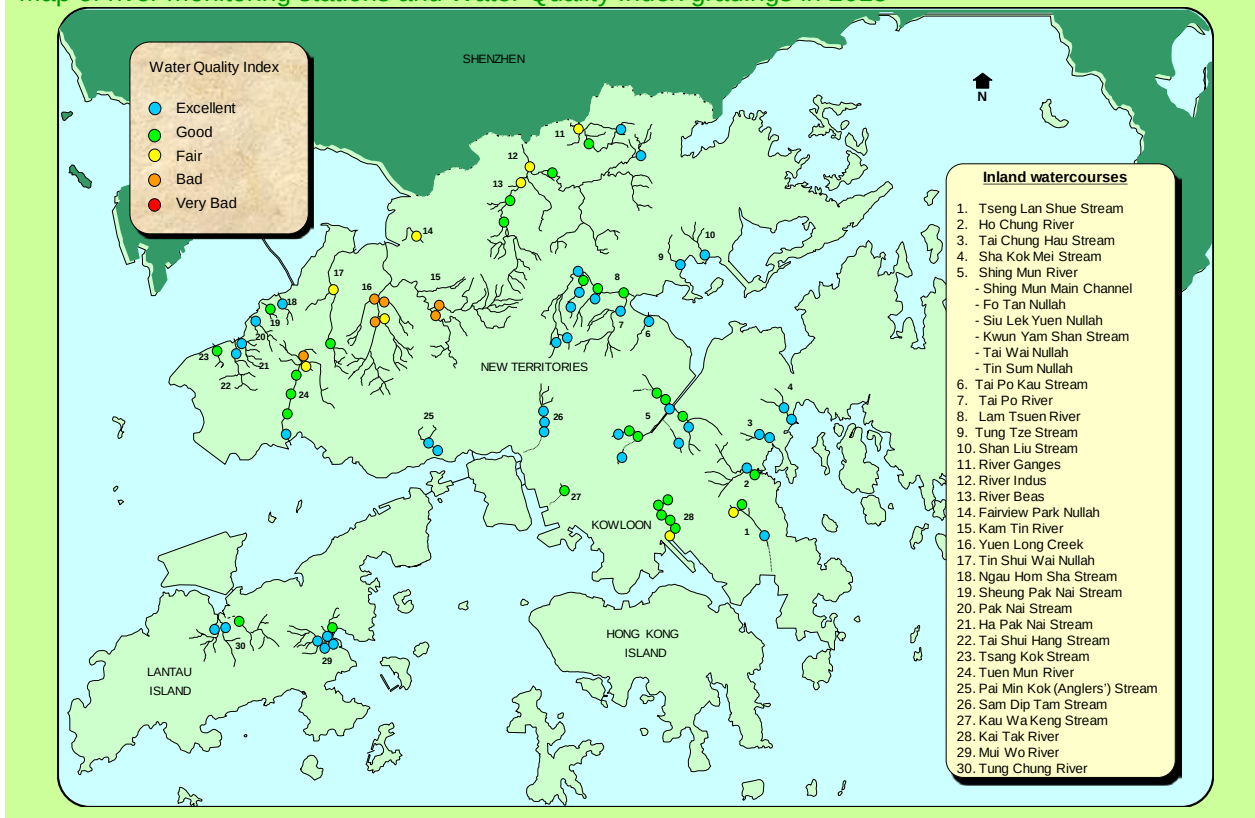
This report summarizes the water quality of major rivers covered by the Environmental Protection Department (EPD)'s river monitoring programme in 2015. The background of the programme, the scientific basis on which it is based, the detailed procedures of sampling, testing, analyses and publication of results can be found in the anniversary edition of "20 Years of River Water Quality Monitoring in Hong Kong 1986-2005". This report and other annual reports from 1998 and onwards can be downloaded from the EPD's web-site:

<http://wqrc.epd.gov.hk/en/water-quality/river-2.aspx>.



EPD river
monitoring staff
conducting sampling
at Shan Liu Stream

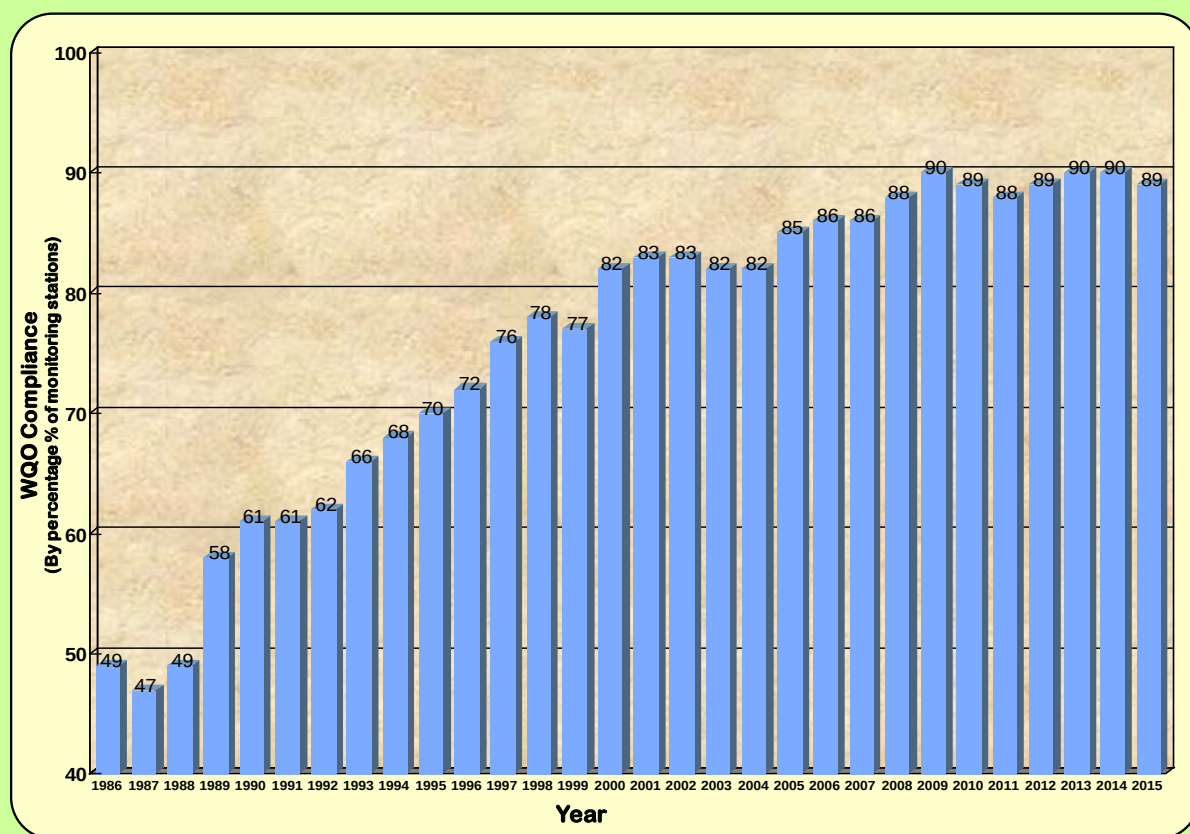
Map of river monitoring stations and Water Quality Index gradings in 2015



The EPD has a comprehensive river water quality monitoring programme in Hong Kong since 1986. In 2015, monitoring took place at 82 stations, covering 30 rivers and streams. The monitoring involved conducting field measurements and collecting water samples for laboratory analyses of over 40 physico-chemical and biological parameters, including organics, nutrients, metals and *E. coli* bacteria.

To assess compliance with the statutory Water Quality Objectives (WQOs), five representative parameters are used: pH, suspended solids, dissolved oxygen, 5-day biochemical oxygen demand, and chemical oxygen demand. The overall compliance rate in 2015 for all the stations was 89%, compared with 90% in 2014, and indicated that the overall water quality of Hong Kong rivers in 2015 continued to be well. The good compliance rates in recent years were the result of implementation of pollution control legislation, including the Water Pollution Control Ordinance and the Livestock Waste Control Scheme introduced under the Waste Disposal Ordinance, and the extension of the sewerage network to local villages under the Sewerage Master Plans and the gradual connection of the village houses to new sewers.

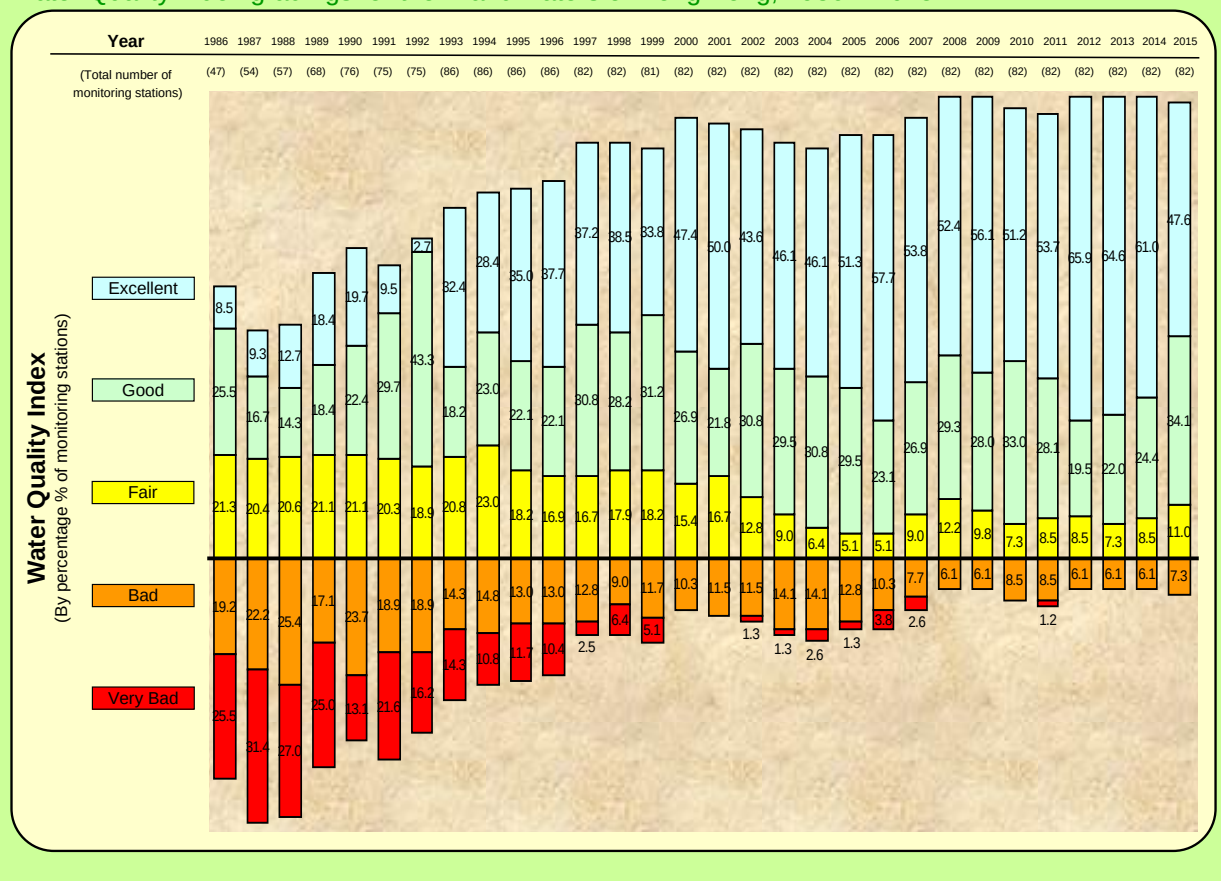
WQO compliance in the inland waters of Hong Kong, 1986 – 2015



Similarly, the Water Quality Index (WQI), which grades the general health of inland water courses by assessing the three parameters that reflect a river's level of organic pollution: dissolved oxygen, 5-day biochemical oxygen demand and ammonia-nitrogen content, continued to perform well in recent years. In 2015, 82% of the river monitoring stations were graded "Good" or "Excellent". The majority of the monitoring stations in Lantau Island, the Eastern New Territories, Southwestern New Territories and Kowloon were in these categories, indicating that the organic pollution loading from their catchments was steadily decreasing.

7% of the monitoring stations were graded 'Bad' in 2015 and no station was graded 'Very Bad'. Most of the 'Bad' stations are located along watercourses in the Northwestern New Territories.

Water Quality Index gradings for the inland waters of Hong Kong, 1986 – 2015



Compared with 2014, 16 stations moved down a grade and 1 station moved up a grade in WQI grading in 2015. These minor changes are considered to be within the normal range of natural fluctuations over the past 10 years.

River or Stream	2014 WQI grading	2015 WQI grading
Sheung Pak Nai Stream & Tseng Kok Stream (DB5, DB8)	Excellent	Good
River Indus (IN1)	Good	Fair
Kai Tak River (KN1)	Good	Fair
Kai Tak River (KN2, KN5, KN7)	Excellent	Good
Kam Tin River (KT1)	Fair	Bad
Mui Wo River (MW5)	Excellent	Good
River Beas (RB1)	Excellent	Good
Tuen Mun River (TN2)	Good	Fair
Tuen Mun River (TN3, TN4)	Excellent	Good
Fo Tan Nullah (TR17L)	Excellent	Good
Tai Wai Nullah (TR19C)	Excellent	Good
Siu Lek Yuen Nullah (TR23A)	Excellent	Good
Tung Tze Stream (TR6)	Good	Excellent



Tung Tze Stream improved a grade to 'Excellent' water quality in 2015.

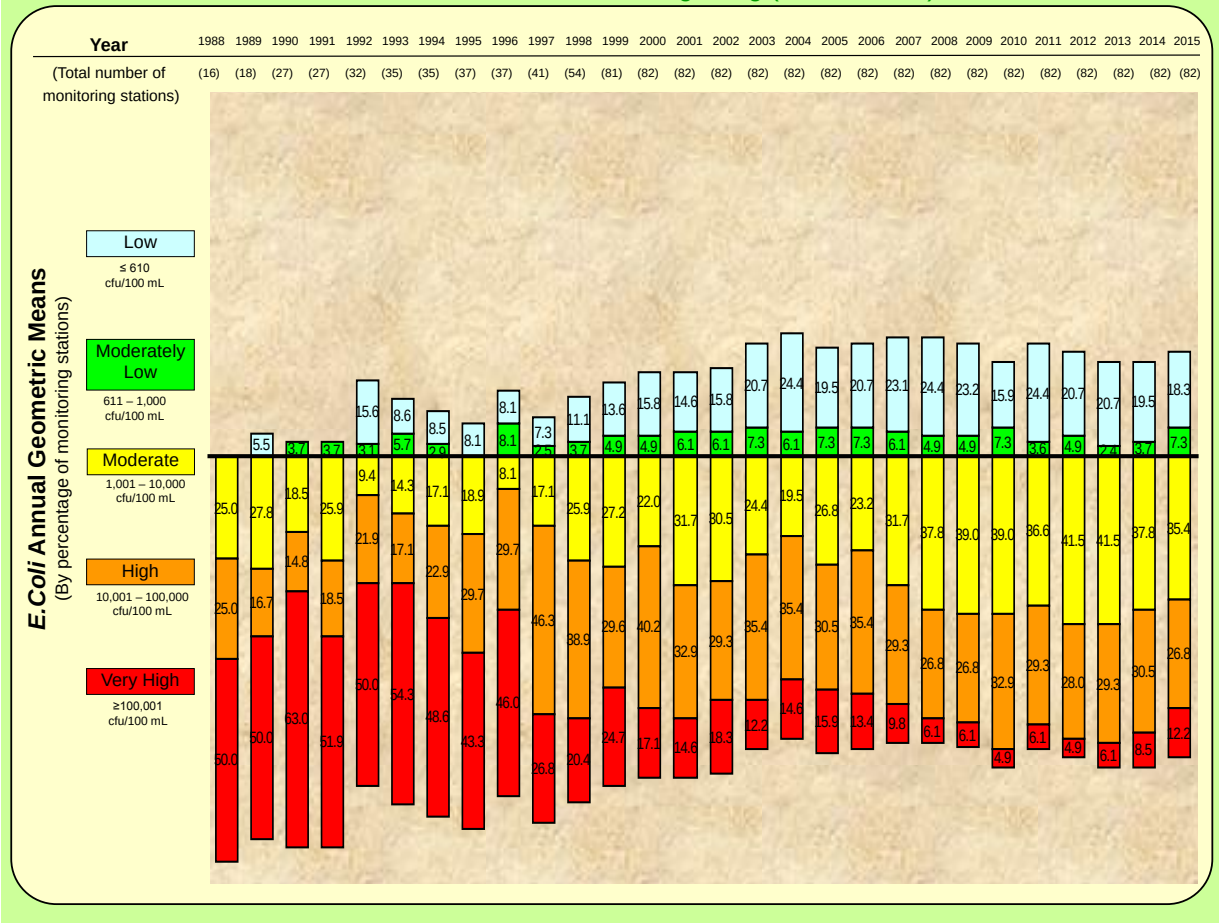
In terms of *E. coli*, the situation was comparable to 2014 with 26% of the stations having 'Low' or 'Moderately Low' levels of *E. coli* (i.e. less than or equal to 1,000 cfu/100 mL) in 2015 while 39% of the stations recorded 'High' or 'Very High' (i.e. over 10,000 cfu/100mL) levels.¹

Most of the stations with 'High' and 'Very High' levels of *E. coli* are located in the northwestern part of the New Territories (e.g. Sheung Pak Nai Stream, Yuen Long Creek, Kam Tin River, Tin Sui Wai Nullah and Fairview Park Nullah), Northern District (e.g. downstream station of River Indus and River Ganges), the Eastern New Territories (e.g. Ho Chung River, Tai Chung Hau Stream, Sha Kok Mei Stream, Fo Tan Nullah, two stations of Lam Tsuen River and upstream stations of Tseng Lan Shue Stream), Southwestern New Territories (e.g. Kau Wa Keng Stream and two upstream stations of Tuen Mun River), and Central and Eastern Kowloon (e.g. Kai Tak River).

In summary, the 2015 WQO compliance rate and WQI grading of Hong Kong's inland water courses were largely similar to 2014's monitoring results and showed long term improvement as a result of the gradual reduction of pollution loading in our rivers.

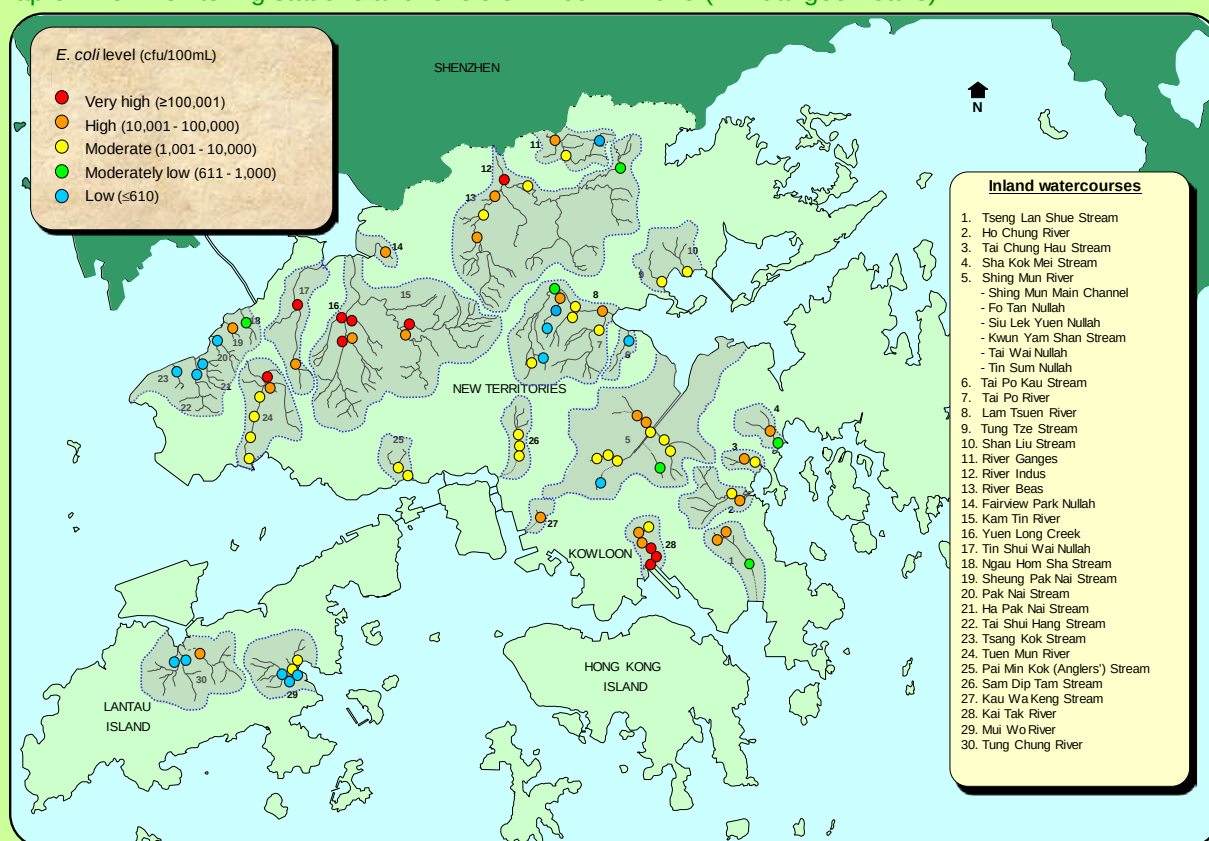
¹ All levels of *E. coli* in this report are reported as annual geometric means (cfu/100mL).

Annual Geomean *E. coli* Levels for Inland Waters of Hong Kong (1988 – 2015)



It should be noted that many of the above sites with high *E. coli* levels in the New Territories are subject to discharges from unsewered village houses, illegal discharges from livestock farms as well as expedient connections in the sewered areas. The Government's continued efforts in enforcement and extension of the public sewer network should be able to reduce the *E. coli* levels at these locations in the future.

Map of river monitoring stations and levels of *E. coli* in 2015 (Annual geomeans)



2. Eastern New Territories rivers

The EPD monitored ten rivers and streams in the Eastern New Territories in 2015. Six are located in the Tolo Harbour and Channel Water Control Zone (WCZ): Shing Mun River, Lam Tsuen River, Tai Po River, Tai Po Kau Stream, Shan Liu Stream and Tung Tze Stream. Within the Port Shelter WCZ, there are Ho Chung River, Sha Kok Mei Stream and Tai Chung Hau Stream. Tseng Lan Shue Stream is located within the Junk Bay WCZ.

The Eastern New Territories had the best river and stream water quality in the territory. In 2015, the overall WQO compliance rate was 97%, similar to 2014². Within the region, three rivers achieved full compliance (100%) with the WQOs in 2015. These are Tai Po Kau Stream and Shan Liu Stream in the Tolo Harbour and Channel WCZ and Sha Kok Mei Stream in the Port Shelter WCZ.

² In this report, the data from 2014 are chosen for short term comparison to illustrate year-to-year fluctuation and data from 1997 are chosen as the baseline for medium term comparison.

Compliance with the Water Quality Objectives in the rivers of the Eastern New Territories in 1997 and in 2015



Although Shing Mun River in the Tolo Harbour and Channel WCZ has not reached 100% WQO compliance in 2015, it showed marked improvement during the last two decades. The WQO compliance rate of Shing Mun River rose from 78% in 1997 to 91% in 2015. Lam Tsuen River's WQO compliance rate was 96% in 2015 while Tai Po River recorded 98% in the same year.

All three rivers in the Port Shelter WCZ achieved a WQO compliance rate of 99% or above in 2015. Ho Chung River showed improvement from a 91% compliance rate in 1997 to 99% in 2015. The relatively unpolluted Tai Chung Hau Stream and Sha Kok Mei Stream have maintained their high compliance rates of 98% or above since 1997.

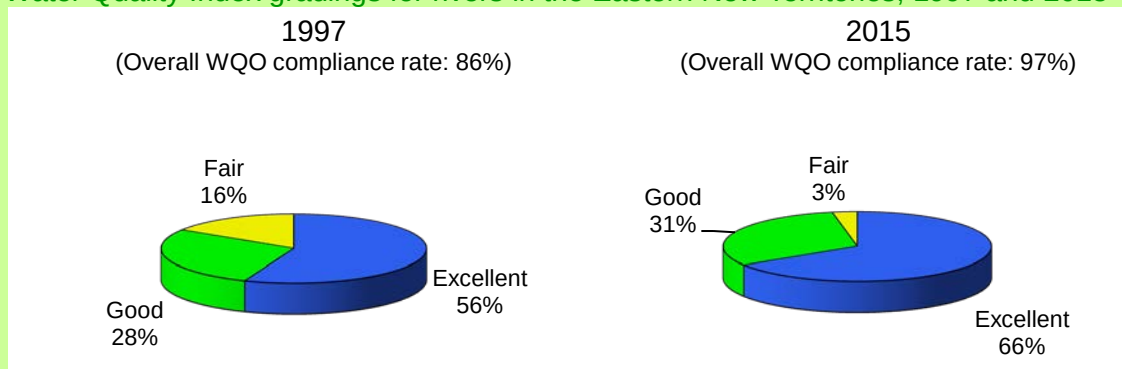
The compliance rate for Tseng Lan Shue Stream, which flows into Junk Bay, improved from 79% in 1997 to 89% in 2015.



The Tai Po Kau Stream is unpolluted and naturally beautiful.

As for the WQI grading, 31 out of 32 (97%) of the river monitoring stations in the Eastern New Territories were graded 'Good' or 'Excellent' in 2015, same as in 2014. The remaining station of Tseng Lan Shue Stream at Tseng Lan Shue Tsuen (JR3) was in the 'Fair' category. The overall improvement was due to the continued enforcement of the pollution control legislation, the implementation of Sewerage Master Plans and extended village sewerage in the catchments.

Water Quality Index gradings for rivers in the Eastern New Territories, 1997 and 2015



In terms of *E. coli*, 9 out of the 32 stations in 2015 had *E. coli* levels of less than 1,000 cfu/100mL which are considered moderately low or low. There were, however, 9 stations still having fairly high levels of *E. coli* of 10,001 – 100,000 cfu/100mL, namely the upstream stations of Tseng Lan Shue Stream (JR3, JR6), the downstream station at Ho Chung River (PR1), the upstream station of Tai Chung Hau Stream (PR8), the upstream station of Sha Kok Mei Stream (PR6), both stations of Fo Tan Nullah (TR17 and TR17L), and two of Lam Tsuen River's stations (TR12I, TR12C). Most of these stations were still receiving discharges from unsewered villages. When the public sewer network is extended to unsewered areas in this part of the New Territories, the water quality of the rivers and streams is expected to improve further.

The *E. coli* level at Shing Mun River Main Channel (TR19I) was 1,300 cfu/100mL in 2015, same as 2014. After the Drainage Services Department (DSD) had rectified the defective sewers which led to overflow of sewage resulting in a moderate *E. coli* level of 4,000 cfu/100ml in 2013, the *E. coli* level in the main channel has gradually decreased. However, we observed a slight elevation of *E. coli* level in the last few months of 2015, due to the collapse of a sewer in the vicinity. An action plan was put in place by DSD to reduce any risk of sewage overflow.

3. Northwestern New Territories rivers

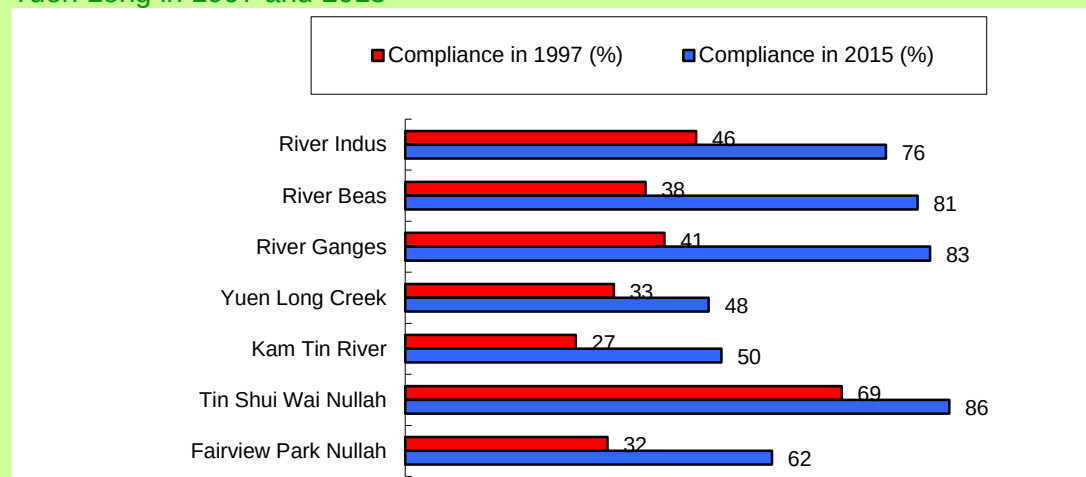
In the Northwestern New Territories, the EPD monitors a total of 13 rivers and streams which flow into Shenzhen River or directly into Deep Bay (Shenzhen Bay). Three of these — Rivers Indus, Beas and Ganges — are major water courses situated in North District. Yuen Long Creek, Kam Tin River, Tin Shui Wai Nullah and Fairview Park Nullah are situated in Yuen Long District, while the six smaller streams are located around the Lau Fau Shan area.

All the monitored rivers in the Northwestern New Territories showed improvement in the WQO compliance rates in the last decade, particularly the Rivers Beas, Indus and Ganges in North District.

In terms of WQO compliance, the upstream station of River Beas (RB1) was 90% in 2015, compared with 93% in 2014. The compliance rates of the midstream station (RB2) and

downstream station (RB3) were 82% and 72% in 2015 as compared with 82% and 70% in 2014 respectively. For River Ganges, the WQO compliance rate of the 3 stations were 100%, 80%, 68% from upstream to downstream (GR3, GR2, GR1) compared with 95%, 85% and 75% in 2014. These changes are considered to be within the normal year to year fluctuations over the past 10 years. Both rivers are still affected by pollution from livestock farms, unsewered villages and small industrial establishments in the catchment. The upstream station (IN3) and midstream station (IN2) of River Indus recorded 95% and 92% compliance in WQO respectively, as compared with 2014's 97% and 95%. The downstream station (IN1) which is subject to backflow from Shenzhen River recorded 42% compliance with the WQO, as compared with 55% in 2014.

Compliance with the Water Quality Objectives in the major rivers of North District and Yuen Long in 1997 and 2015



The compliance rates of stations KT1 and KT2 in Kam Tin River were 68% and 32% in 2015, as compared with 72% and 35% respectively in 2014. Overall, Kam Tin River was still impacted by discharges from livestock farms, unsewered premises and small industrial establishments.

Yuen Long Creek's overall compliance rate in 2015 was 48%, compared with 52% in 2014. The compliance rates for the two upstream stations (YL1 and YL2) were 62% and 60% respectively in 2015 as compared with 58% and 60% in 2014. The compliance rates for the stations in the middle of Yuen Long township (YL3 and YL4) were 33% and 35% in 2015 as compared with 57% and 33% in 2014 respectively. This river was still subject to discharges from livestock farms, unsewered village houses and industrial establishments.

The upstream station of Tin Shui Wai Nullah (TSR2) located near Wo Ping San Tsuen and Tan Kwai Tsuen reached a 90% WQO compliance rate in 2015, as compared with 95% in 2014. The downstream station (TSR1) was 82% in compliance in 2015, same as 2014. The Fairview Park Nullah station (FVR1) achieved 62% WQO compliance in 2015 as compared with 58% in 2014.

The upstream portion of River Ganges is unpolluted.



In 2015, water quality improved upstream along River Ganges: ‘Fair’ for GR1, ‘Good’ for GR2 and ‘Excellent’ for GR3, similar to 2014. The downstream station of River Indus (IN1) which is affected by backflow from Shenzhen River had a ‘Fair’ grading in 2015, compared to ‘Good’ in 2014. The midstream (IN2) and upstream (IN3) stations were ‘Good’ and ‘Excellent’ respectively, similar to 2014. Going downstream along River Beas, the three stations (RB1, RB2, RB3) received ‘Good’, ‘Good’ and ‘Fair’ grading respectively in 2015, compared to ‘Excellent’, ‘Good’ and ‘Fair’ in 2014.

In 2015, both stations, KT1 and KT2, in Kam Tin River attained WQI grading of ‘Bad’, compared with ‘Fair’ and ‘Bad’ respectively in 2014. Yuen Long Creek on the other hand recorded ‘Bad’ and ‘Fair’ grading for the upstream stations YL1 and YL2 in 2015, similar to 2014. Both downstream YL3 and YL4 stations were graded ‘Bad’ in 2015, similar to 2014.

The two stations at Tin Shui Wai Nullah (TSR1 and TSR2) received ‘Fair’ and ‘Good’ grading respectively, similar to 2014. Fairview Park Nullah (FVR1) was graded as ‘Fair’ in 2015, similar to 2014.

The water quality of the smaller streams around the Lau Fau Shan area remained good in 2015. All six streams achieved WQO compliance of 93% or above, with Ha Pak Nai Stream, Tai Shui Hang Stream, Pak Nai Stream and Ngau Hom Sha Stream achieving 100%. Similarly, four streams achieved ‘Excellent’ and two achieved ‘Good’ WQI gradings, compared with all ‘Excellent’ gradings in 2014. In comparison, the overall WQO compliance rates of the other larger water courses in the Northwestern New Territories in 2015 ranged from 48% to 86%, as compared with 52% to 88% in 2014.

The water quality of upstream and midstream stations of the rivers in Northwestern New Territories were mostly better as compared with the downstream stations, probably because pollutants discharged into the water courses tended to accumulate downstream.



The upper portion of Kam Tin River is unpolluted and naturally scenic.

The annual geometric mean levels of *E. coli* in 2015 showed that most of the stations in Northwestern New Territories still contained moderate to very high levels of *E. coli*. However, most of the Lau Fau Shan streams had levels of *E. coli* below 700 cfu/100mL except Sheung Pak Nai Stream (DB5: 15,000 cfu/100mL).

In 2015, the upstream and downstream stations at River Beas (RB1) and (RB3) had higher levels of *E. coli* (12,000 and 19,000 cfu/100mL) compared with the midstream (RB2) (2,000 cfu/100mL), while the upstream station at River Ganges (GR3) had a lower level of *E. coli* (570 cfu/100mL) compared with the midstream and downstream stations (GR2 and GR1) (4,300 cfu/100mL and 35,000 cfu/100mL respectively) in 2015. This suggests that the water quality of these rivers were affected by local pollution sources, including discharge from unsewered villages, livestock farms and expedient connections and the pollutants discharged into water courses tended to accumulate downstream. On the other hand, the elevated *E. coli* level of the downstream River Indus station (IN1) (110,000 cfu/100mL), compared with the upstream (IN3) (740 cfu/100mL) and midstream (IN2) (1,400 cfu/100mL) stations, was probably due to the back flushing effect of the more polluted Shenzhen River.

All of Lau Fau Shan's minor streams achieved 93% or higher Water Quality Objective compliance in 2015.



The *E. coli* levels at Kam Tin River and Fairview Park Nullah remained high in 2015 with the Fairview Park Nullah station recording an *E. coli* level of 29,000 cfu/100mL and Kam Tin River's two stations, KT1 and KT2, recording 78,000 and 300,000 cfu/100mL respectively. The upstream stations of Yuen Long Creek, YL1 and YL2, had *E. coli* levels of 180,000 and 47,000 cfu/100mL and the two mid-streams stations, YL3 and YL4, had *E. coli* levels of 1,000,000 and 2,000,000 cfu/100mL respectively. Tin Shui Wai Nullah's mid-stream and upstream stations, TSR1 and TSR2, registered 300,000 and 24,000 cfu/100mL *E. coli* levels.

In summary, the *E. coli* levels in the major rivers in the Northwestern New Territories were still high largely because of discharge from livestock farms, expedient connections and unsewered villages in the area. The implementation of the Schemes for the Voluntary Surrender of Poultry and Pig Farm Licences has already brought about a reduction of livestock waste loading. The North District Sewerage Master Plan and Yuen Long and Kam Tin Sewerage Master Plan have included plans to provide public sewers to most of the unsewered villages, and the river water quality in the Northwestern New Territories is expected to gradually improve when these schemes are implemented.

4. Lantau Island rivers

Lantau Island is large and less populated. Two major rivers with a total of eight monitoring stations are monitored by the EPD: five along Mui Wo River on the south-eastern side and three at Tung Chung River on the north-western side of the island.

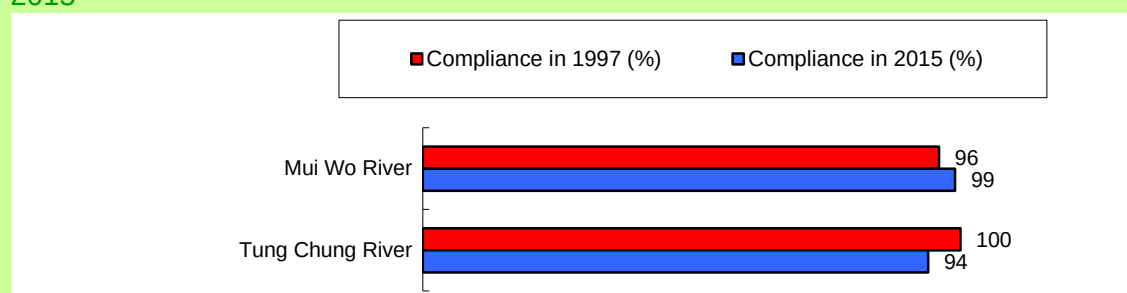
Overall, Mui Wo River and Tung Chung River generally displayed satisfactory water quality. The WQO compliance rates of Mui Wo River and Tung Chung River in 2015 were 99% and 94% respectively, compared to 2014's 100% and 97% respectively.



Tung Chung River achieved 94% WQO compliance rate in 2015.

As for the WQI gradings, all five monitoring stations at Mui Wo River maintained an ‘Excellent’ grading in 2015 except for MW5 of Mui Wo River which was graded ‘Good’. For Tung Chung River, except the downstream station of Tung Chung River (TC3) which was graded ‘Good’ as in 2014, the other two stations were graded ‘Excellent’. The *E. coli* levels of the five monitoring stations at Mui Wo River in 2015 were: 600 cfu/100mL at MW1; 2,300 cfu/100mL at MW2; 210 cfu/100mL at MW3; 540 cfu/100mL at MW4 and 5,400 cfu/100mL at MW5. For Tung Chung River, the *E. coli* level at TC1 was 64 cfu/100mL; 110 cfu/100mL at TC2 and 23,000 cfu/100mL at TC3, suggesting that this station might be affected by discharges or seepage from septic tanks and soakaway systems in the nearby unsewered village houses.

Compliance with the Water Quality Objectives in the rivers of Lantau Island in 1997 and 2015

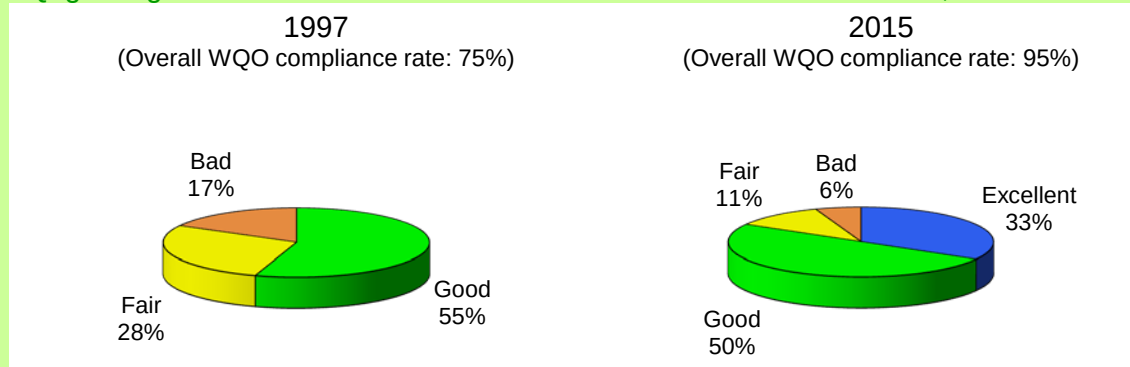


5. Southwestern New Territories and Kowloon rivers

The Southwestern New Territories and Kowloon area runs from Black Point west of Tuen Mun through to the eastern end of Victoria Harbour. The five watercourses being monitored are Tuen Mun River in Tuen Mun, Pai Min Kok Stream and Sam Dip Tam Stream near Tsuen Wan, Kau Wa Keng Stream in Kwai Chung and Kai Tak River located in Kowloon. There was substantial improvement in the water quality of these urban watercourses in the last decade.

In 2015, 15 of the 18 stations (83%) were graded ‘Excellent’ or ‘Good’, as compared with 10 of them graded as ‘Good’ and no ‘Excellent’ station in 1997. The stations with ‘Bad’ water quality grading also decreased from 9 to 1 (only TN1 at Tuen Mun River was graded ‘Bad’ in 2015, as in 2014) during the same period. Similarly, the overall WQO compliance rate of the rivers and streams in the area has risen from 75% in 1997 to 95% in 2015.

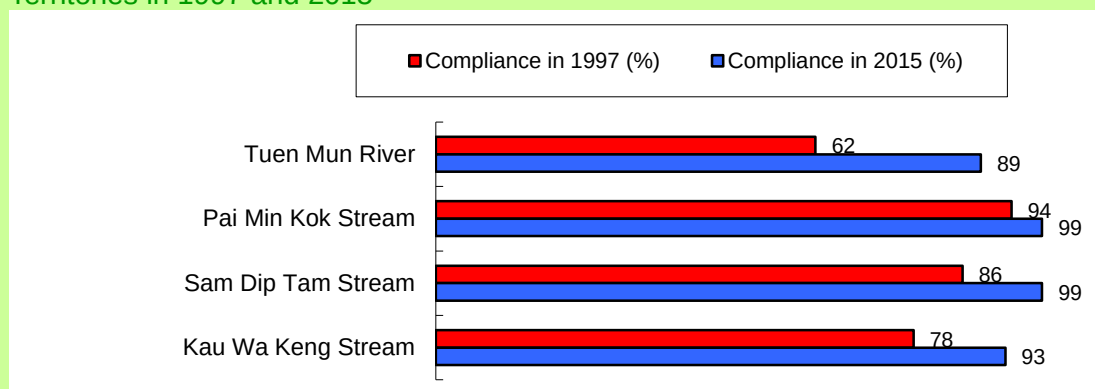
WQI gradings for rivers in the Southwestern New Territories and Kowloon, 1997 and 2015



Tuen Mun River showed marked improvement in the last decade. Its WQO compliance rate rose steadily from 62% in 1997 to 89% in 2015. This improvement was attributed to pollution control efforts as well as implementation of the mitigation measures recommended under the Tuen Mun Sewerage Master Plan. In 2015, the upstream stations (TN1 and TN2) were graded ‘Bad’ and ‘Fair’, the three midstream stations were graded “Good” while the downstream station was graded ‘Excellent’. The upstream station at TN1 was impacted by unsewered villages with an *E. coli* level of 140,000 cfu/100mL in 2015. The *E. coli* levels of the other stations, namely TN2-TN3-TN4-TN5-TN6, remained relatively high at 59,000, 4,000, 8,100, 2,900 and 3,600 cfu/100mL respectively.

Pai Min Kok Stream, still impacted by a few unsewered village houses, achieved an overall WQO compliance rate of 99% in 2015, same as 2014. As in 2014, both the upstream station (AN2) and the downstream station (AN1) were graded ‘Excellent’ in 2015. The *E. coli* level of AN1 and AN2 was, nevertheless, moderately high at 5,400 and 7,000 cfu/100mL respectively.

Compliance with the Water Quality Objectives in the rivers of the Southwestern New Territories in 1997 and 2015



All three monitoring stations (from upstream to downstream: TW1, TW2 and TW3) of Tsuen Wan's Sam Dip Tam Stream achieved 98% or higher compliance with the WQO and all achieved an 'Excellent' WQI grading in 2015, similar to 2014. Their *E. coli* levels were 1,400, 7,300, 3,400 cfu/100mL respectively, suggesting local pollution sources in the mid-stream section of the river.

The monitoring station at Kwai Chung's Kau Wa Keng Stream achieved a WQO compliance rate of 93% in 2015, as compared with 92% in 2014. Its WQI grading was 'Good', similar to 2014. Its *E. coli* level was high at 68,000 cfu/100mL suggesting local pollution sources affecting the stream.

Five stations of Kai Tak River were graded 'Good' in 2015, while the one at downstream (KN1) was graded 'Fair'. The *E. coli* levels from upstream to downstream (KN7, KN5, KN4, KN3, KN2, KN1) were 4,800, 48,000, 94,000, 120,000, 150,000, and 310,000 cfu/100mL in 2015 respectively, compared with 3,200, 22,000, 36,000, 49,000, 42,000, and 190,000 cfu/100mL in 2014. The high *E. coli* level was brought about by a defective foul sewer at the downstream and occasional blockage of a dry weather flow interceptor (DWFI) midstream of Kai Tak River. The water quality of Kai Tak River will further improve once the defective sewer system in the catchment was repaired and upgraded. An action checklist has also been drawn up to upgrade the DWFIs by DSD.

Appendices

Summary of river water quality monitoring stations and sampling frequencies in 2015

Area	Watercourse	Monitoring Station (Number)	Sampling frequency
Eastern New Territories			
Sha Tin	Shing Mun River	TR19I (1)	monthly
	<i>Shing Mun Main Channel</i>	TR23A, TR23L (2)	monthly
	<i>Siu Lek Yuen Nullah</i>	TR17, TR17L (2)	monthly
	<i>Fo Tan Nullah</i>	KY1 (1)	monthly
	<i>Kwun Yam Shan Stream</i>	TR19, TR19A, TR19C (3)	monthly
	<i>Tai Wai Nullah</i>	TR20B (1)	monthly
Tai Po Town Centre	Lam Tsuen River	TR12, TR12B, TR12C, TR12D, TR12E, TR12F, TR12G, TR12H, TR12I (9)	monthly
	Tai Po River	TR13 (1)	monthly
Tai Po Rural Area	Tai Po Kau Stream	TR14 (1)	monthly
	Shan Liu Stream	TR4 (1)	monthly
	Tung Tze Stream	TR6 (1)	monthly
Sai Kung	Ho Chung River	PR1, PR2 (2)	monthly
	Sha Kok Mei Stream	PR5, PR6 (2)	monthly
	Tai Chung Hau Stream	PR7, PR8 (2)	monthly
Tseung Kwan O	Tseng Lan Shue Stream	JR3, JR6, JR11 (3)	monthly
Northwestern New Territories			
North District	River Indus (Ng Tung River)	IN1, IN2, IN3 (3)	monthly
	River Beas (Sheung Yue River)	RB1, RB2, RB3 (3)	monthly
	River Ganges (Ping Yuen River)	GR1, GR2, GR3 (3)	monthly
Yuen Long	Yuen Long Creek	YL1, YL2, YL3, YL4 (4)	monthly
	Kam Tin River	KT1, KT2 (2)	monthly
	Tin Shui Wai Nullah	TSR1, TSR2 (2)	monthly
	Fairview Park Nullah	FVR1 (1)	monthly
Lau Fau Shan	Ha Pak Nai Stream	DB1 (1)	monthly
	Tai Shui Hang Stream	DB2 (1)	monthly
	Pak Nai Stream	DB3 (1)	monthly
	Sheung Pak Nai Stream	DB5 (1)	monthly
	Ngau Hom Sha Stream	DB6 (1)	monthly
	Tseng Kok Stream	DB8 (1)	monthly
Lantau Island			
Mui Wo	Mui Wo River	MW1, MW2, MW3, MW4, MW5 (5)	monthly
Tung Chung	Tung Chung River	TC1, TC2, TC3 (3)	monthly
Southwestern New Territories and Kowloon			
Tuen Mun	Tuen Mun River	TN1, TN2, TN3, TN4, TN5, TN6 (6)	monthly
Tsuen Wan and Kwai Chung	Pai Min Kok (Anglers') Stream	AN1, AN2 (2)	monthly
	Sam Dip Tam Stream	TW1, TW2, TW3 (3)	monthly
	Kau Wa Keng Stream	KW3 (1)	monthly
Kowloon	Kai Tak River	KN1, KN2, KN3, KN4, KN5, KN7 (6)	monthly
Total	30	82	-

River water quality parameters and the methods of analysis (part 1 of 2)

Water Quality Parameter	Reporting Limit and Unit	Analytical Method ¹ / Analyst
Physical Chemical Properties		
Water temperature	0.1 °C	Multi-parameter water quality data logger, model YSI-6820 / On-site measurement / EPD
Dissolved Oxygen	0.1 mg/L, 1%	
pH	0.1	
Conductivity	1 µS/cm	
Turbidity	0.1 NTU	
Flow	1 L/s	Electromagnetic flow meter, model Flo-mate 2000 / On-site measurement / EPD
Solid Contents		
Suspended Solids	0.5 mg/L	In-house method GL-PH-23, based on APHA ² 22ed 2540 D / Government Laboratory
Total Solids	0.5 mg/L	In-house method GL-PH-19, based on APHA 22ed 2540 F / Government Laboratory
Total Volatile Solids	0.5 mg/L	In-house method GL-PH-19, based on APHA 20ed 2540 E / Government Laboratory
Aggregate Organics		
5-Day Biochemical Oxygen Demand (BOD ₅)	1 mg/L	In-house method based on APHA 18ed 5210 B / EPD
Chemical Oxygen Demand (COD)	2 mg/L	In-house method GL-OR-38 & GL-OR-39, based on ASTM ³ D1252-06, Method B or APHA 22ed 5220 C & D / Government Laboratory
Total Organic Carbon (TOC)	1 mg/L	In-house method GL-OR-32, based on APHA 20ed 5310 B / Government Laboratory
Faecal Bacteria		
<i>E. coli</i>	1 cfu /100 mL	In-house method ⁴ , membrane filtration with CHROMagar Liquid ECC medium / EPD
Faecal Coliforms	1 cfu / 100 mL	
Nutrients		
Ammonia-Nitrogen	0.005 mg/L	In-house method GL-IN-15, based on APHA 22ed 4500-NH ₃ / Government Laboratory
Nitrite-Nitrogen	0.002 mg/L	In-house method GL-IN-18, based on APHA 22ed 4500-NO ₂ ⁻ B (FIA) / Government Laboratory
Nitrate-Nitrogen	0.002 mg/L	In-house method GL-IN-18, based on APHA 22ed 4500-NO ₃ ⁻ F & I (FIA) / Government Laboratory
Total Kjeldahl Nitrogen	0.05 mg/L	In-house method GL-IN-14 & GL-IN-15, based on ASTM D3590-11 B (FIA) & APHA 20ed 4500-N A&D (FIA) / Government Laboratory
Ortho-phosphate Phosphorus	0.002 mg/L	In-house method GL-IN-16, based on ASTM D515-88 A (FIA) / Government Laboratory
Total Phosphorus	0.02 mg/L	In-house method GL-IN-14 & GL-IN-16, based on ASTM D515-88 B (FIA) & APHA 20ed 4500-P G (FIA) / Government Laboratory
Molybdate-Reactive Silica	0.05 mg/L	In-house method GL-IN-17, based on APHA 20ed 4500-SiO ₂ C&E (FIA) / Government Laboratory

Reference notes:

1. Mention of brand names of commercial products does not constitute or imply endorsement or recommendation by the Environmental Protection Department.
2. APHA - American Public Health Association: *Standard Methods for the Examination of Water and Wastewater*.
3. ASTM - *Annual Book of American Society for the Testing and Materials Standards*, Vol. 11.01 & 11.02.
4. i) Ho, B.S.W. and Tam, T.Y. (1997). Enumeration of *E. coli* in environmental waters and wastewater using a chromogenic medium. *Wat. Sci. Tech.*, **35**, 409-413.
ii) DoE and DHSS (1983). "The bacteriological examination of drinking water supplies 1982. Report on Public Health and Medical Subjects No. 71. Methods for the Examination of Waters and Associated Materials". Department of Environment, Department of Health and Social Security, Public Health Laboratory Service, H.M.S.O. London.

River water quality parameters and the methods of analysis (part 2 of 2)

Water Quality Parameter	Reporting Limit and Unit	Analytical Method ¹ / Analyst
Metals		
Aluminium	50 µg/L	In-house method GL-TE-63, based on APHA 22ed 3111, 3112, 3113, 3114 & 3120 / Government Laboratory
Antimony	1 µg/L	
Arsenic	1 µg/L	
Barium	1 µg/L	
Beryllium	1 µg/L	
Boron	50 µg/L	
Cadmium	0.1 µg/L	
Chromium	1 µg/L	
Copper	1 µg/L	
Iron	50 µg/L	
Lead	1 µg/L	
Manganese	10 µg/L	
Mercury	1 µg/L	
Molybdenum	2 µg/L	
Nickel	1 µg/L	
Silver	1 µg/L	
Thallium	1 µg/L	
Vanadium	2 µg/L	
Zinc	10 µg/L	
Industrial and Commercial Pollutants		
Cyanide	0.01 mg/L	In-house method GL-IN-42, based on ASTM D 2036-09 or APHA 22ed 4500-CN / Government Laboratory
Fluoride	0.2 mg/L	In-house method GL-IN-47, based on APHA 22ed 4500-F ⁻ C & G (Ion Selective Electrode) and ASTM D1179-99 B (FIA) / Government Laboratory
Anionic Surfactants (as Manoxol OT)	0.05 mg/L	In-house method GL-OR-30, based on BS 6068, Section 2.23 (1994) (Colorimetric) & In-house method GL-OR-65, based on Abbott, D.C. "Analyst", Vol.87, p.286(1962) & S. Motomizu et al., "Analyst" Vol.113, p.747(1988) (FIA) / Government Laboratory
Oil and Grease	0.5 mg/L	In-house method GL-OR-26, based on APHA 22ed 5520 B / Government Laboratory
Sulphide Contents		
Free Hydrogen Sulphide	0.01 mg/L	In-house method GL-IN-46, based on APHA 22ed 4500S ²⁻ D (Colorimetric) / Government Laboratory
Sulphide	0.02 mg/L	
Plant Pigments		
Chlorophyll-a	0.2 µg/L	In house method GL-OR-34, based on APHA 20ed 10200H 2 (spectrophotometric) / Government Laboratory
Pheo-Pigment	0.2 µg/L	

Reference notes:

1. Mention of brand names of commercial products does not constitute or imply endorsement or recommendation by the Environmental Protection Department.
2. APHA - American Public Health Association: Standard Methods for the Examination of Water and Wastewater.
3. ASTM - Annual Book of American Society for the Testing and Materials Standards, Vol. 11.01 & 11.02.

Key Water Quality Objectives (WQOs) for river monitoring stations in the Eastern New Territories

Watercourse	Monitoring station	Key Water Quality Objective				
		pH range	Maximum 5-Day Biochemical Oxygen Demand (mg/L)	Maximum Chemical Oxygen Demand (mg/L)	Maximum Annual Median Suspended Solids (mg/L)	Minimum Dissolved Oxygen (mg/L)
Tolo Harbour and Channel Water Control Zone						
Shing Mun River	KY1	6.5 - 8.5	3	15	20	4
	TR17	6.5 - 8.5	5	30	20	4
	TR17L	6.5 - 8.5	5	30	20	4
	TR19	6.5 - 8.5	5	30	20	4
	TR19A	6.5 - 8.5	5	30	20	4
	TR19C	6.5 - 8.5	5	30	20	4
	TR19I	6.0 - 9.0	5	30	25	4
	TR20B	6.5 - 8.5	5	30	20	4
	TR23A	6.5 - 8.5	3	15	20	4
	TR23L	6.5 - 8.5	3	15	20	4
Lam Tsuen River	TR12	6.5 - 8.5	3	15	20	4
	TR12B	6.5 - 8.5	3	15	20	4
	TR12C	6.5 - 8.5	3	15	20	4
	TR12D	6.5 - 8.5	3	15	20	4
	TR12E	6.5 - 8.5	3	15	20	4
	TR12F	6.5 - 8.5	3	15	20	4
	TR12G	6.5 - 8.5	3	15	20	4
	TR12H	6.5 - 8.5	3	15	20	4
	TR12I	6.0 - 9.0	5	30	25	4
Tai Po River	TR13	6.5 - 8.5	5	30	20	4
Tai Po Kau Stream	TR14	6.0 - 9.0	5	30	25	4
Shan Liu Stream	TR4	6.0 - 9.0	5	30	25	4
Tung Tze Stream	TR6	6.0 - 9.0	5	30	25	4
Port Shelter Water Control Zone						
Ho Chung River	PR1	6.5 - 8.5	5	30	25	4
	PR2	6.5 - 8.5	5	30	25	4
Sha Kok Mei Stream	PR5	6.0 - 9.0	5	30	25	4
	PR6	6.0 - 9.0	5	30	25	4
Tai Chung Hau Stream	PR7	6.0 - 9.0	5	30	25	4
	PR8	6.0 - 9.0	5	30	25	4
Junk Bay Water Control Zone						
Tseng Lan Shue Stream	JR3	6.0 - 9.0	5	30	25	4
	JR6	6.0 - 9.0	5	30	25	4
	JR11	6.0 - 9.0	5	30	25	4

* The WQO compliance for suspended solids is based on annual median value, while WQO compliance for other parameters is based on individual measurements.

Key Water Quality Objectives (WQOs) for river monitoring stations in the Northwestern New Territories

Watercourse	Monitoring station	Key Water Quality Objective				
		pH range	Maximum 5-Day Biochemical Oxygen Demand (mg/L)	Maximum Chemical Oxygen Demand (mg/L)	Maximum Annual Median Suspended Solids (mg/L)	Minimum Dissolved Oxygen (mg/L)
Deep Bay Water Control Zone						
River Indus	IN1	6.5 - 8.5	3	15	20	4
	IN2	6.5 - 8.5	3	15	20	4
	IN3	6.5 - 8.5	3	15	20	4
River Beas	RB1	6.5 - 8.5	3	15	20	4
	RB2	6.5 - 8.5	3	15	20	4
	RB3	6.5 - 8.5	3	15	20	4
River Ganges	GR1	6.5 - 8.5	3	15	20	4
	GR2	6.5 - 8.5	3	15	20	4
	GR3	6.5 - 8.5	3	15	20	4
Yuen Long Creek	YL1	6.5 - 8.5	3	15	20	4
	YL2	6.5 - 8.5	3	15	20	4
	YL3	6.5 - 8.5	5	30	20	4
	YL4	6.5 - 8.5	5	30	20	4
Kam Tin River	KT1	6.5 - 8.5	3	15	20	4
	KT2	6.5 - 8.5	3	15	20	4
Tin Shui Wai Nullah	TSR1	6.0 - 9.0	5	30	20	4
	TSR2	6.0 - 9.0	5	30	20	4
Fairview Park Nullah	FVR1	6.0 - 9.0	5	30	20	4
Ha Pak Nai Stream	DB1	6.0 - 9.0	5	30	20	4
Tai Shui Hang Stream	DB2	6.0 - 9.0	5	30	20	4
Pak Nai Stream	DB3	6.0 - 9.0	5	30	20	4
Sheung Pak Nai Stream	DB5	6.0 - 9.0	5	30	20	4
Ngau Hom Sha Stream	DB6	6.0 - 9.0	5	30	20	4
Tsang Kok Stream	DB8	6.0 - 9.0	5	30	20	4

* The WQO compliance for suspended solids is based on annual median value, while WQO compliance for other parameters is based on individual measurements.

Key Water Quality Objectives (WQOs) for river monitoring stations on Lantau Island

Watercourse	Monitoring station	Key Water Quality Objective				
		pH range	Maximum 5-Day Biochemical Oxygen Demand (mg/L)	Maximum Chemical Oxygen Demand (mg/L)	Maximum Annual Median Suspended Solids (mg/L)	Minimum Dissolved Oxygen (mg/L)
Southern Water Control Zone						
Mui Wo River	MW1	6.5 - 8.5	5	30	20	4
	MW2	6.5 - 8.5	5	30	20	4
	MW3	6.5 - 8.5	5	30	20	4
	MW4	6.5 - 8.5	5	30	20	4
	MW5	6.0 - 9.0	5	30	25	4
North Western Water Control Zone						
Tung Chung River	TC1	6.0 - 9.0	5	30	25	4
	TC2	6.0 - 9.0	5	30	25	4
	TC3	6.0 - 9.0	5	30	25	4

* The WQO compliance for suspended solids is based on annual median value, while WQO compliance for other parameters is based on individual measurements.

Key Water Quality Objectives (WQOs) for river monitoring stations in the Southwestern New Territories and Kowloon

Watercourse	Monitoring station	Key Water Quality Objective				
		pH range	Maximum 5-Day Biochemical Oxygen Demand (mg/L)	Maximum Chemical Oxygen Demand (mg/L)	Maximum Annual Median Suspended Solids (mg/L)	Minimum Dissolved Oxygen (mg/L)
North Western Water Control Zone						
Tuen Mun River	TN1	6.0 - 9.0	5	30	25	4
	TN2	6.5 - 8.5	3	15	20	4
	TN3	6.0 - 9.0	5	30	25	4
	TN4	6.0 - 9.0	5	30	25	4
	TN5	6.0 - 9.0	5	30	25	4
	TN6	6.0 - 9.0	5	30	25	4
Western Buffer Water Control Zone						
Pai Min Kok (Anglers') Stream	AN1	6.0 - 9.0	5	30	25	4
	AN2	6.0 - 9.0	5	30	25	4
Victoria Harbour Water Control Zone						
Sam Dip Tam Stream	TW1	6.0 - 9.0	5	30	25	4
	TW2	6.0 - 9.0	5	30	25	4
	TW3	6.0 - 9.0	5	30	25	4
Kau Wa Keng Stream	KW3	6.0 - 9.0	5	30	25	4
Kai Tak River	KN1	Not applicable				
	KN2					
	KN3					
	KN4					
	KN5					
	KN7					

* The WQO compliance for suspended solids is based on annual median value, while WQO compliance for other parameters is based on individual measurements.

Summary of water quality monitoring data for Shing Mun River (Main Channel and Siu Lek Yuen Nullah) in 2015

Parameter	Unit	Shing Mun Main Channel	Siu Lek Yuen Nullah	
		TR19I	TR23L	TR23A
Dissolved oxygen	mg/L	7.0 (5.6 - 8.3)	8.7 (7.8 - 10.4)	6.2 (4.7 - 9.1)
pH		7.9 (7.6 - 8.1)	8.7 (8.1 - 10.4)	7.6 (7.2 - 8.1)
Suspended solids	mg/L	3 (<1 - 26)	1 (<1 - 110)	3 (<1 - 10)
5-day Biochemical Oxygen Demand	mg/L	3 (2 - 5)	<1 (<1 - 3)	2 (<1 - 5)
Chemical Oxygen Demand	mg/L	13 (8 - 22)	3 (<2 - 5)	8 (5 - 17)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
Faecal coliforms	cfu/ 100mL	16,000 (900 - 560,000)	30,000 (4,200 - 130,000)	21,000 (2,500 - 160,000)
<i>E. coli</i>	cfu/ 100mL	1,300 (100 - 80,000)	1,800 (200 - 12,000)	3,400 (440 - 65,000)
Ammonia-nitrogen	mg/L	0.15 (0.05 - 0.37)	0.02 (<0.01 - 0.07)	0.31 (0.08 - 1.20)
Nitrate-nitrogen	mg/L	0.18 (0.03 - 0.75)	0.20 (0.06 - 0.49)	0.24 (0.08 - 0.48)
Total Kjeldahl Nitrogen	mg/L	0.53 (0.38 - 1.20)	0.25 (0.12 - 0.57)	0.70 (0.45 - 1.80)
Ortho-phosphate	mg/L	<0.01 (<0.01 - 0.03)	<0.01 (<0.01 - 0.02)	0.02 (<0.01 - 0.09)
Total phosphorus	mg/L	0.06 (0.03 - 0.16)	0.02 (<0.02 - 0.13)	0.08 (0.03 - 0.16)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	51 (<50 - 103)	71 (<50 - 1,534)	100 (<50 - 660)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - 0.2)	<0.1 (<0.1 - 0.1)
Chromium	µg/L	3 (<1 - 6)	<1 (<1 - 3)	3 (<1 - 5)
Copper	µg/L	6 (2 - 10)	1 (<1 - 12)	5 (1 - 11)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - 5)	<1 (<1 - 2)
Zinc	µg/L	17 (12 - 36)	16 (12 - 57)	22 (14 - 97)
Flow	L/s	NM	29 (9 - 66)	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. cfu - colony forming unit.
 5. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 6. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Shing Mun River (Fo Tan Nullah and Kwun Yam Shan Stream) in 2015

Parameter	Unit	<u>Fo Tan Nullah</u>		<u>Kwun Yam Shan Stream</u>
		TR17	TR17L	KY1
Dissolved oxygen	mg/L	8.9 (6.3 - 9.6)	6.1 (4.8 - 8.8)	8.3 (7.7 - 9.8)
pH		8.4 (7.3 - 9.2)	7.7 (7.5 - 7.9)	8.0 (7.7 - 8.4)
Suspended solids	mg/L	9 (1 - 44)	3 (1 - 14)	3 (1 - 190)
5-day Biochemical Oxygen Demand	mg/L	14 (3 - 81)	3 (<1 - 8)	<1 (<1 - 9)
Chemical Oxygen Demand	mg/L	18 (5 - 110)	13 (2 - 18)	4 (<2 - 9)
Oil & grease	mg/L	<0.5 (<0.5 - 5.9)	<0.5 (<0.5 - 0.5)	<0.5 (<0.5 - <0.5)
Faecal coliforms	cfu/ 100mL	88,000 (<100 - 3,500,000)	92,000 (8,800 - 2,300,000)	4,200 (510 - 190,000)
<i>E. coli</i>	cfu/ 100mL	14,000 (<100 - 300,000)	11,000 (420 - 1,400,000)	700 (74 - 26,000)
Ammonia-nitrogen	mg/L	0.25 (0.02 - 4.70)	0.33 (0.19 - 2.10)	0.02 (<0.01 - 3.40)
Nitrate-nitrogen	mg/L	0.62 (<0.01 - 0.97)	0.29 (0.14 - 0.70)	0.61 (0.27 - 5.90)
Total Kjeldahl nitrogen	mg/L	1.65 (0.70 - 16.00)	0.77 (0.33 - 3.40)	0.25 (0.13 - 3.80)
Ortho-phosphate	mg/L	0.05 (<0.01 - 1.50)	0.02 (0.01 - 0.15)	0.08 (0.04 - 1.30)
Total phosphorus	mg/L	0.22 (0.05 - 3.20)	0.08 (0.05 - 0.27)	0.12 (0.06 - 1.50)
Total sulphide	mg/L	<0.02 (<0.02 - 0.07)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	116 (<50 - 337)	81 (<50 - 138)	68 (<50 - 697)
Cadmium	µg/L	0.1 (<0.1 - 0.2)	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - 0.2)
Chromium	µg/L	1 (<1 - 12)	2 (<1 - 5)	<1 (<1 - <1)
Copper	µg/L	6 (3 - 16)	5 (3 - 7)	2 (<1 - 9)
Lead	µg/L	1 (<1 - 29)	<1 (<1 - 2)	<1 (<1 - 15)
Zinc	µg/L	51 (27 - 890)	21 (16 - 55)	19 (<10 - 56)
Flow	L/s	56 (28 - 150)	NM	6 (1 - 262)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. cfu - colony forming unit.
 5. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 6. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Shing Mun River (Tai Wai Nullah and Tin Sum Nullah) in 2015

Parameter	Unit	Tai Wai Nullah		Tin Sum Nullah	
		TR19A	TR19C	TR19	TR20B
Dissolved oxygen	mg/L	8.9 (7.8 - 10.7)	9.0 (8.0 - 12.8)	9.8 (8.2 - 14.0)	8.2 (7.7 - 9.6)
pH		8.4 (7.4 - 9.0)	8.0 (7.4 - 9.3)	8.3 (7.4 - 9.6)	7.9 (7.6 - 11.3)
Suspended solids	mg/L	2 (<1 - 48)	4 (<1 - 79)	3 (<1 - 39)	6 (1 - 57)
5-day Biochemical Oxygen Demand	mg/L	1 (<1 - 2)	2 (<1 - 5)	2 (<1 - 8)	<1 (<1 - 3)
Chemical Oxygen Demand	mg/L	5 (<2 - 11)	6 (3 - 11)	6 (4 - 12)	4 (<2 - 10)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - 0.9)	<0.5 (<0.5 - <0.5)
Faecal coliforms	cfu/ 100mL	12,000 (190 - 250,000)	38,000 (6,200 - 1,300,000)	65,000 (6,600 - 750,000)	7 (<1 - 15,000)
<i>E. coli</i>	cfu/ 100mL	2,700 (29 - 240,000)	4,500 (720 - 29,000)	4,300 (380 - 44,000)	4 (<1 - 1,000)
Ammonia-nitrogen	mg/L	0.06 (0.03 - 0.09)	0.05 (<0.01 - 0.12)	0.05 (0.01 - 0.10)	0.21 (0.02 - 2.70)
Nitrate-nitrogen	mg/L	0.94 (0.64 - 1.60)	0.88 (0.69 - 1.30)	0.85 (0.61 - 1.30)	2.00 (0.42 - 9.20)
Total Kjeldahl nitrogen	mg/L	0.37 (0.16 - 0.70)	0.51 (0.28 - 0.71)	0.55 (0.30 - 0.72)	0.49 (<0.05 - 4.70)
Ortho-phosphate	mg/L	0.02 (<0.01 - 0.05)	0.02 (<0.01 - 0.04)	0.02 (<0.01 - 0.04)	0.01 (<0.01 - 0.02)
Total phosphorus	mg/L	0.05 (<0.02 - 0.07)	0.05 (0.04 - 0.10)	0.05 (0.04 - 0.11)	<0.02 (<0.02 - 0.04)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - 0.03)	<0.02 (<0.02 - 0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	102 (69 - 430)	98 (65 - 348)	110 (58 - 282)	454 (143 - 2,498)
Cadmium	µg/L	0.5 (0.1 - 2.1)	0.2 (<0.1 - 0.3)	0.1 (<0.1 - 0.2)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 2)	<1 (<1 - 4)	<1 (<1 - 4)	3 (<1 - 15)
Copper	µg/L	2 (<1 - 4)	4 (2 - 7)	4 (3 - 10)	4 (2 - 9)
Lead	µg/L	1 (<1 - 5)	<1 (<1 - 8)	1 (<1 - 6)	1 (<1 - 3)
Zinc	µg/L	50 (27 - 125)	31 (24 - 46)	30 (24 - 46)	16 (<10 - 40)
Flow	L/s	25 (12 - 1,416)	53 (20 - 2,216)	67 (33 - 1,524)	69 (22 - 206)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. cfu - colony forming unit.
 5. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 6. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Lam Tsuen River in 2015 (Part 1 of 3)

Parameter	Unit	<u>Lam Tsuen River</u>		
		TR12H	TR12D	TR12C
Dissolved oxygen	mg/L	8.2 (7.6 - 10.0)	8.3 (7.6 - 9.8)	7.6 (4.7 - 8.7)
pH		7.4 (7.1 - 7.6)	7.3 (7.0 - 7.6)	7.4 (7.0 - 7.7)
Suspended solids	mg/L	1 (<1 - 4)	<1 (<1 - 13)	4 (<1 - 9)
5-day Biochemical Oxygen Demand	mg/L	<1 (<1 - 1)	<1 (<1 - <1)	6 (1 - 9)
Chemical Oxygen Demand	mg/L	<2 (<2 - 4)	<2 (<2 - 3)	9 (7 - 17)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - 0.5)
Faecal coliforms	cfu/ 100mL	8,300 (2,900 - 25,000)	4,100 (1,200 - 13,000)	140,000 (30,000 - 870,000)
<i>E. coli</i>	cfu/ 100mL	1,300 (230 - 3,200)	220 (34 - 800)	24,000 (9,300 - 120,000)
Ammonia-nitrogen	mg/L	0.06 (0.01 - 0.11)	0.02 (0.01 - 0.03)	1.25 (0.39 - 2.20)
Nitrate-nitrogen	mg/L	0.68 (0.56 - 1.40)	0.38 (0.21 - 0.94)	1.15 (0.91 - 1.50)
Total Kjeldahl Nitrogen	mg/L	0.25 (0.11 - 0.36)	0.15 (<0.05 - 0.25)	2.30 (0.97 - 3.30)
Ortho-phosphate	mg/L	0.04 (0.01 - 0.05)	0.01 (<0.01 - 0.02)	0.20 (0.11 - 0.26)
Total phosphorus	mg/L	0.05 (0.03 - 0.06)	<0.02 (<0.02 - 0.03)	0.29 (0.16 - 0.42)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - 0.07)
Aluminium	µg/L	<50 (<50 - 67)	<50 (<50 - 116)	54 (<50 - 127)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - 0.2)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - 1)	<1 (<1 - 1)
Copper	µg/L	<1 (<1 - 6)	1 (<1 - 2)	1 (<1 - 2)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - 1)	<1 (<1 - 2)
Zinc	µg/L	17 (14 - 57)	32 (23 - 78)	21 (16 - 48)
Flow	L/s	62 (18 - 243)	30 (9 - 116)	84 (63 - 186)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. cfu - colony forming unit.
 5. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 6. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Lam Tsuen River in 2015 (Part 2 of 3)

Parameter	Unit	<u>Lam Tsuen River</u>		
		TR12G	TR12F	TR12B
Dissolved oxygen	mg/L	7.7 (6.7 - 9.6)	8.3 (7.8 - 10.2)	8.6 (7.8 - 10.4)
pH		7.5 (6.9 - 7.8)	7.6 (7.1 - 7.8)	7.4 (7.0 - 7.7)
Suspended solids	mg/L	1 (<1 - 9)	4 (<1 - 22)	2 (<1 - 3)
5-day Biochemical Oxygen Demand	mg/L	<1 (<1 - <1)	<1 (<1 - 6)	<1 (<1 - 1)
Chemical Oxygen Demand	mg/L	3 (<2 - 7)	4 (<2 - 7)	3 (<2 - 5)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
Faecal coliforms	cfu/ 100mL	3,400 (350 - 16,000)	7,800 (1,700 - 51,000)	6,300 (830 - 24,000)
<i>E. coli</i>	cfu/ 100mL	140 (27 - 6,200)	480 (210 - 4,700)	650 (280 - 1,500)
Ammonia-nitrogen	mg/L	0.01 (<0.01 - 0.03)	0.03 (0.02 - 0.19)	0.04 (<0.01 - 0.07)
Nitrate-nitrogen	mg/L	0.04 (<0.01 - 0.34)	0.35 (0.31 - 0.83)	0.87 (0.51 - 1.90)
Total Kjeldahl nitrogen	mg/L	0.19 (<0.05 - 0.38)	0.28 (0.12 - 0.50)	0.29 (0.13 - 0.67)
Ortho-phosphate	mg/L	0.02 (<0.01 - 0.03)	0.04 (0.01 - 0.12)	0.05 (0.02 - 0.08)
Total phosphorus	mg/L	0.04 (<0.02 - 0.07)	0.08 (0.04 - 0.19)	0.06 (0.04 - 0.15)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	<50 (<50 - 120)	62 (<50 - 153)	51 (<50 - 96)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - 1)	<1 (<1 - 2)
Copper	µg/L	<1 (<1 - <1)	<1 (<1 - 2)	1 (<1 - 3)
Lead	µg/L	<1 (<1 - 2)	<1 (<1 - 4)	<1 (<1 - 2)
Zinc	µg/L	13 (<10 - 20)	20 (10 - 32)	18 (10 - 34)
Flow	L/s	28 (9 - 44)	30 (2 - 72)	143 (117 - 672)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. cfu - colony forming unit.
 5. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 6. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Lam Tsuen River (Part 3 of 3) and Tai Po River in 2015

Parameter	Unit	Lam Tsuen River			Tai Po River
		TR12E	TR12	TR12I	TR13
Dissolved oxygen	mg/L	8.2 (7.5 - 9.7)	8.4 (7.6 - 9.8)	5.4 (4.7 - 7.9)	6.9 (5.5 - 8.5)
pH		7.6 (7.4 - 7.8)	8.0 (7.6 - 8.3)	7.3 (7.1 - 7.5)	7.3 (7.0 - 7.6)
Suspended solids	mg/L	2 (<1 - 8)	6 (1 - 22)	3 (1 - 11)	2 (<1 - 10)
5-day Biochemical Oxygen Demand	mg/L	<1 (<1 - 1)	5 (1 - 18)	2 (1 - 6)	1 (<1 - 5)
Chemical Oxygen Demand	mg/L	3 (<2 - 5)	8 (4 - 16)	9 (3 - 17)	7 (4 - 17)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
Faecal coliforms	cfu/ 100mL	9,700 (4,400 - 32,000)	14,000 (5,000 - 37,000)	150,000 (40,000 - 510,000)	34,000 (1,500 - 170,000)
<i>E. coli</i>	cfu/ 100mL	2,400 (1,000 - 7,200)	1,900 (550 - 12,000)	45,000 (4,300 - 160,000)	9,600 (1,200 - 48,000)
Ammonia-nitrogen	mg/L	0.06 (0.03 - 0.24)	1.55 (0.16 - 4.60)	0.77 (0.16 - 1.60)	0.25 (0.13 - 0.47)
Nitrate-nitrogen	mg/L	0.79 (0.40 - 1.10)	1.25 (0.48 - 1.80)	0.68 (0.27 - 1.60)	0.53 (0.34 - 0.70)
Total Kjeldahl nitrogen	mg/L	0.24 (0.07 - 0.34)	3.05 (0.50 - 6.70)	1.25 (0.55 - 2.20)	0.52 (0.33 - 0.86)
Ortho-phosphate	mg/L	0.03 (<0.01 - 0.05)	0.28 (0.07 - 0.56)	0.10 (0.06 - 0.16)	0.05 (0.04 - 0.09)
Total phosphorus	mg/L	0.04 (0.02 - 0.09)	0.36 (0.09 - 0.68)	0.17 (0.10 - 0.23)	0.10 (0.06 - 0.12)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	116 (<50 - 654)	85 (54 - 152)	50 (<50 - 152)	<50 (<50 - 173)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - 0.3)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 1)	<1 (<1 - <1)	2 (<1 - 6)	2 (<1 - 4)
Copper	µg/L	<1 (<1 - 1)	2 (<1 - 3)	3 (<1 - 6)	3 (2 - 5)
Lead	µg/L	<1 (<1 - 2)	<1 (<1 - 3)	<1 (<1 - <1)	<1 (<1 - 2)
Zinc	µg/L	19 (11 - 33)	21 (13 - 35)	15 (11 - 57)	16 (10 - 25)
Flow	L/s	134 (82 - 167)	48 (14 - 464)	NM	120 (31 - 153)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. cfu - colony forming unit.
 5. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 6. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Tai Po Kau Stream, Shan Liu Stream and Tung Tze Stream in 2015

Parameter	Unit	<u>Tai Po Kau Stream</u>	<u>Shan Liu Stream</u>	<u>Tung Tze Stream</u>
		TR14	TR4	TR6
Dissolved oxygen	mg/L	7.8 (6.7 - 9.5)	8.1 (4.7 - 9.2)	6.5 (4.6 - 8.1)
pH		7.3 (6.9 - 7.6)	7.8 (7.2 - 8.3)	7.7 (7.2 - 8.0)
Suspended solids	mg/L	1 (<1 - 3)	5 (2 - 15)	7 (2 - 21)
5-day Biochemical Oxygen Demand	mg/L	<1 (<1 - 3)	1 (<1 - 2)	2 (<1 - 5)
Chemical Oxygen Demand	mg/L	4 (<2 - 9)	5 (<2 - 7)	11 (6 - 33)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
Faecal coliforms	cfu/ 100mL	1,800 (81 - 10,000)	10,000 (830 - 63,000)	5,700 (340 - 36,000)
<i>E. coli</i>	cfu/ 100mL	260 (38 - 1,400)	1,600 (200 - 9,000)	2,900 (80 - 20,000)
Ammonia-nitrogen	mg/L	0.06 (0.02 - 0.21)	0.14 (<0.01 - 0.47)	0.40 (0.07 - 2.20)
Nitrate-nitrogen	mg/L	0.19 (0.08 - 0.41)	0.63 (0.39 - 2.10)	0.17 (<0.01 - 0.63)
Total Kjeldahl nitrogen	mg/L	0.26 (0.16 - 0.88)	0.37 (0.28 - 1.10)	0.74 (0.10 - 3.20)
Ortho-phosphate	mg/L	0.01 (<0.01 - 0.05)	0.07 (0.04 - 0.18)	0.04 (<0.01 - 0.23)
Total phosphorus	mg/L	0.03 (<0.02 - 0.11)	0.10 (0.08 - 0.20)	0.11 (0.03 - 0.29)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	<50 (<50 - 122)	60 (<50 - 180)	74 (<50 - 517)
Cadmium	µg/L	<0.1 (<0.1 - 0.3)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	1 (<1 - 4)	<1 (<1 - 1)	2 (<1 - 4)
Copper	µg/L	1 (<1 - 5)	1 (<1 - 2)	5 (2 - 7)
Lead	µg/L	<1 (<1 - 2)	<1 (<1 - <1)	<1 (<1 - 3)
Zinc	µg/L	17 (10 - 36)	16 (11 - 27)	22 (11 - 46)
Flow	L/s	30 (14 - 71)	51 (26 - 83)	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. cfu - colony forming unit.
 5. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 6. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Ho Chung River in 2015

Parameter	Unit	<u>Ho Chung River</u>	
		PR1	PR2
Dissolved oxygen	mg/L	6.9 (5.7 - 7.8)	8.2 (7.7 - 8.8)
pH		7.1 (6.9 - 7.4)	7.5 (7.1 - 7.7)
Suspended solids	mg/L	4 (<1 - 24)	3 (1 - 7)
5-day Biochemical Oxygen Demand	mg/L	3 (1 - 6)	<1 (<1 - 1)
Chemical Oxygen Demand	mg/L	10 (5 - 17)	3 (<2 - 4)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
Faecal coliforms	cfu/ 100mL	89,000 (31,000 - 270,000)	8,000 (2,800 - 26,000)
<i>E. coli</i>	cfu/ 100mL	27,000 (11,000 - 67,000)	1,100 (350 - 4,800)
Ammonia-nitrogen	mg/L	1.60 (0.25 - 3.70)	0.05 (0.03 - 0.12)
Nitrate-nitrogen	mg/L	0.30 (0.19 - 0.56)	0.32 (0.14 - 0.53)
Total Kjeldahl nitrogen	mg/L	3.05 (0.83 - 5.20)	0.25 (0.17 - 0.47)
Ortho-phosphate	mg/L	0.15 (0.04 - 0.33)	0.01 (<0.01 - 0.02)
Total phosphorus	mg/L	0.23 (0.11 - 0.64)	0.03 (<0.02 - 0.04)
Total sulphide	mg/L	<0.02 (<0.02 - 0.03)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	77 (<50 - 148)	66 (<50 - 115)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	1 (<1 - 3)	<1 (<1 - 3)
Copper	µg/L	3 (1 - 5)	1 (<1 - 15)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - 1)
Zinc	µg/L	22 (14 - 79)	17 (11 - 24)
Flow	L/s	NM	335 (210 - 520)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. cfu - colony forming unit.
 5. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 6. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Sha Kok Mei Stream in 2015

Parameter	Unit	Sha Kok Mei Stream	
		PR5	PR6
Dissolved oxygen	mg/L	7.4 (6.5 - 8.5)	8.4 (7.9 - 9.4)
pH		7.5 (7.1 - 7.9)	7.4 (7.0 - 8.0)
Suspended solids	mg/L	2 (<1 - 8)	2 (<1 - 10)
5-day Biochemical Oxygen Demand	mg/L	1 (<1 - 2)	1 (<1 - 2)
Chemical Oxygen Demand	mg/L	5 (4 - 7)	4 (<2 - 6)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
Faecal coliforms	cfu/ 100mL	2,900 (<10 - 79,000)	31,000 (15,000 - 83,000)
<i>E. coli</i>	cfu/ 100mL	800 (<10 - 48,000)	11,000 (3,400 - 57,000)
Ammonia-nitrogen	mg/L	0.16 (0.05 - 0.54)	0.15 (0.04 - 0.37)
Nitrate-nitrogen	mg/L	0.91 (0.69 - 1.20)	1.65 (1.20 - 2.50)
Total Kjeldahl nitrogen	mg/L	0.59 (0.30 - 1.10)	0.47 (0.07 - 0.81)
Ortho-phosphate	mg/L	0.08 (0.05 - 0.12)	0.08 (0.05 - 0.16)
Total phosphorus	mg/L	0.12 (0.07 - 0.21)	0.11 (0.06 - 0.18)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	<50 (<50 - 304)	50 (<50 - 115)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 2)	<1 (<1 - 1)
Copper	µg/L	1 (<1 - 3)	1 (<1 - 4)
Lead	µg/L	<1 (<1 - 2)	<1 (<1 - <1)
Zinc	µg/L	18 (11 - 26)	19 (<10 - 45)
Flow	L/s	70 (52 - 464)	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. cfu - colony forming unit.
 5. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 6. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Tai Chung Hau Stream in 2015

Parameter	Unit	Tai Chung Hau Stream	
		PR7	PR8
Dissolved oxygen	mg/L	8.5 (7.6 - 9.4)	7.9 (7.2 - 9.0)
pH		7.5 (7.3 - 7.7)	7.8 (7.2 - 8.2)
Suspended solids	mg/L	3 (<1 - 14)	3 (<1 - 18)
5-day Biochemical Oxygen Demand	mg/L	1 (<1 - 2)	3 (<1 - 6)
Chemical Oxygen Demand	mg/L	5 (3 - 9)	4 (3 - 8)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
Faecal coliforms	cfu/ 100mL	28,000 (8,500 - 88,000)	34,000 (8,800 - 97,000)
<i>E. coli</i>	cfu/ 100mL	8,600 (3,300 - 39,000)	10,000 (2,500 - 73,000)
Ammonia-nitrogen	mg/L	0.10 (0.04 - 0.35)	0.12 (0.02 - 0.56)
Nitrate-nitrogen	mg/L	0.72 (0.44 - 1.30)	1.10 (0.54 - 1.50)
Total Kjeldahl Nitrogen	mg/L	0.53 (0.21 - 1.10)	0.53 (0.28 - 1.50)
Ortho-phosphate	mg/L	0.06 (0.02 - 0.09)	0.06 (0.01 - 0.11)
Total phosphorus	mg/L	0.09 (0.04 - 0.15)	0.11 (0.06 - 0.19)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	87 (<50 - 313)	59 (<50 - 626)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 3)	<1 (<1 - 1)
Copper	µg/L	2 (<1 - 4)	1 (<1 - 23)
Lead	µg/L	<1 (<1 - 2)	<1 (<1 - 2)
Zinc	µg/L	20 (<10 - 51)	18 (<10 - 38)
Flow	L/s	90 (12 - 198)	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. cfu - colony forming unit.
 5. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 6. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Tseng Lan Shue Stream in 2015

Parameter	Unit	Tseng Lan Shue Stream		
		JR3	JR6	JR11
Dissolved oxygen	mg/L	5.9 (5.3 - 7.5)	7.7 (7.2 - 8.5)	8.6 (7.9 - 9.8)
pH		7.2 (6.9 - 7.3)	7.4 (7.3 - 7.6)	7.5 (7.0 - 7.9)
Suspended solids	mg/L	8 (1 - 40)	4 (2 - 10)	2 (<1 - 6)
5-day Biochemical Oxygen Demand	mg/L	10 (3 - 17)	5 (2 - 25)	1 (<1 - 2)
Chemical Oxygen Demand	mg/L	16 (8 - 35)	11 (6 - 25)	6 (3 - 11)
Oil & grease	mg/L	<0.5 (<0.5 - 1.5)	<0.5 (<0.5 - 1.3)	<0.5 (<0.5 - <0.5)
Faecal coliforms	cfu/ 100mL	130,000 (50,000 - 600,000)	110,000 (49,000 - 320,000)	5,800 (620 - 47,000)
<i>E. coli</i>	cfu/ 100mL	60,000 (30,000 - 170,000)	45,000 (12,000 - 110,000)	880 (270 - 10,000)
Ammonia-nitrogen	mg/L	5.10 (0.71 - 16.00)	0.25 (0.05 - 1.40)	0.07 (0.02 - 0.22)
Nitrate-nitrogen	mg/L	1.25 (0.49 - 1.90)	2.35 (0.77 - 3.00)	3.35 (1.00 - 5.40)
Total Kjeldahl nitrogen	mg/L	8.35 (1.40 - 19.00)	1.40 (0.53 - 2.90)	0.52 (<0.05 - 1.00)
Ortho-phosphate	mg/L	0.59 (0.16 - 1.20)	0.40 (0.12 - 0.54)	0.42 (0.12 - 0.68)
Total phosphorus, SP	mg/L	0.77 (0.29 - 1.50)	0.61 (0.19 - 0.77)	0.48 (0.16 - 0.74)
Total sulphide	mg/L	<0.02 (<0.02 - 0.05)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - 0.03)
Aluminium	µg/L	133 (67 - 322)	101 (57 - 221)	<50 (<50 - 178)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - 0.2)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Copper	µg/L	3 (2 - 5)	3 (2 - 4)	2 (<1 - 7)
Lead	µg/L	1 (<1 - 3)	<1 (<1 - 2)	<1 (<1 - 1)
Zinc	µg/L	29 (22 - 40)	29 (16 - 80)	20 (13 - 26)
Flow	L/s	NM	NM	64 (44 - 284)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. cfu - colony forming unit.
 5. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 6. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for River Indus in 2015

Parameter	Unit	River Indus		
		IN1	IN2	IN3
Dissolved oxygen	mg/L	5.6 (3.1 - 7.8)	7.2 (4.4 - 8.5)	8.2 (7.3 - 9.4)
pH		7.2 (6.9 - 7.5)	7.3 (7.0 - 7.7)	7.6 (7.1 - 9.9)
Suspended solids	mg/L	37 (16 - 70)	5 (3 - 110)	4 (<1 - 65)
5-day Biochemical Oxygen Demand	mg/L	6 (2 - 47)	3 (1 - 5)	1 (<1 - 3)
Chemical Oxygen Demand	mg/L	20 (12 - 27)	8 (5 - 16)	7 (3 - 11)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
Faecal coliforms	cfu/ 100mL	290,000 (25,000 - 1,300,000)	11,000 (310 - 320,000)	7,900 (90 - 71,000)
<i>E. coli</i>	cfu/ 100mL	110,000 (15,000 - 800,000)	1,400 (100 - 16,000)	740 (50 - 4,900)
Ammonia-nitrogen	mg/L	2.55 (0.10 - 6.50)	0.51 (0.08 - 2.50)	0.08 (0.03 - 0.13)
Nitrate-nitrogen	mg/L	2.85 (0.79 - 5.40)	1.20 (0.64 - 1.90)	0.81 (0.45 - 1.90)
Total Kjeldahl nitrogen	mg/L	5.70 (0.92 - 9.80)	1.25 (0.65 - 3.40)	0.53 (0.36 - 0.94)
Ortho-phosphate	mg/L	0.25 (0.15 - 0.45)	0.12 (0.05 - 0.17)	0.12 (0.07 - 0.18)
Total phosphorus	mg/L	0.54 (0.35 - 0.79)	0.19 (0.15 - 0.30)	0.16 (0.10 - 0.22)
Total sulphide	mg/L	<0.02 (<0.02 - 0.03)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	282 (170 - 626)	77 (<50 - 440)	134 (<50 - 869)
Cadmium	µg/L	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	2 (<1 - 3)	<1 (<1 - 1)	4 (<1 - 34)
Copper	µg/L	5 (3 - 10)	2 (2 - 6)	2 (<1 - 4)
Lead	µg/L	3 (2 - 11)	1 (<1 - 9)	<1 (<1 - 2)
Zinc	µg/L	41 (16 - 65)	19 (15 - 44)	18 (<10 - 41)
Flow	L/s	NM	NM	19 (12 - 1,335)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. cfu - colony forming unit.
 5. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 6. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for River Beas in 2015

Parameter	Unit	River Beas		
		RB1	RB2	RB3
Dissolved oxygen	mg/L	9.5 (8.1 - 11.4)	7.8 (6.5 - 8.8)	7.9 (4.3 - 9.6)
pH		7.8 (7.2 - 8.8)	7.4 (7.0 - 7.5)	7.4 (7.1 - 7.6)
Suspended solids	mg/L	5 (2 - 18)	5 (3 - 54)	17 (5 - 64)
5-day Biochemical Oxygen Demand	mg/L	2 (<1 - 11)	4 (2 - 12)	6 (3 - 13)
Chemical Oxygen Demand	mg/L	10 (7 - 21)	11 (7 - 22)	16 (8 - 38)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
Faecal coliforms	cfu/ 100mL	59,000 (9,000 - 540,000)	20,000 (2,000 - 210,000)	100,000 (4,800 - 890,000)
<i>E. coli</i>	cfu/ 100mL	12,000 (1,800 - 75,000)	2,000 (160 - 28,000)	19,000 (2,000 - 220,000)
Ammonia-nitrogen	mg/L	0.13 (0.04 - 2.10)	0.81 (0.13 - 3.60)	1.45 (0.09 - 3.50)
Nitrate-nitrogen	mg/L	0.98 (0.54 - 1.70)	0.71 (0.36 - 1.90)	0.86 (0.63 - 2.90)
Total Kjeldahl nitrogen	mg/L	1.09 (0.33 - 3.60)	2.00 (0.95 - 6.20)	3.40 (0.83 - 7.30)
Ortho-phosphate	mg/L	0.27 (0.16 - 0.38)	0.19 (0.11 - 0.43)	0.22 (0.10 - 0.35)
Total phosphorus	mg/L	0.41 (0.23 - 0.56)	0.40 (0.22 - 0.80)	0.41 (0.27 - 0.72)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - 0.03)	<0.02 (<0.02 - 0.03)
Aluminium	µg/L	92 (<50 - 633)	54 (<50 - 271)	147 (62 - 402)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - 2)
Copper	µg/L	2 (1 - 4)	2 (1 - 5)	4 (3 - 12)
Lead	µg/L	<1 (<1 - 5)	<1 (<1 - 4)	2 (<1 - 8)
Zinc	µg/L	18 (13 - 34)	18 (14 - 53)	45 (15 - 236)
Flow	L/s	105 (56 - 4,650)	83 (63 - 6,780)	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. cfu - colony forming unit.
 5. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 6. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for River Ganges in 2015

Parameter	Unit	River Ganges		
		GR1	GR2	GR3
Dissolved oxygen	mg/L	8.0 (6.1 - 11.4)	6.3 (4.9 - 7.9)	7.9 (6.5 - 8.8)
pH		7.5 (6.9 - 8.0)	7.2 (6.9 - 7.4)	7.4 (7.0 - 7.7)
Suspended solids	mg/L	14 (4 - 140)	4 (2 - 85)	5 (<1 - 16)
5-day Biochemical Oxygen Demand	mg/L	13 (2 - 220)	3 (1 - 15)	<1 (<1 - 2)
Chemical Oxygen Demand	mg/L	21 (8 - 300)	16 (6 - 28)	4 (<2 - 8)
Oil & grease	mg/L	<0.5 (<0.5 - 0.7)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
Faecal coliforms	cfu/ 100mL	67,000 (5,600 - 780,000)	23,000 (1,100 - 170,000)	6,100 (160 - 130,000)
<i>E. coli</i>	cfu/ 100mL	35,000 (1,500 - 760,000)	4,300 (700 - 50,000)	570 (45 - 11,000)
Ammonia-nitrogen	mg/L	6.65 (0.31 - 61.00)	0.72 (0.02 - 14.00)	0.09 (0.02 - 0.24)
Nitrate-nitrogen	mg/L	1.05 (<0.01 - 2.70)	1.05 (0.14 - 2.10)	0.57 (0.24 - 1.40)
Total Kjeldahl nitrogen	mg/L	10.35 (1.30 - 93.00)	2.50 (0.60 - 18.00)	0.50 (0.25 - 1.30)
Ortho-phosphate	mg/L	1.39 (0.23 - 5.00)	0.28 (0.19 - 0.52)	<0.01 (<0.01 - <0.01)
Total phosphorus	mg/L	1.87 (0.35 - 6.90)	0.49 (0.30 - 0.94)	0.03 (<0.02 - 0.11)
Total sulphide	mg/L	<0.02 (<0.02 - 0.03)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	81 (<50 - 742)	79 (<50 - 323)	63 (<50 - 270)
Cadmium	µg/L	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - 0.2)
Chromium	µg/L	<1 (<1 - 1)	<1 (<1 - 2)	<1 (<1 - <1)
Copper	µg/L	3 (2 - 7)	2 (1 - 9)	2 (<1 - 4)
Lead	µg/L	<1 (<1 - 6)	1 (<1 - 5)	<1 (<1 - 2)
Zinc	µg/L	27 (17 - 48)	29 (14 - 45)	22 (13 - 77)
Flow	L/s	12 (2 - 924)	21 (19 - 768)	53 (45 - 495)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. cfu - colony forming unit.
 5. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 6. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Yuen Long Creek in 2015 (Part 1 of 2)

Parameter	Unit	Yuen Long Creek	
		YL1	YL2
Dissolved oxygen	mg/L	5.5 (4.0 - 7.9)	6.0 (4.7 - 11.4)
pH		7.4 (7.1 - 7.6)	7.3 (7.1 - 7.9)
Suspended solids	mg/L	18 (4 - 91)	4 (<1 - 17)
5-day Biochemical Oxygen Demand	mg/L	25 (8 - 41)	7 (4 - 12)
Chemical Oxygen Demand	mg/L	26 (15 - 61)	26 (18 - 35)
Oil & grease	mg/L	0.6 (<0.5 - 3.3)	0.6 (<0.5 - 2.7)
Faecal coliforms	cfu/ 100mL	400,000 (110,000 - 970,000)	81,000 (24,000 - 490,000)
<i>E. coli</i>	cfu/ 100mL	180,000 (71,000 - 660,000)	47,000 (9,000 - 340,000)
Ammonia-nitrogen	mg/L	7.60 (1.20 - 18.00)	10.00 (2.00 - 14.00)
Nitrate-nitrogen	mg/L	0.46 (<0.01 - 1.10)	2.45 (1.00 - 9.40)
Total Kjeldahl nitrogen	mg/L	11.50 (3.20 - 23.00)	12.50 (4.40 - 19.00)
Ortho-phosphate	mg/L	1.20 (0.46 - 2.50)	2.55 (0.59 - 3.80)
Total phosphorus	mg/L	1.80 (0.77 - 3.00)	2.80 (0.85 - 4.00)
Total sulphide	mg/L	0.03 (<0.02 - 0.13)	<0.02 (<0.02 - 0.04)
Aluminium	µg/L	133 (99 - 371)	51 (<50 - 221)
Cadmium	µg/L	0.1 (<0.1 - 0.2)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 2)	<1 (<1 - 1)
Copper	µg/L	7 (5 - 11)	3 (2 - 8)
Lead	µg/L	2 (2 - 15)	<1 (<1 - 2)
Zinc	µg/L	53 (39 - 91)	32 (25 - 50)
Flow	L/s	155 (51 - 344)	14 (10 - 44)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. cfu - colony forming unit.
 5. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 6. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Yuen Long Creek in 2015 (Part 2 of 2)

Parameter	Unit	Yuen Long Creek	
		YL3	YL4
Dissolved oxygen	mg/L	3.9 (2.0 - 8.3)	4.6 (3.0 - 6.7)
pH		7.5 (7.3 - 9.7)	7.3 (7.0 - 7.6)
Suspended solids	mg/L	34 (11 - 130)	50 (11 - 98)
5-day Biochemical Oxygen Demand	mg/L	58 (29 - 131)	140 (18 - 210)
Chemical Oxygen Demand	mg/L	48 (17 - 190)	107 (28 - 210)
Oil & grease	mg/L	1.4 (<0.5 - 6.6)	6.2 (<0.5 - 18.0)
Faecal coliforms	cfu/ 100mL	2,400,000 (920,000 - 7,400,000)	5,000,000 (1,300,000 - 12,000,000)
<i>E. coli</i>	cfu/ 100mL	1,000,000 (260,000 - 2,500,000)	2,000,000 (510,000 - 5,100,000)
Ammonia-nitrogen	mg/L	6.15 (2.20 - 12.00)	7.40 (1.60 - 8.40)
Nitrate-nitrogen	mg/L	0.01 (<0.01 - 1.40)	<0.01 (<0.01 - 0.59)
Total Kjeldahl nitrogen	mg/L	10.00 (4.40 - 23.00)	12.00 (3.20 - 29.00)
Ortho-phosphate	mg/L	0.92 (0.28 - 3.00)	0.43 (0.06 - 0.75)
Total phosphorus	mg/L	1.35 (0.58 - 3.70)	1.35 (0.59 - 2.00)
Total sulphide	mg/L	0.08 (<0.02 - 0.10)	0.12 (<0.02 - 0.33)
Aluminium	µg/L	165 (79 - 1,384)	193 (139 - 372)
Cadmium	µg/L	<0.1 (<0.1 - 0.2)	<0.1 (<0.1 - 0.1)
Chromium	µg/L	<1 (<1 - 2)	1 (<1 - 2)
Copper	µg/L	8 (4 - 23)	6 (3 - 9)
Lead	µg/L	3 (1 - 17)	3 (1 - 9)
Zinc	µg/L	59 (31 - 228)	64 (40 - 100)
Flow	L/s	443 (110 - 700)	104 (96 - 171)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. cfu - colony forming unit.
 5. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 6. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Kam Tin River in 2015

Parameter	Unit	Kam Tin River	
		KT1	KT2
Dissolved oxygen	mg/L	5.8 (4.0 - 6.7)	4.3 (2.4 - 6.6)
pH		7.2 (7.1 - 7.6)	7.7 (7.3 - 9.0)
Suspended solids	mg/L	10 (3 - 38)	52 (<1 - 370)
5-day Biochemical Oxygen Demand	mg/L	12 (5 - 33)	31 (15 - 65)
Chemical Oxygen Demand	mg/L	16 (10 - 60)	28 (16 - 73)
Oil & grease	mg/L	<0.5 (<0.5 - 2.6)	<0.5 (<0.5 - 1.8)
Faecal coliforms	cfu/ 100mL	370,000 (170,000 - 880,000)	520,000 (91,000 - 2,600,000)
<i>E. coli</i>	cfu/ 100mL	78,000 (25,000 - 280,000)	300,000 (77,000 - 2,200,000)
Ammonia-nitrogen	mg/L	5.15 (2.00 - 9.70)	8.75 (3.80 - 13.00)
Nitrate-nitrogen	mg/L	0.69 (0.09 - 1.10)	0.32 (<0.01 - 0.70)
Total Kjeldahl nitrogen	mg/L	7.15 (3.40 - 19.00)	13.00 (6.90 - 19.00)
Ortho-phosphate	mg/L	0.85 (0.46 - 1.90)	1.30 (0.96 - 2.30)
Total phosphorus	mg/L	1.15 (0.73 - 3.00)	1.80 (1.40 - 3.40)
Total sulphide	mg/L	<0.02 (<0.02 - 0.20)	0.03 (<0.02 - 0.09)
Aluminium	µg/L	80 (59 - 285)	288 (148 - 4,539)
Cadmium	µg/L	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - 0.5)
Chromium	µg/L	<1 (<1 - 2)	1 (<1 - 6)
Copper	µg/L	17 (6 - 66)	8 (4 - 54)
Lead	µg/L	1 (<1 - 6)	2 (<1 - 146)
Zinc	µg/L	50 (23 - 118)	49 (19 - 159)
Flow	L/s	261 (81 - 540)	278 (135 - 576)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. cfu - colony forming unit.
 5. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 6. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Tin Shui Wai Nullah and Fairview Park Nullah in 2015

Parameter	Unit	Tin Shui Wai Nullah		Fairview Park Nullah
		TSR1	TSR2	FVR1
Dissolved oxygen	mg/L	5.1 (3.2 - 9.1)	10.8 (7.1 - 13.0)	6.3 (5.4 - 14.1)
pH		7.5 (7.2 - 8.5)	8.4 (7.4 - 9.6)	7.5 (7.2 - 8.8)
Suspended solids	mg/L	7 (2 - 11)	3 (2 - 21)	30 (7 - 47)
5-day Biochemical Oxygen Demand	mg/L	9 (4 - 41)	4 (<1 - 8)	6 (4 - 20)
Chemical Oxygen Demand	mg/L	16 (9 - 64)	7 (3 - 19)	20 (13 - 45)
Oil & grease	mg/L	<0.5 (<0.5 - 1.5)	<0.5 (<0.5 - 0.6)	<0.5 (<0.5 - 0.6)
Faecal coliforms	cfu/ 100mL	1,200,000 (130,000 - 11,000,000)	74,000 (3,500 - 860,000)	73,000 (16,000 - 280,000)
<i>E. coli</i>	cfu/ 100mL	300,000 (35,000 - 4,300,000)	24,000 (1,300 - 120,000)	29,000 (8,500 - 200,000)
Ammonia-nitrogen	mg/L	3.45 (0.45 - 5.80)	0.48 (0.05 - 1.60)	2.55 (0.33 - 3.20)
Nitrate-nitrogen	mg/L	0.38 (<0.01 - 1.50)	0.87 (0.59 - 1.60)	0.82 (0.36 - 1.80)
Total Kjeldahl nitrogen	mg/L	5.20 (1.40 - 11.00)	0.90 (0.36 - 4.00)	4.10 (2.80 - 5.20)
Ortho-phosphate	mg/L	0.31 (0.11 - 0.68)	0.10 (<0.01 - 0.24)	0.38 (0.15 - 0.65)
Total phosphorus	mg/L	0.47 (0.19 - 1.00)	0.15 (0.05 - 0.48)	0.59 (0.47 - 0.91)
Total sulphide	mg/L	<0.02 (<0.02 - 0.19)	<0.02 (<0.02 - <0.02)	0.02 (<0.02 - 0.04)
Aluminium	µg/L	107 (<50 - 183)	143 (75 - 330)	250 (153 - 405)
Cadmium	µg/L	<0.1 (<0.1 - 0.2)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - 1.3)
Chromium	µg/L	<1 (<1 - 2)	<1 (<1 - 2)	2 (<1 - 3)
Copper	µg/L	4 (2 - 6)	2 (<1 - 9)	4 (3 - 5)
Lead	µg/L	1 (<1 - 2)	1 (<1 - 2)	2 (1 - 3)
Zinc	µg/L	33 (23 - 50)	20 (<10 - 36)	37 (23 - 50)
Flow	L/s	NM	34 (20 - 58)	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. cfu - colony forming unit.
 5. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 6. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Ha Pak Nai Stream, Pak Nai Stream and Sheung Pak Nai Stream in 2015

Parameter	Unit	Ha Pak Nai Stream	Pak Nai Stream	Sheung Pak Nai Stream
		DB1	DB3	DB5
Dissolved oxygen	mg/L	8.5 (7.5 - 10.4)	8.0 (7.2 - 10.3)	8.2 (6.8 - 10.4)
pH		7.6 (7.4 - 8.6)	7.2 (7.0 - 7.5)	7.3 (7.0 - 8.5)
Suspended solids	mg/L	2 (1 - 5)	2 (1 - 11)	5 (2 - 1,300)
5-day Biochemical Oxygen Demand	mg/L	<1 (<1 - 1)	<1 (<1 - 1)	3 (<1 - 7)
Chemical Oxygen Demand	mg/L	<2 (<2 - 4)	<2 (<2 - 4)	4 (<2 - 27)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
Faecal coliforms	cfu/ 100mL	1,300 (550 - 6,500)	4,100 (440 - 34,000)	32,000 (880 - 200,000)
<i>E. coli</i>	cfu/ 100mL	77 (9 - 470)	420 (110 - 2,100)	15,000 (290 - 190,000)
Ammonia-nitrogen	mg/L	0.02 (<0.01 - 0.05)	0.03 (0.02 - 0.10)	0.79 (0.06 - 3.40)
Nitrate-nitrogen	mg/L	0.22 (0.17 - 0.60)	0.21 (0.15 - 0.40)	0.24 (0.19 - 0.47)
Total Kjeldahl nitrogen	mg/L	0.15 (0.06 - 0.59)	0.18 (0.11 - 0.43)	2.00 (0.42 - 11.00)
Ortho-phosphate	mg/L	<0.01 (<0.01 - <0.01)	<0.01 (<0.01 - 0.03)	0.06 (<0.01 - 0.31)
Total phosphorus	mg/L	<0.02 (<0.02 - 0.07)	<0.02 (<0.02 - 0.04)	0.11 (0.02 - 0.42)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - 0.04)
Aluminium	µg/L	119 (<50 - 282)	54 (<50 - 102)	64 (<50 - 5,299)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - 0.7)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - 2)	<1 (<1 - 2)
Copper	µg/L	<1 (<1 - 1)	<1 (<1 - 1)	1 (<1 - 151)
Lead	µg/L	<1 (<1 - 1)	<1 (<1 - 2)	1 (<1 - 850)
Zinc	µg/L	14 (<10 - 21)	15 (<10 - 30)	17 (12 - 128)
Flow	L/s	13 (5 - 32)	10 (5 - 45)	24 (7 - 30)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. cfu - colony forming unit.
 5. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 6. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Ngau Hom Sha Stream, Tai Shui Hang Stream and Tsang Kok Stream in 2015

Parameter	Unit	<u>Ngau Hom Sha Stream</u>	<u>Tai Shui Hang Stream</u>	<u>Tsang Kok Stream</u>
		DB6	DB2	DB8
Dissolved oxygen	mg/L	7.3 (5.1 - 9.0)	8.5 (7.0 - 10.6)	9.4 (7.0 - 11.4)
pH		6.9 (6.6 - 7.2)	7.9 (7.6 - 8.1)	7.8 (7.5 - 9.3)
Suspended solids	mg/L	9 (2 - 18)	2 (1 - 9)	5 (2 - 160)
5-day Biochemical Oxygen Demand	mg/L	1 (<1 - 3)	<1 (<1 - 2)	<1 (<1 - 18)
Chemical Oxygen Demand	mg/L	9 (4 - 12)	<2 (<2 - 6)	5 (<2 - 24)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
Faecal coliforms	cfu/ 100mL	5,700 (1,200 - 29,000)	1,600 (350 - 10,000)	4,600 (650 - 120,000)
<i>E. coli</i>	cfu/ 100mL	700 (72 - 9,000)	200 (27 - 3,300)	240 (21 - 33,000)
Ammonia-nitrogen	mg/L	0.22 (0.10 - 0.97)	0.02 (<0.01 - 0.16)	0.03 (0.02 - 0.19)
Nitrate-nitrogen	mg/L	0.24 (0.03 - 0.66)	0.19 (0.13 - 0.77)	0.89 (<0.01 - 2.60)
Total Kjeldahl nitrogen	mg/L	0.66 (0.41 - 1.20)	0.18 (0.07 - 0.39)	0.29 (0.17 - 3.20)
Ortho-phosphate	mg/L	0.03 (<0.01 - 0.07)	<0.01 (<0.01 - 0.01)	<0.01 (<0.01 - <0.01)
Total phosphorus	mg/L	0.16 (0.07 - 0.47)	<0.02 (<0.02 - 0.05)	<0.02 (<0.02 - 1.20)
Total sulphide	mg/L	<0.02 (<0.02 - 0.04)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - 0.03)
Aluminium	µg/L	54 (<50 - 1,428)	145 (<50 - 177)	162 (72 - 4,590)
Cadmium	µg/L	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - 0.6)
Chromium	µg/L	<1 (<1 - 4)	<1 (<1 - 7)	<1 (<1 - 5)
Copper	µg/L	2 (<1 - 5)	1 (<1 - 17)	2 (<1 - 62)
Lead	µg/L	<1 (<1 - 5)	<1 (<1 - 2)	1 (<1 - 88)
Zinc	µg/L	26 (16 - 42)	18 (<10 - 31)	25 (14 - 213)
Flow	L/s	5 (2 - 24)	240 (110 - 290)	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. cfu - colony forming unit.
 5. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 6. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Mui Wo River in 2015 (Part 1 of 2)

Parameter	Unit	Mui Wo River		
		MW1	MW2	MW3
Dissolved oxygen	mg/L	8.0 (7.5 - 10.0)	8.1 (7.0 - 10.7)	8.6 (7.7 - 11.6)
pH		7.5 (6.8 - 7.9)	7.5 (6.7 - 7.8)	7.5 (6.8 - 7.8)
Suspended solids	mg/L	1 (<1 - 5)	3 (1 - 15)	2 (<1 - 7)
5-day Biochemical Oxygen Demand	mg/L	<1 (<1 - 3)	<1 (<1 - 3)	<1 (<1 - 2)
Chemical Oxygen Demand	mg/L	5 (<2 - 7)	7 (3 - 11)	<2 (<2 - 4)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
Faecal coliforms	cfu/ 100mL	7,800 (620 - 45,000)	12,000 (1,000 - 48,000)	4,300 (380 - 27,000)
<i>E. coli</i>	cfu/ 100mL	600 (100 - 2,700)	2,300 (150 - 10,000)	210 (38 - 3,200)
Ammonia-nitrogen	mg/L	0.05 (0.02 - 0.24)	0.18 (0.03 - 1.20)	0.02 (0.01 - 0.06)
Nitrate-nitrogen	mg/L	0.36 (0.17 - 0.83)	0.25 (0.14 - 0.51)	0.42 (0.08 - 0.76)
Total Kjeldahl nitrogen	mg/L	0.32 (0.16 - 0.59)	0.47 (0.25 - 1.80)	0.17 (0.11 - 0.50)
Ortho-phosphate	mg/L	0.08 (0.03 - 0.16)	0.05 (0.02 - 0.16)	0.06 (0.02 - 0.09)
Total phosphorus	mg/L	0.10 (0.05 - 0.17)	0.11 (0.06 - 0.21)	0.09 (0.03 - 0.14)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	68 (<50 - 154)	69 (<50 - 221)	70 (<50 - 86)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - 0.2)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 3)	1 (<1 - 4)	<1 (<1 - 2)
Copper	µg/L	2 (<1 - 6)	1 (<1 - 6)	1 (<1 - 2)
Lead	µg/L	<1 (<1 - 1)	<1 (<1 - 4)	<1 (<1 - 2)
Zinc	µg/L	19 (13 - 37)	18 (12 - 30)	21 (12 - 26)
Flow	L/s	18 (10 - 450)	NM	14 (6 - 744)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. cfu - colony forming unit.
 5. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 6. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Mui Wo River in 2015 (Part 2 of 2)

Parameter	Unit	Mui Wo River	
		MW4	MW5
Dissolved oxygen	mg/L	7.5 (7.0 - 9.3)	7.6 (6.5 - 9.1)
pH		7.0 (6.8 - 7.8)	7.2 (6.8 - 7.8)
Suspended solids	mg/L	4 (1 - 95)	6 (1 - 48)
5-day Biochemical Oxygen Demand	mg/L	<1 (<1 - 7)	2 (<1 - 9)
Chemical Oxygen Demand	mg/L	7 (3 - 14)	9 (4 - 15)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
Faecal coliforms	cfu/ 100mL	2,500 (130 - 52,000)	27,000 (940 - 130,000)
<i>E. coli</i>	cfu/ 100mL	540 (41 - 4,000)	5,400 (260 - 21,000)
Ammonia-nitrogen	mg/L	0.21 (0.06 - 1.50)	0.57 (0.07 - 1.50)
Nitrate-nitrogen	mg/L	0.34 (0.11 - 0.70)	0.21 (0.10 - 0.53)
Total Kjeldahl nitrogen	mg/L	0.50 (0.29 - 1.80)	0.91 (0.40 - 2.50)
Ortho-phosphate	mg/L	0.05 (<0.01 - 0.19)	0.06 (<0.01 - 0.09)
Total phosphorus	mg/L	0.12 (0.04 - 0.32)	0.13 (0.05 - 0.26)
Total sulphide	mg/L	<0.02 (<0.02 - 0.04)	<0.02 (<0.02 - 0.03)
Aluminium	µg/L	90 (<50 - 310)	93 (51 - 283)
Cadmium	µg/L	<0.1 (<0.1 - 0.3)	<0.1 (<0.1 - 0.1)
Chromium	µg/L	2 (<1 - 4)	1 (<1 - 3)
Copper	µg/L	2 (1 - 7)	2 (<1 - 4)
Lead	µg/L	<1 (<1 - 5)	1 (<1 - 9)
Zinc	µg/L	27 (18 - 44)	20 (11 - 42)
Flow	L/s	55 (30 - 415)	56 (16 - 708)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. cfu - colony forming unit.
 5. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 6. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Tung Chung River in 2015

Parameter	Unit	Tung Chung River		
		TC1	TC2	TC3
Dissolved oxygen	mg/L	7.7 (6.9 - 9.4)	8.4 (7.7 - 9.6)	8.2 (7.6 - 9.5)
pH		7.0 (6.4 - 7.6)	7.4 (6.7 - 8.3)	7.8 (7.3 - 8.2)
Suspended solids	mg/L	<1 (<1 - 11)	2 (<1 - 9)	2 (<1 - 10)
5-day Biochemical Oxygen Demand	mg/L	<1 (<1 - 6)	<1 (<1 - 9)	7 (<1 - 23)
Chemical Oxygen Demand	mg/L	<2 (<2 - 11)	4 (<2 - 10)	8 (3 - 11)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - 0.8)
Faecal coliforms	cfu/ 100mL	1,700 (130 - 37,000)	5,400 (420 - 63,000)	90,000 (37,000 - 540,000)
<i>E. coli</i>	cfu/ 100mL	64 (11 - 2,900)	110 (15 - 3,100)	23,000 (3,100 - 60,000)
Ammonia-nitrogen	mg/L	0.02 (<0.01 - 0.05)	0.02 (0.01 - 0.04)	0.60 (0.05 - 2.40)
Nitrate-nitrogen	mg/L	0.04 (0.02 - 0.35)	0.01 (<0.01 - 0.25)	0.09 (0.04 - 0.45)
Total Kjeldahl nitrogen	mg/L	0.17 (0.10 - 0.53)	0.21 (0.12 - 0.44)	0.97 (0.18 - 4.10)
Ortho-phosphate	mg/L	<0.01 (<0.01 - 0.03)	<0.01 (<0.01 - 0.01)	0.06 (<0.01 - 0.29)
Total phosphorus	mg/L	<0.02 (<0.02 - 0.06)	<0.02 (<0.02 - 0.05)	0.11 (<0.02 - 0.40)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - 0.02)	<0.02 (<0.02 - 0.02)
Aluminium	µg/L	<50 (<50 - 308)	<50 (<50 - 263)	<50 (<50 - 328)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 2)	<1 (<1 - 2)	<1 (<1 - 1)
Copper	µg/L	<1 (<1 - 2)	<1 (<1 - 2)	1 (<1 - 26)
Lead	µg/L	<1 (<1 - 1)	<1 (<1 - 2)	<1 (<1 - 2)
Zinc	µg/L	16 (<10 - 24)	13 (<10 - 23)	16 (11 - 32)
Flow	L/s	59 (13 - 408)	67 (30 - 456)	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. cfu - colony forming unit.
 5. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 6. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Tuen Mun River in 2015 (Part 1 of 2)

Parameter	Unit	Tuen Mun River		
		TN1	TN2	TN3
Dissolved oxygen	mg/L	5.4 (3.9 - 7.0)	8.3 (4.2 - 10.0)	6.1 (4.4 - 7.3)
pH		7.8 (7.5 - 8.2)	7.4 (7.0 - 7.7)	7.6 (7.2 - 8.0)
Suspended solids	mg/L	11 (5 - 55)	5 (3 - 22)	3 (1 - 11)
5-day Biochemical Oxygen Demand	mg/L	31 (9 - 132)	7 (4 - 13)	2 (1 - 8)
Chemical Oxygen Demand	mg/L	25 (19 - 44)	13 (6 - 19)	9 (4 - 17)
Oil & grease	mg/L	0.7 (<0.5 - 2.5)	<0.5 (<0.5 - 0.7)	<0.5 (<0.5 - <0.5)
Faecal coliforms	cfu/ 100mL	710,000 (180,000 - 2,600,000)	150,000 (79,000 - 310,000)	44,000 (1,600 - 360,000)
<i>E. coli</i>	cfu/ 100mL	140,000 (61,000 - 310,000)	59,000 (15,000 - 110,000)	4,000 (470 - 39,000)
Ammonia-nitrogen	mg/L	6.35 (0.68 - 8.80)	2.90 (0.38 - 5.50)	0.37 (0.05 - 0.83)
Nitrate-nitrogen	mg/L	0.30 (<0.01 - 1.30)	1.40 (0.76 - 2.30)	0.52 (0.18 - 1.80)
Total Kjeldahl nitrogen	mg/L	9.90 (6.70 - 13.00)	6.40 (0.91 - 9.50)	0.76 (0.45 - 1.10)
Ortho-phosphate	mg/L	0.60 (0.29 - 0.84)	0.52 (0.08 - 0.77)	0.03 (0.01 - 0.08)
Total phosphorus	mg/L	0.85 (0.43 - 1.30)	0.63 (0.13 - 0.87)	0.07 (0.04 - 0.17)
Total sulphide	mg/L	0.04 (<0.02 - 0.09)	<0.02 (<0.02 - 0.05)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	80 (<50 - 231)	143 (62 - 347)	122 (61 - 269)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 1)	<1 (<1 - 1)	3 (<1 - 6)
Copper	µg/L	4 (2 - 6)	3 (2 - 4)	5 (3 - 9)
Lead	µg/L	<1 (<1 - 2)	1 (<1 - 7)	<1 (<1 - 3)
Zinc	µg/L	24 (21 - 45)	31 (22 - 48)	20 (<10 - 45)
Flow	L/s	153 (45 - 250)	8 (4 - 84)	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. cfu - colony forming unit.
 5. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 6. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Tuen Mun River in 2015 (Part 2 of 2)

Parameter	Unit	Tuen Mun River		
		TN4	TN5	TN6
Dissolved oxygen	mg/L	6.1 (3.8 - 7.5)	6.2 (4.8 - 7.6)	6.1 (4.5 - 7.9)
pH		7.6 (7.3 - 8.1)	7.6 (7.4 - 8.1)	7.4 (7.0 - 8.0)
Suspended solids	mg/L	3 (2 - 22)	3 (2 - 12)	2 (<1 - 16)
5-day Biochemical Oxygen Demand	mg/L	2 (1 - 12)	3 (<1 - 8)	2 (<1 - 6)
Chemical Oxygen Demand	mg/L	8 (5 - 33)	7 (5 - 21)	8 (5 - 11)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
Faecal coliforms	cfu/ 100mL	61,000 (7,600 - 1,500,000)	28,000 (3,500 - 290,000)	38,000 (6,300 - 3,400,000)
<i>E. coli</i>	cfu/ 100mL	8,100 (900 - 160,000)	2,900 (280 - 40,000)	3,600 (450 - 710,000)
Ammonia-nitrogen	mg/L	0.41 (0.10 - 1.10)	0.37 (0.10 - 0.99)	0.34 (0.09 - 0.57)
Nitrate-nitrogen	mg/L	0.44 (<0.01 - 1.50)	0.46 (0.28 - 1.80)	0.47 (0.26 - 1.40)
Total Kjeldahl nitrogen	mg/L	0.94 (0.60 - 1.60)	0.83 (0.53 - 1.50)	0.83 (0.40 - 1.20)
Ortho-phosphate	mg/L	0.02 (0.01 - 0.11)	0.03 (0.01 - 0.11)	0.03 (0.01 - 0.08)
Total phosphorus	mg/L	0.08 (0.05 - 0.23)	0.07 (0.04 - 0.19)	0.07 (0.04 - 0.18)
Total sulphide	mg/L	<0.02 (<0.02 - 0.03)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - 0.03)
Aluminium	µg/L	150 (74 - 467)	122 (72 - 274)	108 (51 - 303)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - 0.1)
Chromium	µg/L	2 (<1 - 7)	2 (<1 - 6)	2 (<1 - 6)
Copper	µg/L	5 (2 - 9)	6 (3 - 9)	5 (3 - 10)
Lead	µg/L	<1 (<1 - 5)	<1 (<1 - 3)	<1 (<1 - 4)
Zinc	µg/L	22 (11 - 32)	19 (12 - 24)	21 (14 - 32)
Flow	L/s	NM	NM	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. cfu - colony forming unit.
 5. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 6. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Pai Min Kok Stream and Kau Wa Keng Stream in 2015

Parameter	Unit	Pai Min Kok Stream		Kau Wa Keng Stream
		AN1	AN2	KW3
Dissolved oxygen	mg/L	8.5 (7.6 - 9.5)	9.1 (7.8 - 11.4)	8.8 (7.8 - 11.3)
pH		7.7 (7.5 - 8.3)	8.0 (7.7 - 9.0)	7.3 (7.2 - 8.0)
Suspended solids	mg/L	2 (<1 - 8)	2 (<1 - 4)	3 (1 - 13)
5-day Biochemical Oxygen Demand	mg/L	1 (<1 - 6)	<1 (<1 - 3)	3 (<1 - 15)
Chemical Oxygen Demand	mg/L	11 (5 - 21)	5 (4 - 25)	8 (5 - 23)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
Faecal coliforms	cfu/ 100mL	29,000 (2,700 - 330,000)	23,000 (2,900 - 110,000)	140,000 (45,000 - 680,000)
<i>E. coli</i>	cfu/ 100mL	5,400 (1,100 - 44,000)	7,000 (1,900 - 74,000)	68,000 (28,000 - 320,000)
Ammonia-nitrogen	mg/L	0.06 (0.03 - 1.10)	0.03 (0.01 - 0.11)	0.37 (0.24 - 1.10)
Nitrate-nitrogen	mg/L	1.05 (0.59 - 1.70)	0.73 (0.35 - 1.60)	2.10 (1.60 - 2.50)
Total Kjeldahl nitrogen	mg/L	0.63 (0.34 - 2.50)	0.47 (0.24 - 1.20)	0.94 (0.56 - 2.20)
Ortho-phosphate	mg/L	0.05 (0.02 - 0.12)	0.06 (0.03 - 0.15)	0.09 (0.06 - 0.21)
Total phosphorus	mg/L	0.09 (0.05 - 0.18)	0.10 (0.05 - 0.19)	0.19 (0.09 - 0.29)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - 0.04)
Aluminium	µg/L	86 (<50 - 230)	71 (<50 - 254)	132 (68 - 331)
Cadmium	µg/L	0.1 (<0.1 - 0.2)	<0.1 (<0.1 - <0.1)	1.1 (0.5 - 2.1)
Chromium	µg/L	1 (<1 - 4)	<1 (<1 - <1)	1 (<1 - 5)
Copper	µg/L	5 (3 - 27)	2 (<1 - 7)	4 (3 - 8)
Lead	µg/L	<1 (<1 - 1)	<1 (<1 - 2)	2 (1 - 5)
Zinc	µg/L	30 (20 - 44)	21 (14 - 62)	96 (58 - 194)
Flow	L/s	NM	4 (1 - 13)	20 (13 - 35)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. cfu - colony forming unit.
 5. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 6. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Sam Dip Tam Stream in 2015

Parameter	Unit	Sam Dip Tam Stream		
		TW1	TW2	TW3
Dissolved oxygen	mg/L	8.2 (7.4 - 9.5)	8.8 (8.0 - 10.3)	8.5 (7.8 - 9.7)
pH		7.8 (7.5 - 8.0)	8.0 (7.7 - 8.2)	8.1 (7.5 - 8.6)
Suspended solids	mg/L	2 (<1 - 4)	2 (<1 - 3)	2 (<1 - 8)
5-day Biochemical Oxygen Demand	mg/L	<1 (<1 - 50)	1 (<1 - 25)	<1 (<1 - 2)
Chemical Oxygen Demand	mg/L	3 (<2 - 5)	4 (<2 - 6)	4 (2 - 7)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
Faecal coliforms	cfu/ 100mL	8,100 (1,100 - 58,000)	27,000 (6,900 - 120,000)	16,000 (2,500 - 150,000)
<i>E. coli</i>	cfu/ 100mL	1,400 (240 - 7,000)	7,300 (1,300 - 89,000)	3,400 (900 - 32,000)
Ammonia-nitrogen	mg/L	0.02 (<0.01 - 0.05)	0.11 (0.05 - 0.81)	0.06 (0.03 - 0.12)
Nitrate-nitrogen	mg/L	0.91 (0.78 - 1.30)	1.15 (0.94 - 1.60)	1.55 (1.20 - 2.50)
Total Kjeldahl nitrogen	mg/L	0.22 (0.11 - 0.34)	0.46 (0.24 - 1.40)	0.35 (0.15 - 0.50)
Ortho-phosphate	mg/L	0.05 (0.04 - 0.07)	0.11 (0.07 - 0.23)	0.12 (0.09 - 0.19)
Total phosphorus	mg/L	0.07 (0.06 - 0.09)	0.13 (0.09 - 0.25)	0.15 (0.11 - 0.22)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	<50 (<50 - 128)	<50 (<50 - 74)	<50 (<50 - 57)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - 2)	<1 (<1 - 1)
Copper	µg/L	2 (1 - 3)	2 (1 - 4)	2 (1 - 4)
Lead	µg/L	1 (<1 - 4)	<1 (<1 - 2)	<1 (<1 - 1)
Zinc	µg/L	19 (12 - 32)	23 (16 - 68)	18 (12 - 31)
Flow	L/s	NM	36 (5 - 51)	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. cfu - colony forming unit.
 5. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 6. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Kai Tak River in 2015 (Part 1 of 2)

Parameter	Unit	Kai Tak River		
		KN1	KN2	KN3
Dissolved oxygen	mg/L	5.3 (3.1 - 6.4)	6.9 (5.7 - 8.9)	7.0 (5.3 - 9.2)
pH		7.1 (6.7 - 7.4)	7.4 (7.1 - 7.9)	7.3 (7.1 - 8.0)
Suspended solids	mg/L	4 (2 - 52)	11 (5 - 21)	12 (5 - 23)
5-day Biochemical Oxygen Demand	mg/L	5 (2 - 19)	6 (2 - 76)	6 (2 - 23)
Chemical Oxygen Demand	mg/L	22 (13 - 39)	23 (13 - 35)	23 (16 - 36)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - 0.7)	<0.5 (<0.5 - 0.6)
Faecal coliforms	cfu/ 100mL	770,000 (120,000 - 2,600,000)	340,000 (64,000 - 1,600,000)	310,000 (57,000 - 2,300,000)
<i>E. coli</i>	cfu/ 100mL	310,000 (43,000 - 1,700,000)	150,000 (35,000 - 710,000)	120,000 (31,000 - 510,000)
Ammonia-nitrogen	mg/L	1.35 (0.57 - 4.20)	0.74 (0.29 - 3.40)	0.70 (0.26 - 3.10)
Nitrate-nitrogen	mg/L	3.00 (0.73 - 5.00)	3.75 (1.10 - 5.10)	3.75 (1.10 - 5.10)
Total Kjeldahl nitrogen	mg/L	2.75 (1.60 - 7.80)	2.40 (1.30 - 4.10)	2.15 (1.20 - 4.50)
Ortho-phosphate	mg/L	1.35 (0.19 - 1.80)	1.65 (0.33 - 2.30)	1.70 (0.34 - 2.20)
Total phosphorus	mg/L	1.65 (0.33 - 2.00)	1.90 (0.55 - 2.60)	1.95 (0.53 - 2.50)
Total sulphide	mg/L	0.03 (<0.02 - 0.27)	<0.02 (<0.02 - 0.03)	<0.02 (<0.02 - 0.03)
Aluminium	µg/L	80 (<50 - 415)	133 (64 - 567)	148 (50 - 494)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	2 (<1 - 4)	2 (<1 - 4)	2 (1 - 4)
Copper	µg/L	6 (3 - 12)	9 (4 - 17)	9 (5 - 16)
Lead	µg/L	<1 (<1 - 6)	<1 (<1 - 5)	1 (<1 - 5)
Zinc	µg/L	30 (18 - 39)	36 (26 - 57)	40 (27 - 74)
Flow	L/s	NM	NM	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. cfu - colony forming unit.
 5. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 6. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

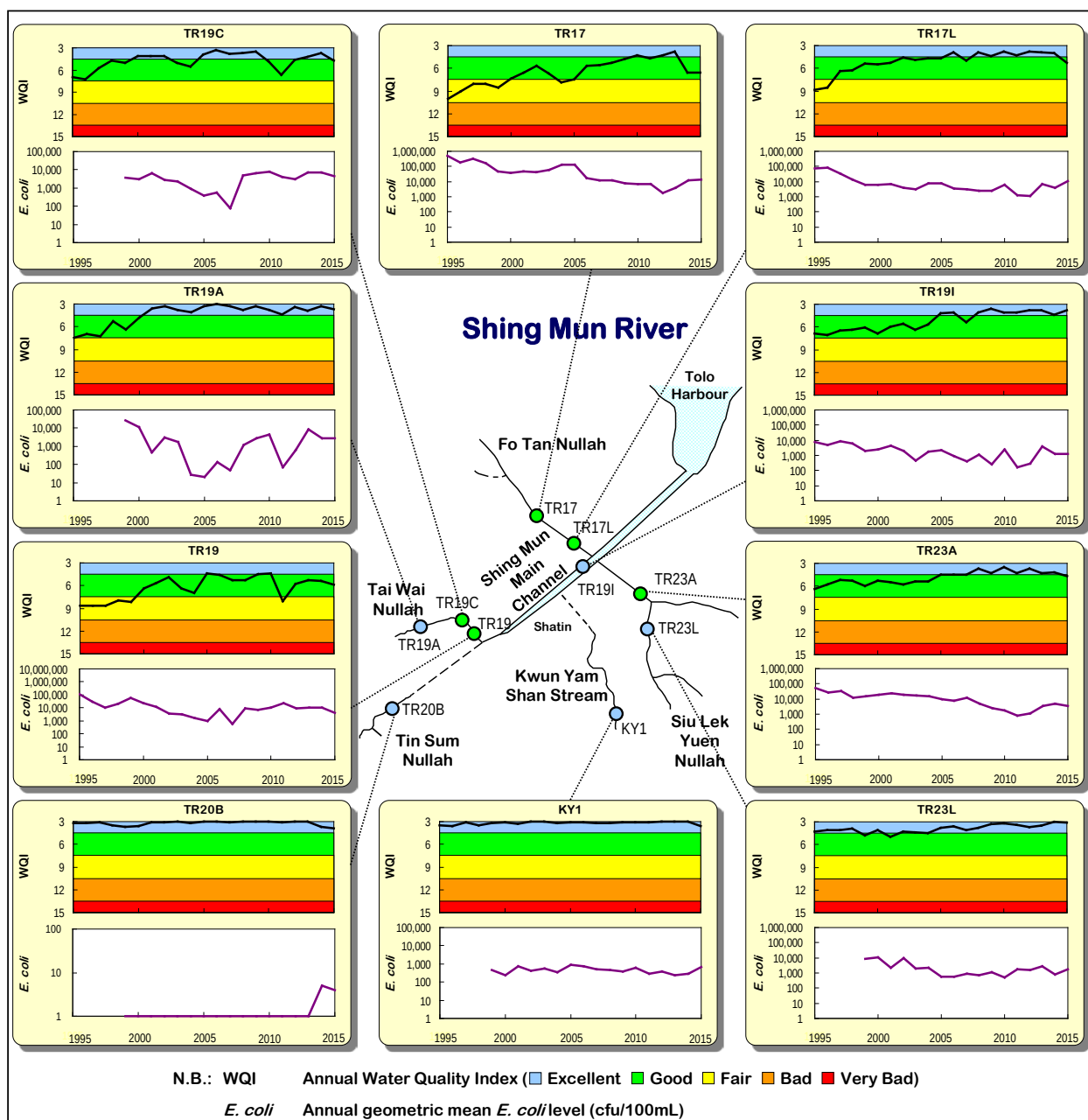
Summary of water quality monitoring data for Kai Tak River in 2015 (Part 2 of 2)

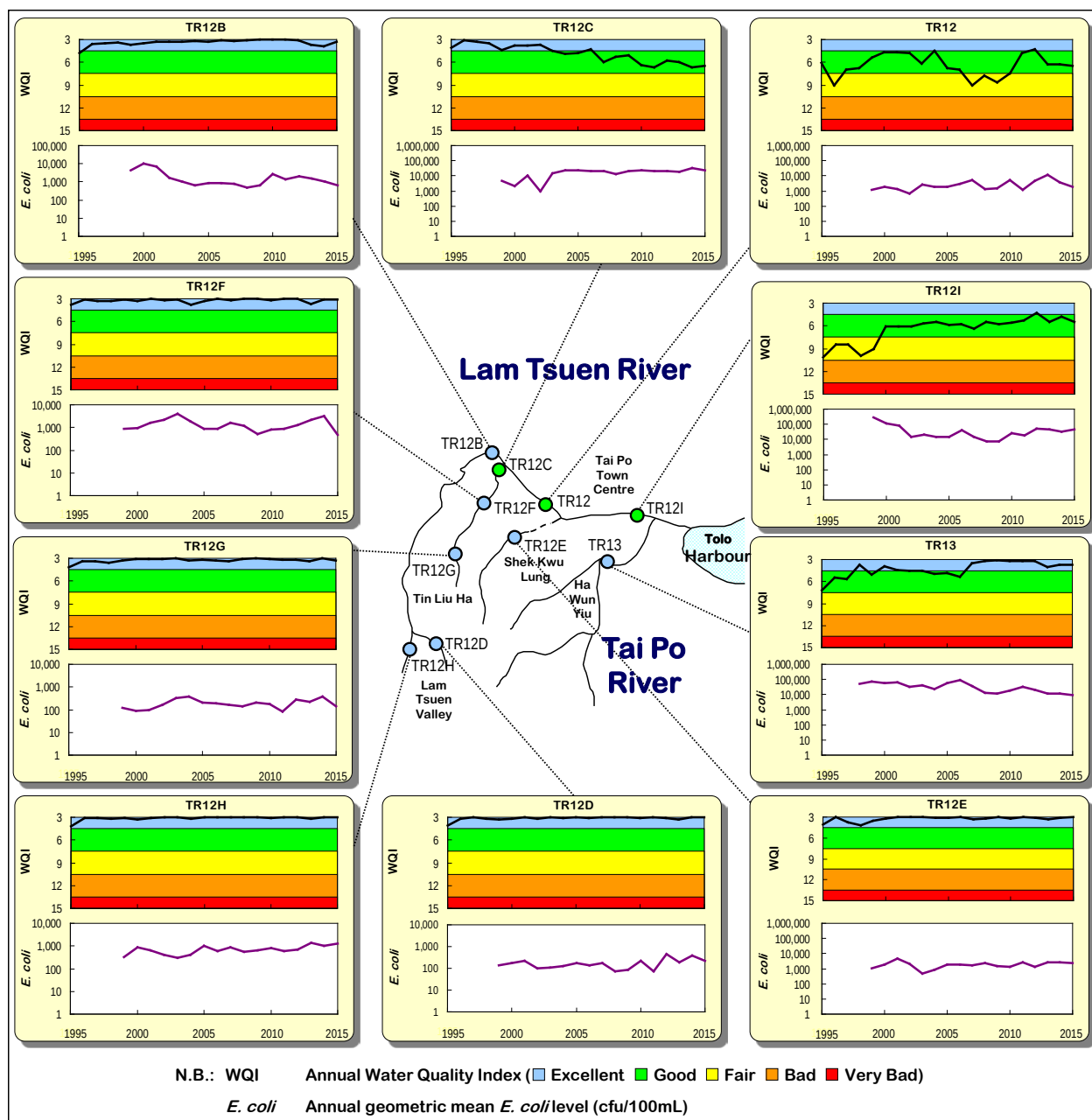
Parameter	Unit	Kai Tak River		
		KN4	KN5	KN7
Dissolved oxygen	mg/L	7.3 (6.2 - 8.3)	7.4 (6.1 - 8.3)	7.5 (6.7 - 8.4)
pH		7.3 (6.7 - 7.9)	7.3 (6.8 - 7.8)	7.2 (6.8 - 7.5)
Suspended solids	mg/L	13 (5 - 43)	9 (4 - 29)	6 (2 - 250)
5-day Biochemical Oxygen Demand	mg/L	9 (2 - 36)	7 (2 - 78)	5 (1 - 26)
Chemical Oxygen Demand	mg/L	25 (18 - 36)	27 (16 - 40)	25 (17 - 31)
Oil & grease	mg/L	<0.5 (<0.5 - 0.8)	<0.5 (<0.5 - 0.5)	<0.5 (<0.5 - 0.6)
Faecal coliforms	cfu/ 100mL	260,000 (64,000 - 2,400,000)	120,000 (31,000 - 960,000)	10,000 (2,200 - 1,300,000)
<i>E. coli</i>	cfu/ 100mL	94,000 (17,000 - 940,000)	48,000 (12,000 - 460,000)	4,800 (1,100 - 100,000)
Ammonia-nitrogen	mg/L	0.69 (0.16 - 2.80)	0.62 (0.16 - 2.40)	0.35 (0.09 - 2.80)
Nitrate-nitrogen	mg/L	4.05 (1.30 - 5.10)	3.85 (1.50 - 5.40)	4.25 (0.89 - 5.60)
Total Kjeldahl nitrogen	mg/L	2.45 (1.40 - 4.00)	2.15 (1.10 - 3.90)	1.90 (1.20 - 4.60)
Ortho-phosphate	mg/L	1.65 (0.52 - 1.90)	1.65 (0.72 - 1.90)	1.60 (0.16 - 2.70)
Total phosphorus	mg/L	1.90 (0.81 - 2.50)	1.90 (0.85 - 2.20)	1.80 (0.95 - 3.40)
Total sulphide	mg/L	<0.02 (<0.02 - 0.06)	<0.02 (<0.02 - 0.02)	<0.02 (<0.02 - 0.02)
Aluminium	µg/L	103 (<50 - 524)	105 (<50 - 449)	<50 (<50 - 852)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - 0.1)
Chromium	µg/L	2 (<1 - 4)	2 (<1 - 3)	2 (<1 - 3)
Copper	µg/L	11 (5 - 18)	9 (5 - 15)	10 (4 - 21)
Lead	µg/L	<1 (<1 - 4)	1 (<1 - 2)	<1 (<1 - 11)
Zinc	µg/L	35 (24 - 67)	33 (26 - 72)	35 (28 - 120)
Flow	L/s	NM	NM	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
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 5. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
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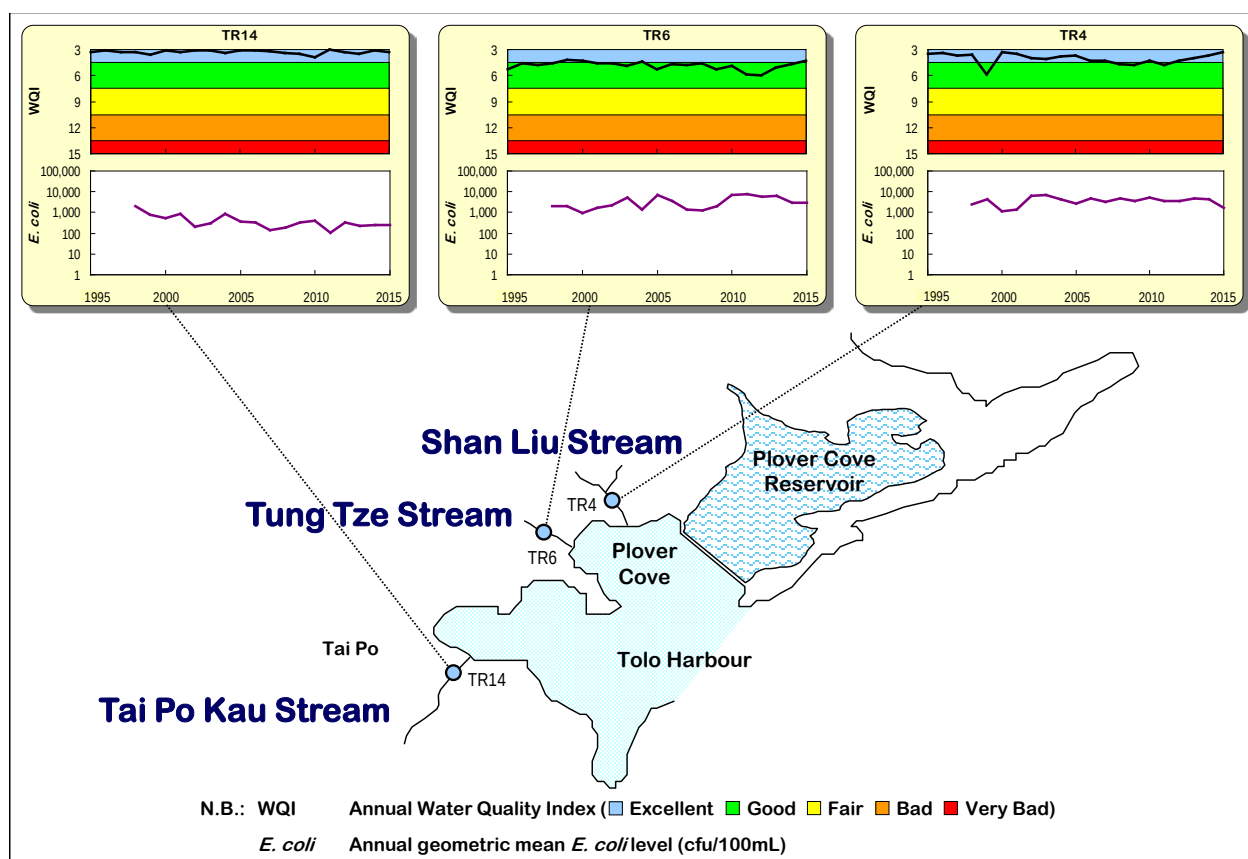
Compliance with the river Water Quality Objectives in 2015

Watercourse	pH	5-Day Biochemical Oxygen Demand	Chemical Oxygen Demand	Dissolved Oxygen	Suspended Solids	Overall Compliance
Eastern New Territories						
Shing Mun River	74%	83%	96%	100%	100%	91%
Lam Tsuen River	100%	83%	98%	100%	100%	96%
Tai Po River	100%	92%	100%	100%	100%	98%
Tai Po Kau Stream	100%	100%	100%	100%	100%	100%
Shan Liu Stream	100%	100%	100%	100%	100%	100%
Tung Tze Stream	100%	100%	92%	100%	100%	98%
Ho Chung River	100%	96%	100%	100%	100%	99%
Sha Kok Mei Stream	100%	100%	100%	100%	100%	100%
Tai Chung Hau Stream	100%	96%	100%	100%	100%	99%
Tseng Lan Shue Stream	100%	53%	94%	100%	100%	89%
Northwestern New Territories						
River Indus	92%	58%	72%	92%	67%	76%
River Beas	97%	36%	72%	100%	100%	81%
River Ganges	100%	53%	61%	100%	100%	83%
Yuen Long Creek	98%	0%	10%	79%	50%	48%
Kam Tin River	96%	0%	21%	83%	50%	50%
Tin Shui Wai Nullah	88%	54%	92%	96%	100%	86%
Fairview Park Nullah	100%	25%	83%	100%	0%	62%
Ha Pak Nai Stream	100%	100%	100%	100%	100%	100%
Tai Shui Hang Stream	100%	100%	100%	100%	100%	100%
Pak Nai Stream	100%	100%	100%	100%	100%	100%
Sheung Pak Nai Stream	100%	75%	100%	100%	100%	95%
Ngau Hom Sha Stream	100%	100%	100%	100%	100%	100%
Tsang Kok Stream	92%	75%	100%	100%	100%	93%
Lantau Island						
Mui Wo River	100%	97%	100%	100%	100%	99%
Tung Chung River	100%	69%	100%	100%	100%	94%
Southwestern New Territories & Kowloon						
Tuen Mun River	100%	57%	90%	97%	100%	89%
Pai Min Kok Stream	100%	96%	100%	100%	100%	99%
Kau Wa Keng Stream	100%	67%	100%	100%	100%	93%
Sam Dip Tam Stream	100%	94%	100%	100%	100%	99%
Kai Tak River	Not applicable					
Average Compliance (All monitoring stations)	96%	72%	88%	98%	93%	89%

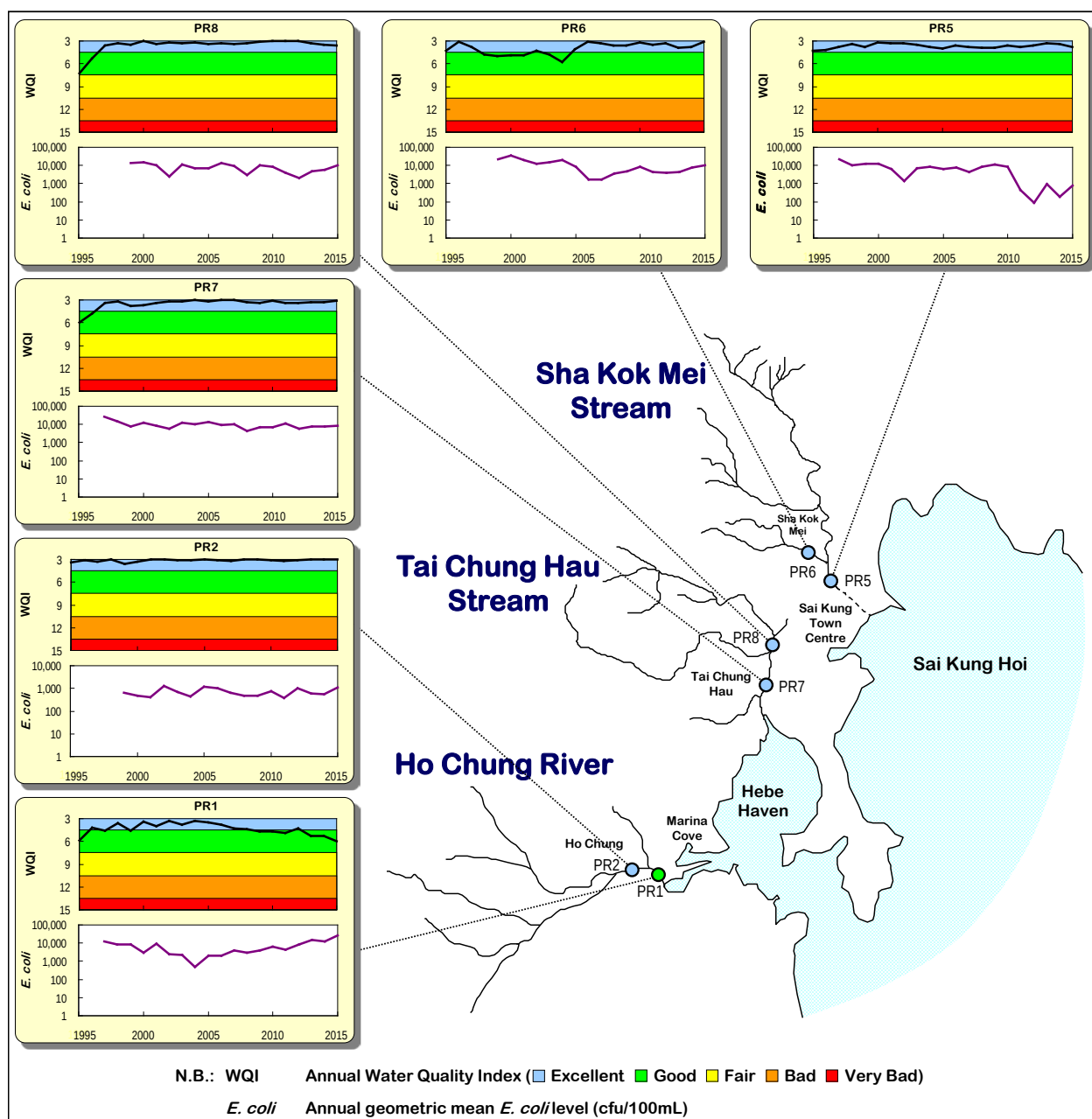
Water Quality Index gradings and *E. coli* levels in Shing Mun River

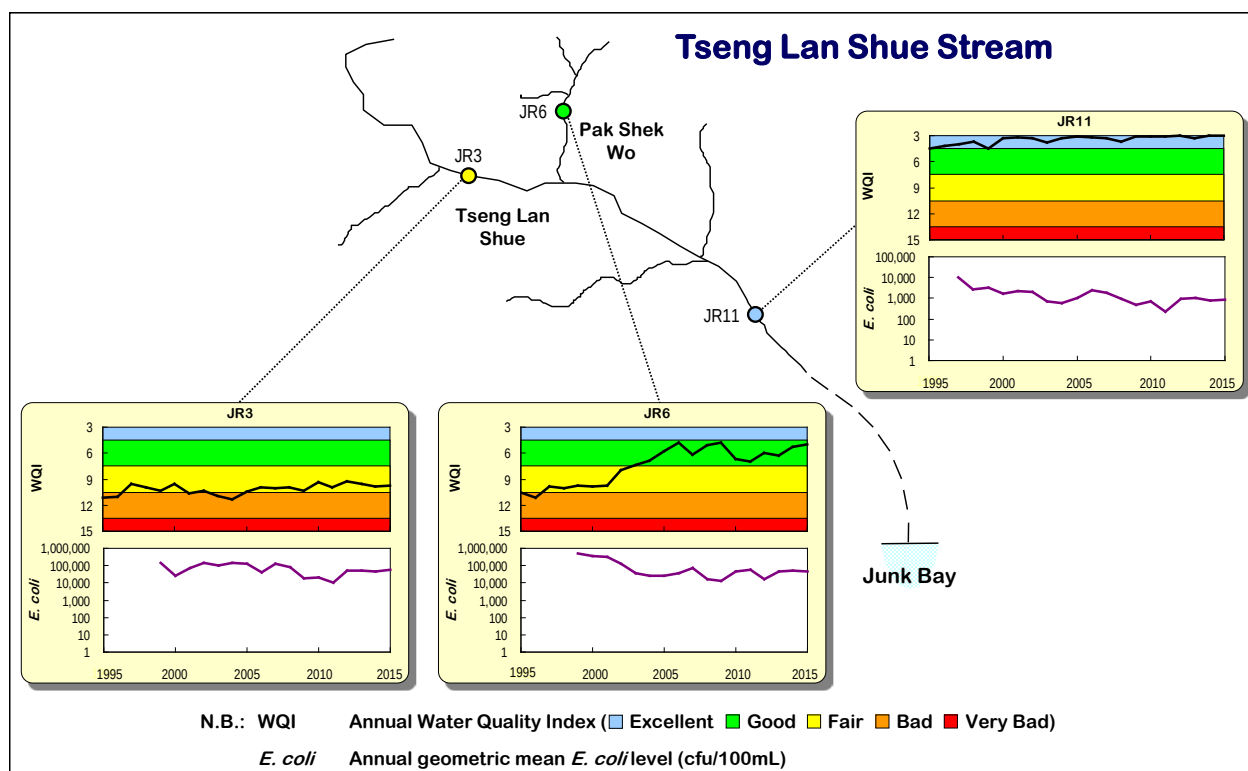
Water Quality Index gradings and *E. coli* levels in Lam Tsuen River and Tai Po River

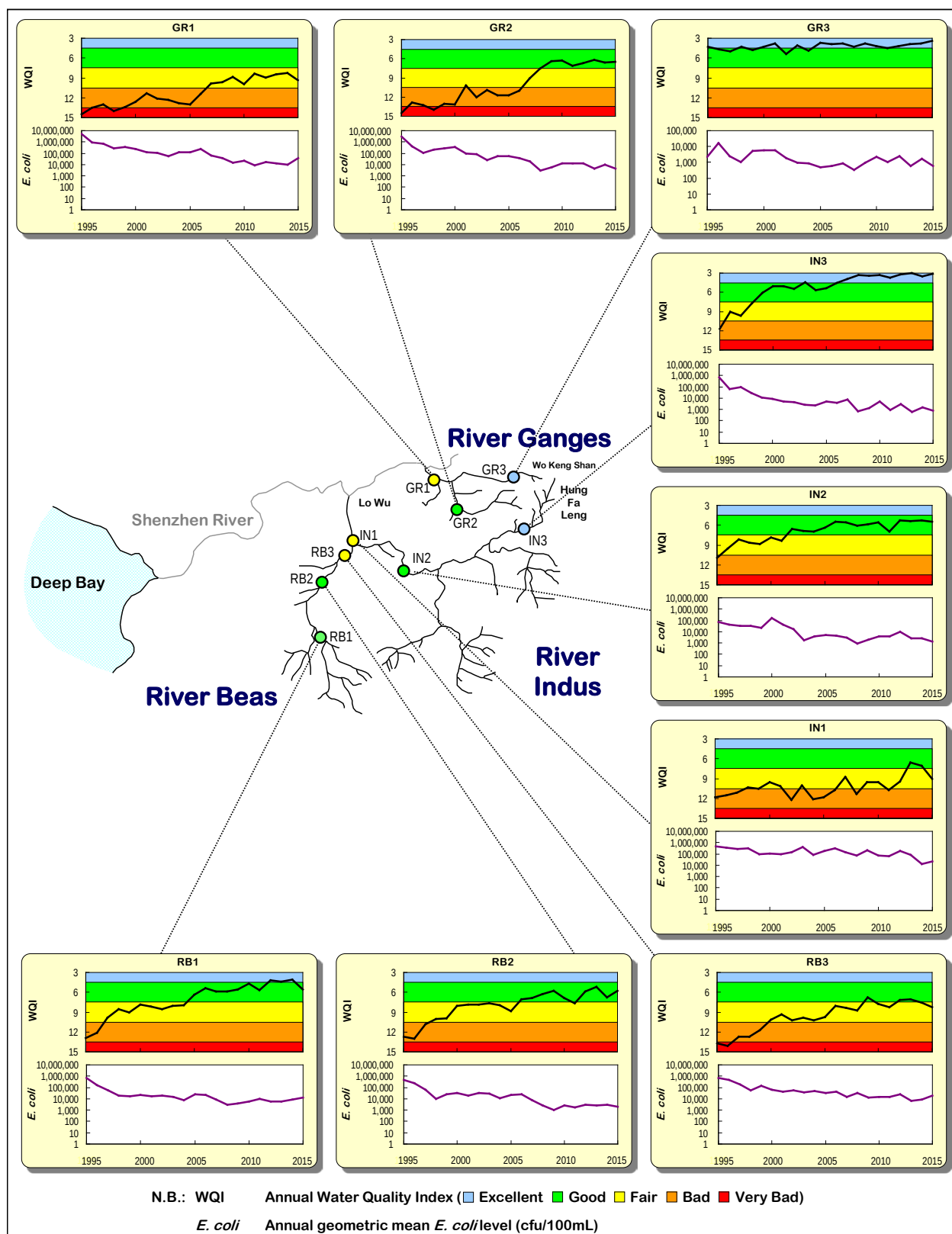
Water Quality Index gradings and *E. coli* levels in Tai Po Kau Stream, Shan Liu Stream and Tung Tze Stream



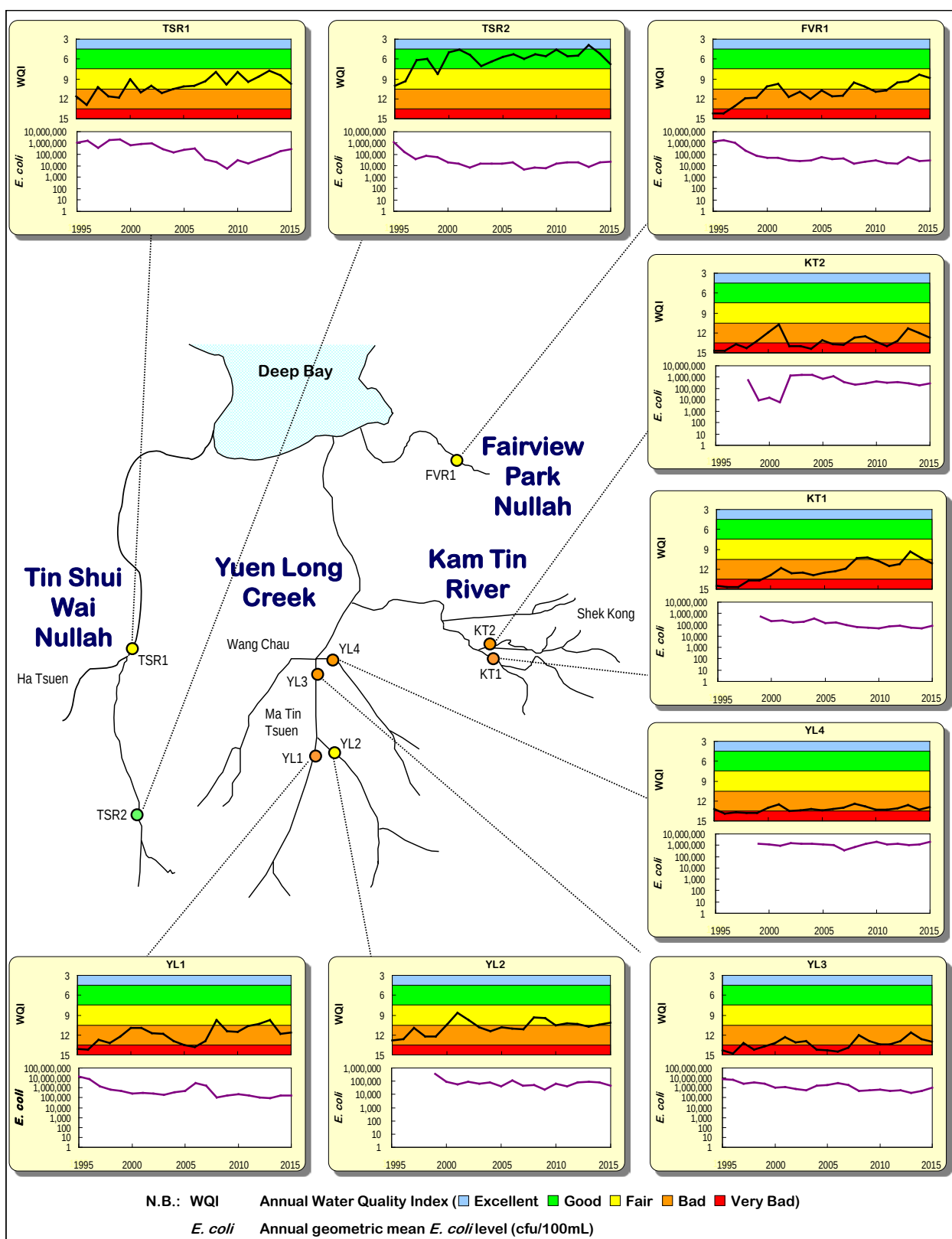
Water Quality Index gradings and *E. coli* levels in Ho Chung River, Tai Chung Hau Stream and Sha Kok Mei Stream



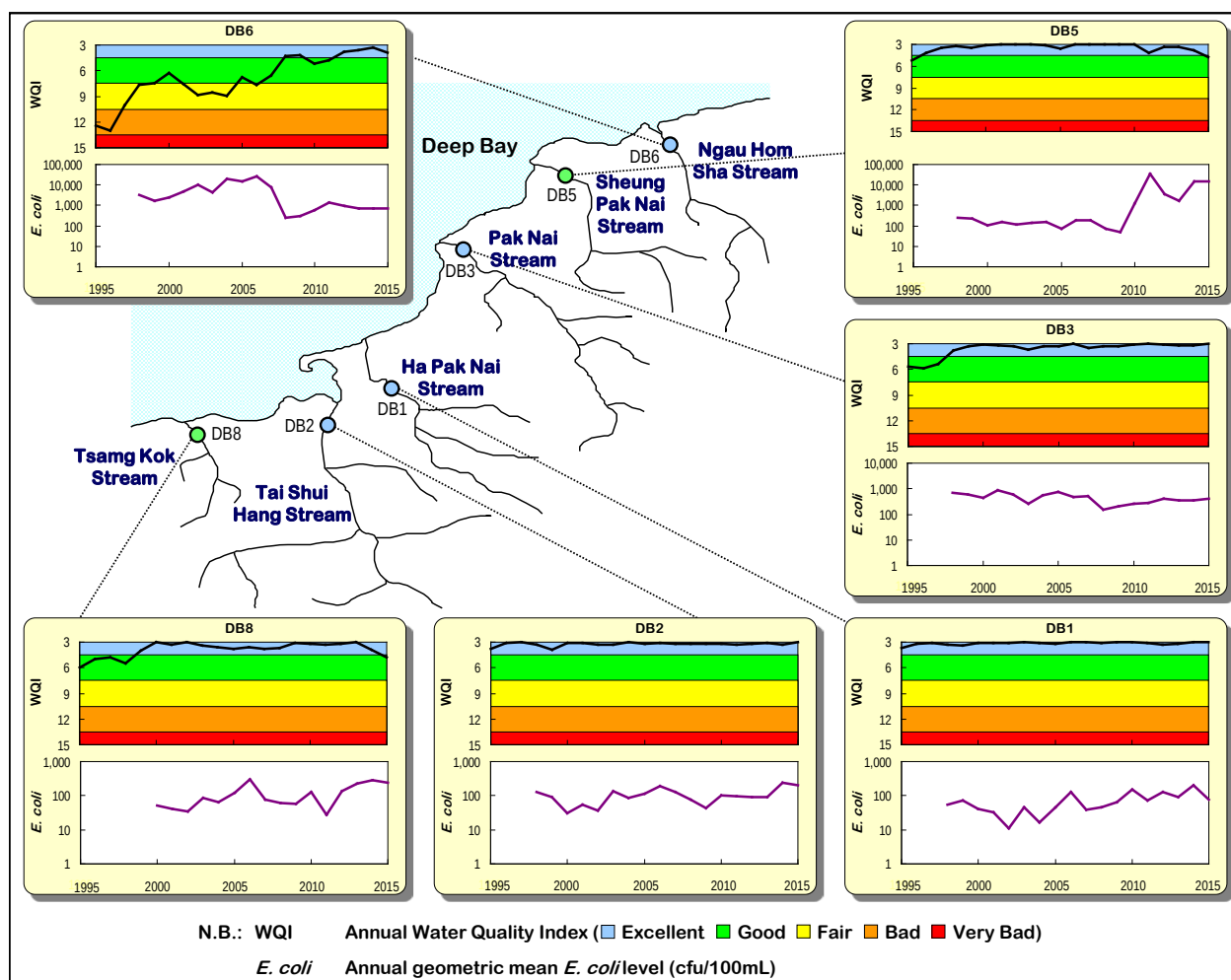
Water Quality Index gradings and *E. coli* levels in Tseng Lan Shue Stream

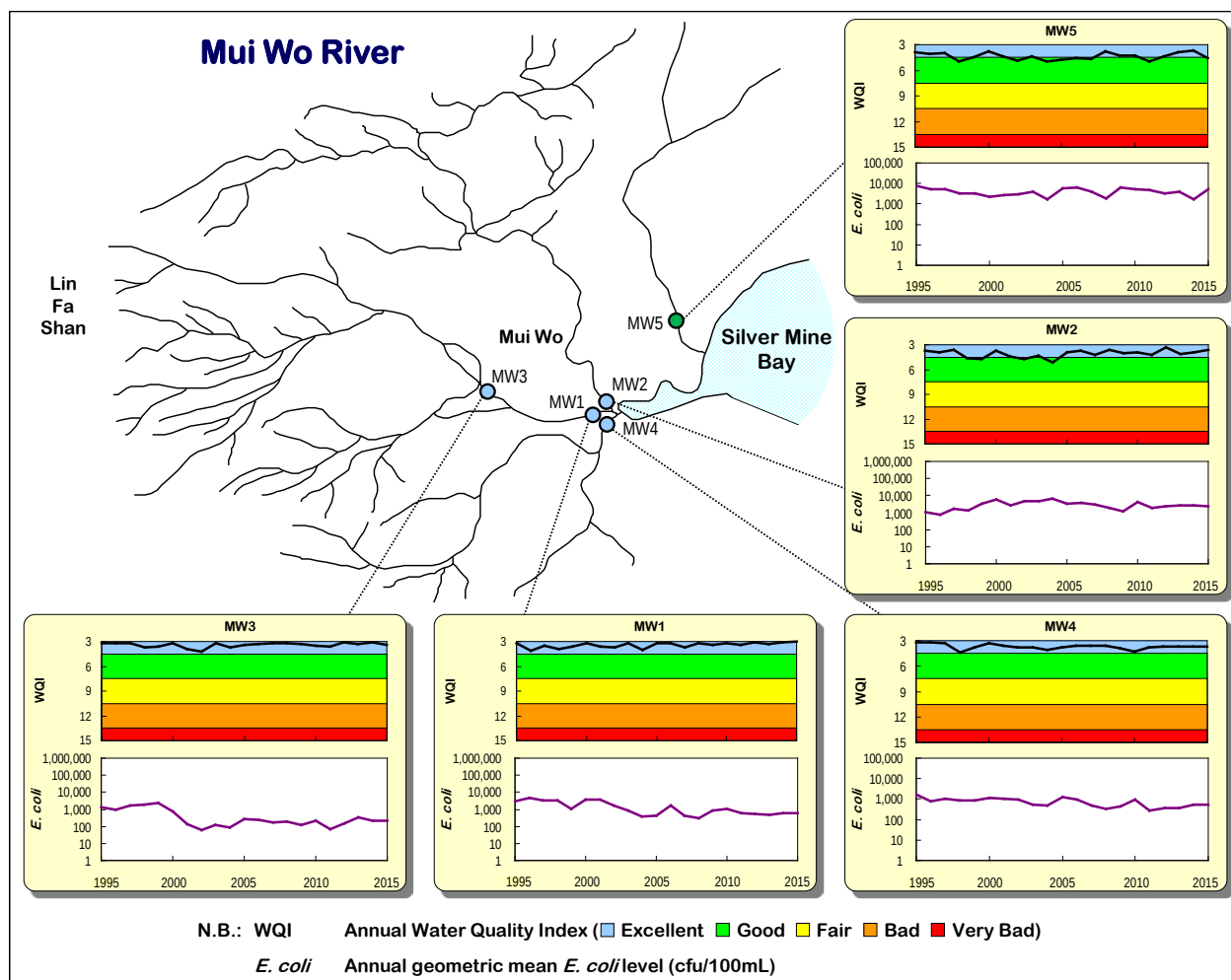
Water Quality Index gradings and *E. coli* levels in River Indus, River Beas and River Ganges

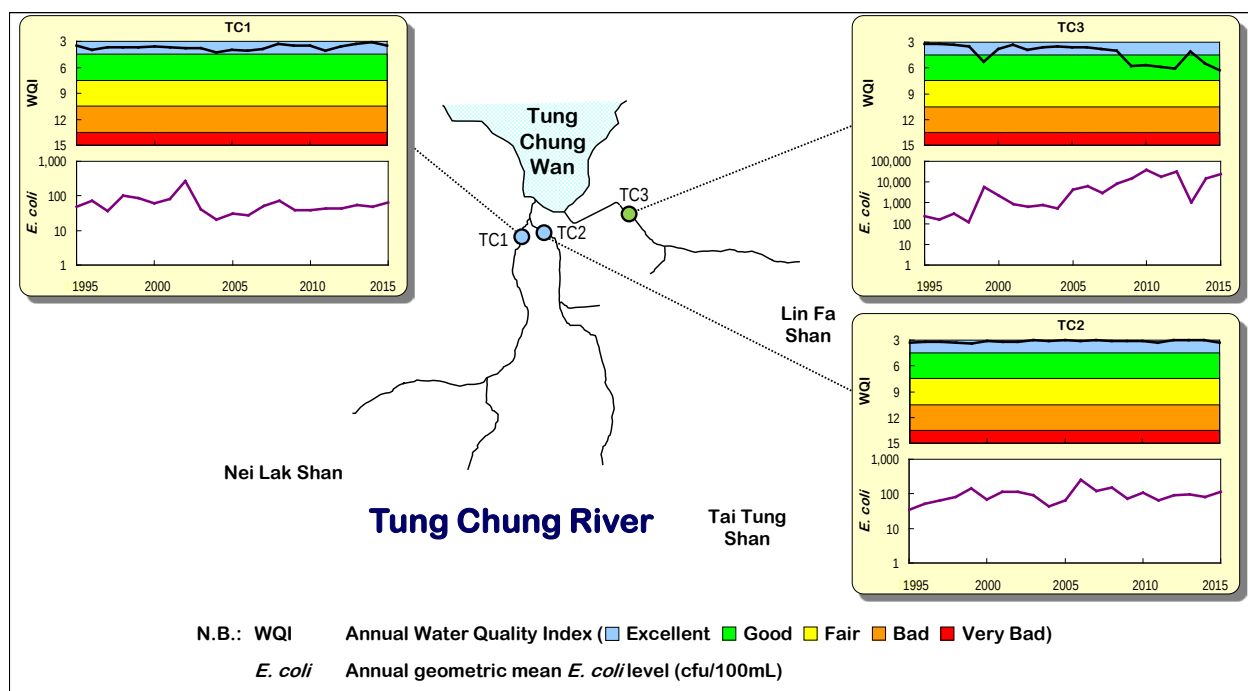
Water Quality Index gradings and *E. coli* levels in Yuen Long Creek, Kam Tin River, Tin Shui Wai Nullah and Fairview Park Nullah

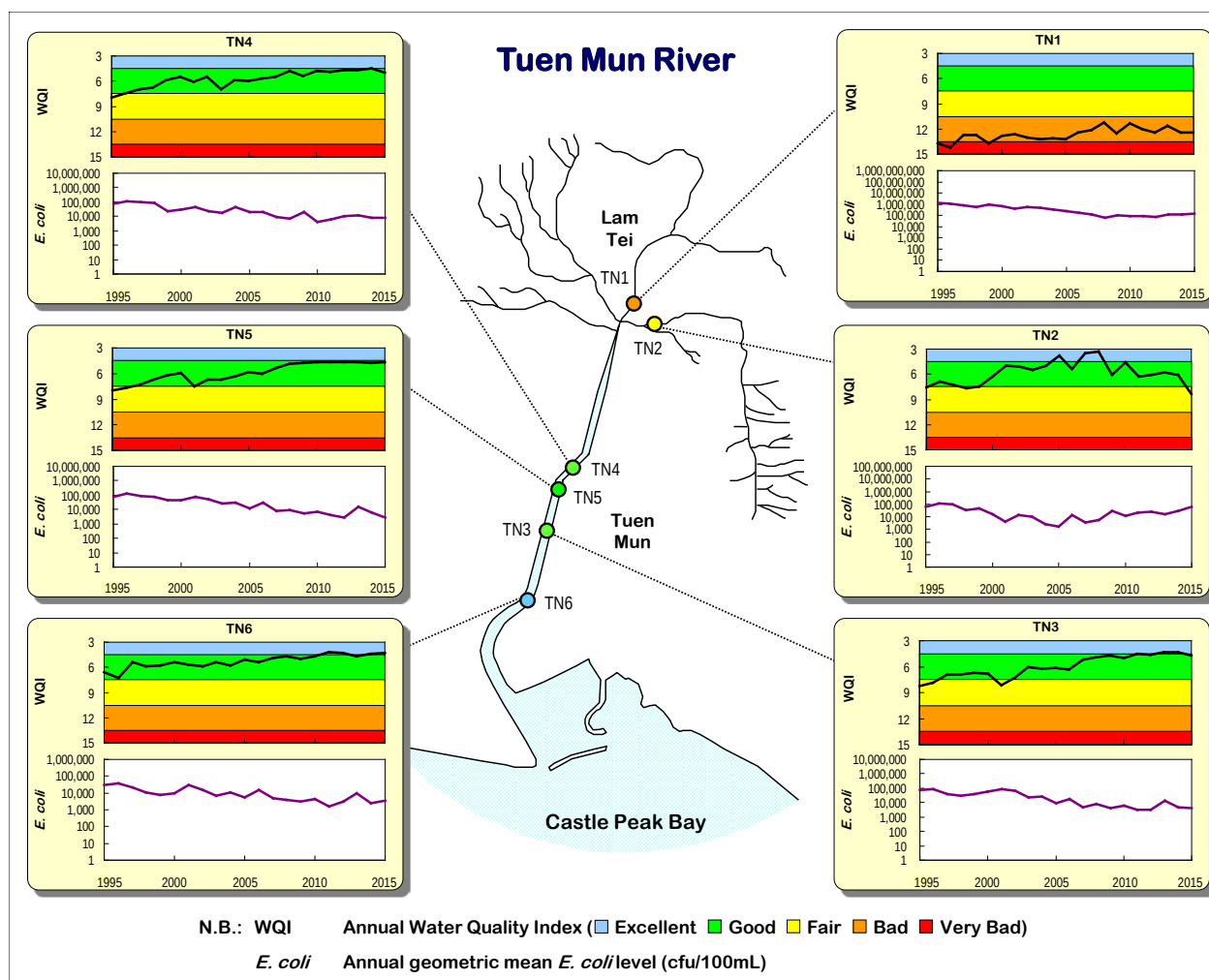


Water Quality Index gradings and *E. coli* levels in Ha Pak Nai Stream, Tai Shui Hang Stream, Pak Nai Stream, Sheung Pak Nai Stream, Ngau Hom Sha Stream and Tsang Kok Stream

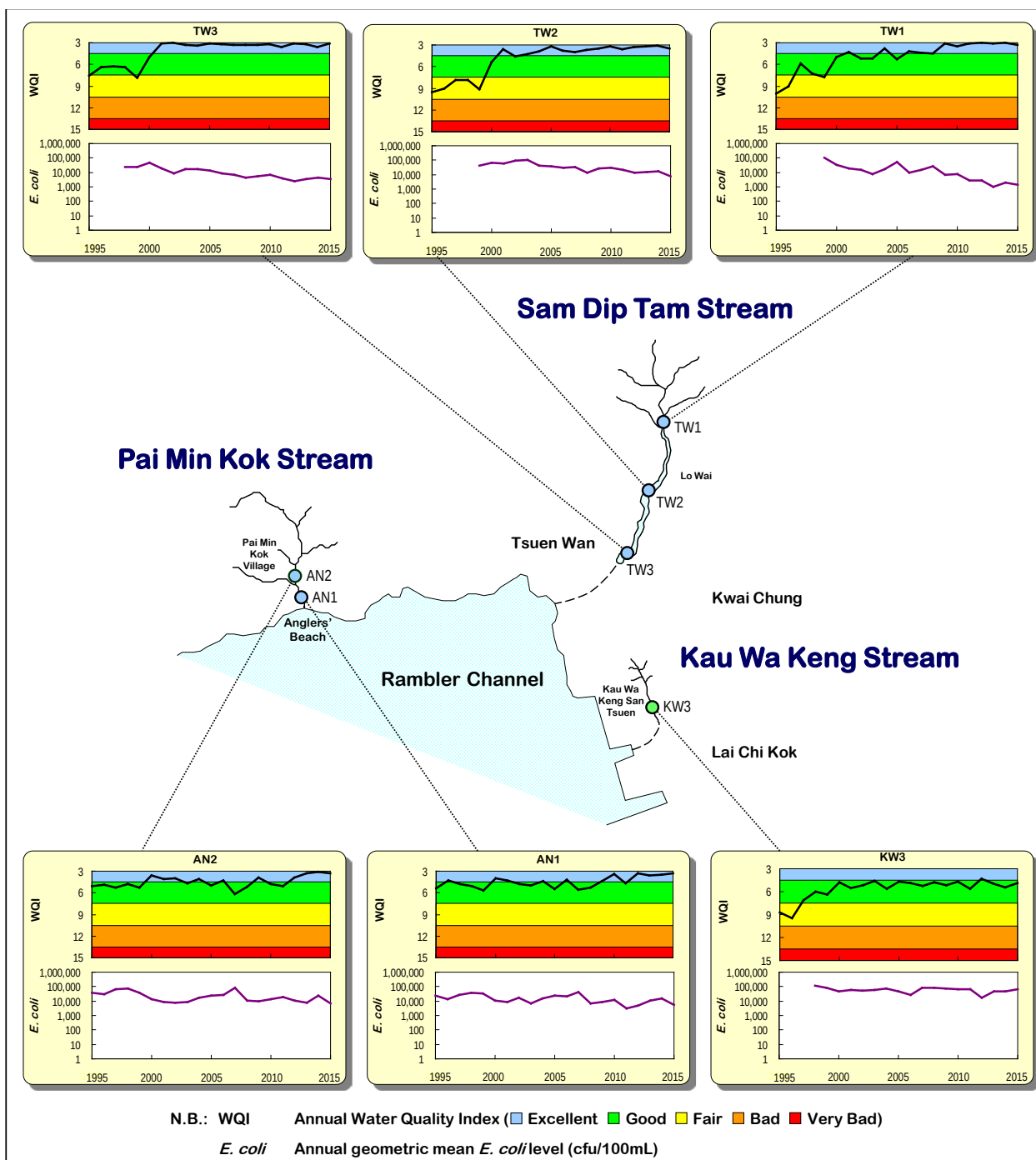


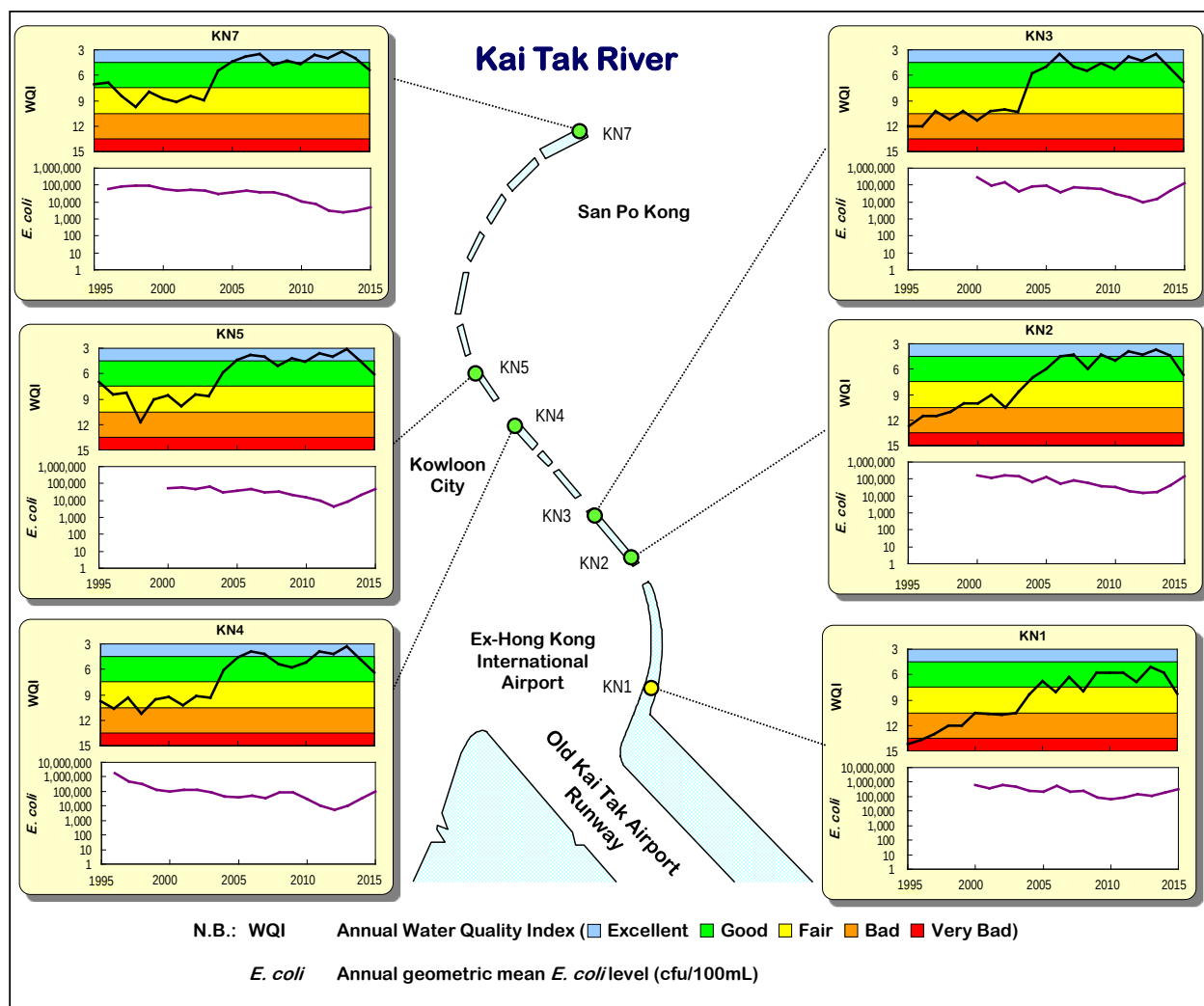
Water Quality Index gradings and *E. coli* levels in Mui Wo River

Water Quality Index gradings and *E. coli* levels in Tung Chung River

Water Quality Index gradings and *E. coli* levels in Tuen Mun River

Water Quality Index gradings and *E. coli* levels in Pai Min Kok Stream, Sam Dip Tam Stream and Kau Wa Keng Stream



Water Quality Index gradings and *E. coli* levels in Kai Tak River

Water quality trends for Shing Mun River (Main Channel, Siu Lek Yuen Nullah and Fo Tan Nullah) as analysed by the Seasonal Kendall Test

Watercourse		Shing Mun River				
		Main Channel	Siu Lek Yuen Nullah		Fo Tan Nullah	
Monitoring station		TR19I	TR23L	TR23A	TR17	TR17L
Monitoring period*		86-15	86-15	86-15	86-15	86-15
Parameter	Unit	Results of the Seasonal Kendall Test				
Dissolved oxygen	mg/L	↗	↗	↗	↗	↗
pH		–	↗	–	↗	↗
Suspended solids	mg/L	↘	↘	↘	↘	↘
BOD ₅	mg/L	↘	↘	↘	↘	↘
COD	mg/L	↘	↘	↘	↘	↘
Oil & grease	mg/L	–	–	–	↘	–
<i>E. coli</i>	cfu/100mL	↘	–	↘	↘	↘
Faecal coliforms	cfu/100mL	↘	↘	↘	↘	↘
Ammonia-nitrogen	mg/L	↘	↘	↘	↘	↘
Nitrate-nitrogen	mg/L	↗	↘	↘	–	↗
Total Kjeldahl nitrogen	mg/L	↘	↘	↘	↘	↘
Ortho-phosphate	mg/L	↘	↘	↘	↘	↘
Total phosphorus	mg/L	↘	↘	↘	↘	↘
Sulphide	mg/L	–	–	–	–	–
Aluminium	µg/L	↘	↘	↘	↘	–
Cadmium	µg/L	–	–	–	–	–
Chromium	µg/L	–	–	–	↘	–
Copper	µg/L	–	–	–	↘	↘
Lead	µg/L	–	–	–	↘	–
Zinc	µg/L	–	–	–	↘	–
Flow	L/s	×	–	×	↗	×

- Notes:
1. – indicates no significant trend was detected at $p < 0.05$.
 2. ↗ represents an increasing trend significant at $p < 0.05$.
 3. ↘ represents a decreasing trend significant at $p < 0.05$.
 4. × indicates no measurement was taken.
 5. * indicates the monitoring period for most of the parameters, a few commenced in different years during the period.

Water quality trends for Shing Mun River (Kwun Yam Shan Stream, Tai Wai Nullah and Tin Sum Nullah) as analysed by the Seasonal Kendall Test

Watercourse		Shing Mun River				
		Kwun Yam Shan Stream	Tai Wai Nullah			Tin Sum Nullah
Monitoring station		KY1	TR19A	TR19C	TR19	TR20B
Monitoring period*		88-15	86-15	86-15	86-15	86-15
Parameter	Unit	Results of the Seasonal Kendall Test				
Dissolved oxygen	mg/L	–	–	↗	↗	–
pH		↗	–	–	↗	↗
Suspended solids	mg/L	–	–	↘	↘	↘
BOD ₅	mg/L	↘	↘	↘	↘	↘
COD	mg/L	↘	↘	↘	↘	↘
Oil & grease	mg/L	–	–	–	–	–
<i>E. coli</i>	cfu/100mL	–	–	–	↘	–
Faecal coliforms	cfu/100mL	–	–	–	↘	–
Ammonia-nitrogen	mg/L	↘	↘	↘	↘	↘
Nitrate-nitrogen	mg/L	↘	↘	↘	↘	↗
Total Kjeldahl nitrogen	mg/L	↘	↘	↘	↘	↘
Ortho-phosphate	mg/L	↘	↘	↘	↘	–
Total phosphorus	mg/L	↘	↘	↘	↘	↘
Sulphide	mg/L	–	–	–	–	–
Aluminium	µg/L	–	–	↘	↘	↘
Cadmium	µg/L	–	↗	–	–	–
Chromium	µg/L	–	–	–	–	↘
Copper	µg/L	–	↘	↘	↘	↘
Lead	µg/L	–	–	↘	↘	–
Zinc	µg/L	–	–	–	↘	–
Flow	L/s	↗	–	↘	↘	–

- Notes:
1. – indicates no significant trend was detected at $p < 0.05$.
 2. ↗ represents an increasing trend significant at $p < 0.05$.
 3. ↘ represents a decreasing trend significant at $p < 0.05$.
 4. × indicates no measurement was taken.
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Water quality trends for Lam Tsuen River and Tai Po River, as analysed by the Seasonal Kendall Test

Watercourse		Lam Tsuen River									Tai Po River
Monitoring station		TR12H	TR12D	TR12G	TR12F	TR12C	TR12B	TR12E	TR12	TR12I	TR13
Monitoring period*		88-15	86-15	86-15	86-15	86-15	86-15	86-15	86-15	89-15	86-15
Parameter	Unit	Results of the Seasonal Kendall Test									
Dissolved oxygen	mg/L	↗	↗	↗	↗	–	↗	↗	↘	↗	↗
pH		↗	↗	↗	↗	↗	–	↗	–	–	–
Suspended solids	mg/L	↘	↘	↘	↘	↘	–	–	–	↘	↘
BOD ₅	mg/L	↘	↘	↘	↘	–	↘	↘	–	↘	↘
COD	mg/L	↘	↘	↘	↘	↘	↘	↘	↘	↘	↘
Oil & grease	mg/L	–	–	–	–	–	–	–	–	–	–
<i>E. coli</i>	cfu/100mL	↗	–	–	–	↗	–	–	–	–	↘
Faecal coliforms	cfu/100mL	↗	–	–	–	↗	–	–	–	↘	↘
Ammonia-nitrogen	mg/L	–	↘	↘	↘	↗	↘	–	↗	↘	↘
Nitrate-nitrogen	mg/L	↘	–	↘	–	↘	↘	–	–	↗	–
Total Kjeldahl nitrogen	mg/L	↘	↘	↘	↘	–	↘	↘	–	↘	↘
Ortho-phosphate	mg/L	↘	↘	↘	↘	↗	↘	↘	↘	↘	↘
Total phosphorus	mg/L	↘	↘	↘	↘	↘	↘	↘	↘	↘	↘
Sulphide	mg/L	–	–	–	–	–	–	–	–	–	–
Aluminium	µg/L	–	–	–	–	↘	–	↗	–	–	–
Cadmium	µg/L	–	–	–	–	–	–	–	–	–	–
Chromium	µg/L	–	–	–	–	–	–	–	–	–	–
Copper	µg/L	–	–	–	–	↘	–	–	–	–	–
Lead	µg/L	–	–	–	–	–	–	–	–	–	–
Zinc	µg/L	–	–	–	–	–	–	–	–	–	–
Flow	L/s	–	↘	–	–	–	–	↗	–	×	–

- Notes:
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 4. × indicates no measurement was taken.
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Water quality trends for Tai Po Kau Stream, Shan Liu Stream and Tung Tze Stream, as analysed by the Seasonal Kendall Test

Watercourse		Tai Po Kau Stream	Shan Liu Stream	Tung Tze Stream
Monitoring station		TR14	TR4	TR6
Monitoring period*		86-15	86-15	86-15
Parameter	Unit	Results of the Seasonal Kendall Test		
Dissolved oxygen	mg/L	↗	–	–
pH		↗	↗	–
Suspended solids	mg/L	↘	–	↘
BOD ₅	mg/L	↘	↗	↘
COD	mg/L	↘	–	↘
Oil & grease	mg/L	–	–	–
<i>E. coli</i>	cfu/100mL	↘	–	↗
Faecal coliforms	cfu/100mL	–	–	–
Ammonia-nitrogen	mg/L	↘	↗	↗
Nitrate-nitrogen	mg/L	↘	↗	↗
Total Kjeldahl nitrogen	mg/L	↘	↗	–
Ortho-phosphate	mg/L	↘	↗	–
Total phosphorus	mg/L	↘	–	↘
Sulphide	mg/L	–	–	–
Aluminium	µg/L	–	–	–
Cadmium	µg/L	–	–	–
Chromium	µg/L	–	–	–
Copper	µg/L	–	–	–
Lead	µg/L	–	–	–
Zinc	µg/L	–	–	–
Flow	L/s	–	↗	×

- Notes:
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 2. ↗ represents an increasing trend significant at $p < 0.05$.
 3. ↘ represents a decreasing trend significant at $p < 0.05$.
 4. × indicates no measurement was taken.
 5. * indicates the monitoring period for most of the parameters, a few commenced in different years during the period.

Water quality trends for Ho Chung River, Sha Kok Mei Stream and Tai Chung Hau Stream, as analysed by the Seasonal Kendall Test

Watercourse		Ho Chung River		Sha Kok Mei Stream		Tai Chung Hau Stream	
Monitoring station		PR1	PR2	PR5	PR6	PR7	PR8
Monitoring period*		86-15	86-15	89-15	89-15	89-15	89-15
Parameter	Unit	Results of the Seasonal Kendall Test					
Dissolved oxygen	mg/L	↗	↗	–	–	↗	↗
pH		–	↗	↗	↗	↗	↗
Suspended solids	mg/L	↘	–	↘	↘	↘	↘
BOD ₅	mg/L	↘	↘	↘	↘	↘	↘
COD	mg/L	↘	↘	↘	↘	↘	↘
Oil & grease	mg/L	–	–	–	–	–	–
<i>E. coli</i>	cfu/100mL	–	–	↘	↘	↘	↘
Faecal coliforms	cfu/100mL	–	–	↘	↘	–	–
Ammonia-nitrogen	mg/L	–	↘	↘	–	↘	↘
Nitrate-nitrogen	mg/L	↗	–	↗	–	–	–
Total Kjeldahl nitrogen	mg/L	↘	↘	↘	↘	↘	↘
Ortho-phosphate	mg/L	↘	↘	↘	↘	↘	↘
Total phosphorus	mg/L	↘	↘	↘	↘	↘	↘
Sulphide	mg/L	–	–	–	–	–	–
Aluminium	µg/L	–	–	–	–	↘	↘
Cadmium	µg/L	–	–	–	–	–	–
Chromium	µg/L	–	–	–	–	–	–
Copper	µg/L	–	–	–	–	↘	↘
Lead	µg/L	–	–	–	–	–	–
Zinc	µg/L	↘	–	–	–	–	–
Flow	L/s	×	↗	↗	×	–	×

- Notes:
1. – indicates no significant trend was detected at $p < 0.05$.
 2. ↗ represents an increasing trend significant at $p < 0.05$.
 3. ↘ represents a decreasing trend significant at $p < 0.05$.
 4. × indicates no measurement was taken.
 5. * indicates the monitoring period for most of the parameters, a few commenced in different years during the period.

Water quality trends for Tseng Lan Shue Stream, as analysed by the Seasonal Kendall Test

Watercourse		Tseng Lan Shue Stream		
Monitoring station		JR3	JR6	JR11
Monitoring period*		86-15	86-15	86-15
Parameter	Unit	Results of the Seasonal Kendall Test		
Dissolved oxygen	mg/L	↗	↗	↗
pH		↗	–	↗
Suspended solids	mg/L	↘	↘	↘
BOD ₅	mg/L	↘	↘	↘
COD	mg/L	↘	↘	↘
Oil & grease	mg/L	–	↘	–
<i>E. coli</i>	cfu/100mL	↘	↘	↘
Faecal coliforms	cfu/100mL	↘	↘	↘
Ammonia-nitrogen	mg/L	–	↘	↘
Nitrate-nitrogen	mg/L	↗	↗	–
Total Kjeldahl nitrogen	mg/L	↘	↘	↘
Ortho-phosphate	mg/L	↘	↘	↘
Total phosphorus	mg/L	↘	↘	↘
Sulphide	mg/L	–	–	–
Aluminium	µg/L	–	–	–
Cadmium	µg/L	–	–	–
Chromium	µg/L	–	–	–
Copper	µg/L	↘	↘	↘
Lead	µg/L	–	↘	–
Zinc	µg/L	↘	↘	–
Flow	L/s	×	×	–

- Notes:
1. – indicates no significant trend was detected at $p < 0.05$.
 2. ↗ represents an increasing trend significant at $p < 0.05$.
 3. ↘ represents a decreasing trend significant at $p < 0.05$.
 4. × indicates no measurement was taken.
 5. * indicates the monitoring period for most of the parameters, a few commenced in different years during the period.

Water quality trends for River Indus, River Beas and River Ganges, as analysed by the Seasonal Kendall Test

Watercourse		River Indus			River Beas			River Ganges		
Monitoring station		IN1	IN2	IN3	RB1	RB2	RB3	GR1	GR2	GR3
Monitoring period*		87-15	87-15	87-15	86-15	86-15	86-15	87-15	87-15	90-15
Parameter	Unit	Results of the Seasonal Kendall Test								
Dissolved oxygen	mg/L	↗	↗	↗	↗	↗	↗	↗	↗	–
pH		↗	↗	↗	↗	↗	↗	↗	↗	↗
Suspended solids	mg/L	–	↘	↘	↘	↘	↘	↘	↘	↘
BOD ₅	mg/L	↘	↘	↘	↘	↘	↘	↘	↘	↘
COD	mg/L	↘	↘	↘	↘	↘	↘	↘	↘	↘
Oil & grease	mg/L	–	–	–	–	–	–	↘	↘	–
<i>E. coli</i>	cfu/100mL	↘	↘	↘	↘	↘	↘	↘	↘	↘
Faecal coliforms	cfu/100mL	↘	↘	↘	↘	↘	↘	↘	↘	–
Ammonia-nitrogen	mg/L	↘	↘	↘	↘	↘	↘	↘	↘	–
Nitrate-nitrogen	mg/L	↗	↗	↗	↗	↗	↗	↗	↗	–
Total Kjeldahl nitrogen	mg/L	↘	↘	↘	↘	↘	↘	↘	↘	↘
Ortho-phosphate	mg/L	↘	↘	↘	↘	↘	↘	↘	↘	↘
Total phosphorus	mg/L	↘	↘	↘	↘	↘	↘	↘	↘	↘
Sulphide	mg/L	↘	–	–	↘	↘	↘	↘	↘	–
Aluminium	µg/L	–	–	–	↘	↘	↘	↘	↘	–
Cadmium	µg/L	–	–	–	–	–	–	–	–	–
Chromium	µg/L	–	–	–	–	–	–	–	–	–
Copper	µg/L	↘	↘	↘	↘	↘	↘	↘	↘	–
Lead	µg/L	↘	↘	–	↘	↘	↘	↘	↘	–
Zinc	µg/L	↘	↘	↘	↘	↘	↘	↘	↘	–
Flow	L/s	×	×	–	↘	↘	×	↘	–	↗

- Notes:
1. – indicates no significant trend was detected at $p < 0.05$.
 2. ↗ represents an increasing trend significant at $p < 0.05$.
 3. ↘ represents a decreasing trend significant at $p < 0.05$.
 4. × indicates no measurement was taken.
 5. * indicates the monitoring period for most of the parameters, a few commenced in different years during the period.

Water quality trends for Yuen Long Creek, Kam Tin River, Tin Shui Wai Nullah and Fairview Park Nullah as analysed by the Seasonal Kendall Test

Watercourse		Yuen Long Creek				Kam Tin River		Tin Shui Wai Nullah		Fairview Park Nullah
Monitoring station		YL1	YL2	YL3	YL4	KT1	KT2	TSR1	TSR2	FVR1
Monitoring period*		86-15	86-15	86-15	86-15	86-15	86-15	92-15	92-15	93-15
Parameter	Unit	Results of the Seasonal Kendall Test								
Dissolved oxygen	mg/L	↗	↗	↗	↗	↗	↗	↗	↗	↗
pH		↗	↗	↗	↗	↗	↗	–	–	↗
Suspended solids	mg/L	↘	↘	↘	↘	↘	↘	↘	↘	–
BOD ₅	mg/L	↘	↘	↘	–	↘	↘	↘	↘	↘
COD	mg/L	↘	↘	↘	↘	↘	↘	↘	↘	↘
Oil & grease	mg/L	↘	↘	–	–	↘	–	–	–	–
<i>E. coli</i>	cfu/100 mL	↘	–	↘	–	↘	–	↘	↘	↘
Faecal coliforms	cfu/100 mL	↘	↘	↘	–	↘	–	↘	↘	↘
Ammonia-nitrogen	mg/L	↘	↘	↘	↘	↘	↘	↘	↘	↘
Nitrate-nitrogen	mg/L	↗	↗	–	↘	↗	–	–	–	–
Total Kjeldahl nitrogen	mg/L	↘	↘	↘	↘	↘	↘	↘	↘	↘
Ortho-phosphate	mg/L	↘	↘	↘	↘	↘	↘	↘	↘	↘
Total phosphorus	mg/L	↘	↘	↘	↘	↘	↘	↘	↘	↘
Sulphide	mg/L	↘	↘	↘	↘	↘	↘	–	–	–
Aluminium	µg/L	–	↘	–	–	↘	–	–	↘	↗
Cadmium	µg/L	↘	–	↘	–	–	–	–	–	–
Chromium	µg/L	–	–	↘	↘	↘	–	–	–	–
Copper	µg/L	↘	↘	↘	↘	–	↘	↘	↘	↘
Lead	µg/L	–	↘	↘	↘	↘	–	↘	↘	–
Zinc	µg/L	↘	↘	↘	↘	–	–	↘	↘	–
Flow	L/s	–	↘	–	↘	↘	–	×	–	×

- Notes:
1. – indicates no significant trend was detected at $p < 0.05$.
 2. ↗ represents an increasing trend significant at $p < 0.05$.
 3. ↘ represents a decreasing trend significant at $p < 0.05$.
 4. × indicates no measurement was taken.
 5. * indicates the monitoring period for most of the parameters, a few commenced in different years during the period.

Water quality trends for Ha Pak Nai Stream, Pak Nai Stream, Sheung Pak Nai Stream, Ngau Hom Sha Stream, Tai Shui Hang Stream and Tsang Kok Stream as analysed by the Seasonal Kendall Test

Watercourse		Ha Pak Nai Stream	Pak Nai Stream	Sheung Pak Nai Stream	Ngau Hom Sha Stream	Tai Shui Hang Stream	Tsang Kok Stream
Monitoring station		DB1	DB3	DB5	DB6	DB2	DB8
Monitoring period*		89-15	89-15	89-15	89-15	89-15	89-15
Parameter	Unit	Results of the Seasonal Kendall Test					
Dissolved oxygen	mg/L	↗	↗	↗	↗	↗	↗
pH		↗	↗	↗	↗	↗	–
Suspended solids	mg/L	↘	↘	↘	↘	–	↘
BOD ₅	mg/L	↘	↘	↘	↘	↘	↘
COD	mg/L	↘	↘	↘	↘	–	↘
Oil & grease	mg/L	–	–	–	–	–	–
<i>E. coli</i>	cfu/100mL	↗	–	↗	↘	–	–
Faecal coliforms	cfu/100mL	↗	–	↗	–	↗	–
Ammonia-nitrogen	mg/L	–	↘	–	↘	–	↘
Nitrate-nitrogen	mg/L	↗	↘	–	↗	↗	↗
Total Kjeldahl nitrogen	mg/L	↘	↘	–	↘	↘	↘
Ortho-phosphate	mg/L	↘	↘	↘	↘	↘	↘
Total phosphorus	mg/L	–	↘	↘	↘	↘	↘
Sulphide	mg/L	–	–	–	–	–	–
Aluminium	µg/L	–	–	–	↘	↗	–
Cadmium	µg/L	–	–	–	–	–	–
Chromium	µg/L	–	–	–	–	–	–
Copper	µg/L	–	–	–	↘	–	–
Lead	µg/L	–	↘	–	↘	–	–
Zinc	µg/L	–	–	–	↘	–	–
Flow	L/s	–	–	–	–	↗	×

- Notes:
1. – indicates no significant trend was detected at $p < 0.05$.
 2. ↗ represents an increasing trend significant at $p < 0.05$.
 3. ↘ represents a decreasing trend significant at $p < 0.05$.
 4. × indicates no measurement was taken.
 5. * indicates the monitoring period for most of the parameters, a few commenced in different years during the period.

Water quality trends for Mui Wo River as analysed by the Seasonal Kendall Test

Watercourse		Mui Wo River				
Monitoring station		MW1	MW2	MW3	MW4	MW5
Monitoring period*		86-15	86-15	86-15	88-15	88-15
Parameter	Unit	Results of the Seasonal Kendal Test				
Dissolved oxygen	mg/L	–	↗	↗	–	–
pH		↗	↗	↗	–	–
Suspended solids	mg/L	↘	↘	↘	–	–
BOD ₅	mg/L	↘	↘	↘	↘	–
COD	mg/L	↘	↘	↘	↘	↘
Oil & grease	mg/L	–	–	–	–	–
<i>E. coli</i>	cfu/100mL	↘	–	↘	↘	–
Faecal coliforms	cfu/100mL	–	↗	–	–	↗
Ammonia-nitrogen	mg/L	–	↗	↘	↗	↗
Nitrate-nitrogen	mg/L	–	–	–	↗	↗
Total Kjeldahl nitrogen	mg/L	↘	–	↘	–	↗
Ortho-phosphate	mg/L	↘	↘	↘	–	–
Total phosphorus	mg/L	↘	↘	↘	↘	–
Sulphide	mg/L	–	–	–	–	–
Aluminium	µg/L	↗	–	↗	–	–
Cadmium	µg/L	–	–	–	–	–
Chromium	µg/L	–	–	–	–	–
Copper	µg/L	–	–	–	–	–
Lead	µg/L	–	–	–	–	–
Zinc	µg/L	–	–	–	–	–
Flow	L/s	↘	×	–	–	–

- Notes:
1. – indicates no significant trend was detected at $p < 0.05$.
 2. ↗ represents an increasing trend significant at $p < 0.05$.
 3. ↘ represents a decreasing trend significant at $p < 0.05$.
 4. × indicates no measurement was taken.
 5. * indicates the monitoring period for most of the parameters, a few commenced in different years during the period.

Water quality trends for Tung Chung River as analysed by the Seasonal Kendall Test

Watercourse		Tung Chung River		
Monitoring station		TC1	TC2	TC3
Monitoring period*		92-15	92-15	92-15
Parameter	Unit	Results of the Seasonal Kendall Test		
Dissolved oxygen	mg/L	–	–	–
pH		–	–	–
Suspended solids	mg/L	–	–	–
BOD ₅	mg/L	↘	–	↗
COD	mg/L	↘	↘	–
Oil & grease	mg/L	–	–	–
<i>E. coli</i>	cfu/100mL	–	–	↗
Faecal coliforms	cfu/100mL	↗	↗	↗
Ammonia-nitrogen	mg/L	–	–	↗
Nitrate-nitrogen	mg/L	–	–	↗
Total Kjeldahl nitrogen	mg/L	–	–	↗
Ortho-phosphate	mg/L	↘	↘	↗
Total phosphorus	mg/L	–	–	↗
Sulphide	mg/L	–	–	–
Aluminium	µg/L	–	–	–
Cadmium	µg/L	–	–	–
Chromium	µg/L	–	–	–
Copper	µg/L	–	–	–
Lead	µg/L	–	–	–
Zinc	µg/L	–	–	–
Flow	L/s	↗	↗	×

- Notes:
1. – indicates no significant trend was detected at $p < 0.05$.
 2. ↗ represents an increasing trend significant at $p < 0.05$.
 3. ↘ represents a decreasing trend significant at $p < 0.05$.
 4. × indicates no measurement was taken.
 5. * indicates the monitoring period for most of the parameters, a few commenced in different years during the period.

Water quality trends for Tuen Mun River as analysed by the Seasonal Kendall Test

Watercourse		Tuen Mun River					
Monitoring station		TN1	TN2	TN3	TN4	TN5	TN6
Monitoring period*		86-15	86-15	86-15	86-15	86-15	86-15
Parameter	Unit	Results of the Seasonal Kendall Test					
Dissolved oxygen	mg/L	↗	↗	↗	↗	↗	↗
pH		–	↗	–	–	↗	↘
Suspended solids	mg/L	↘	↘	↘	↘	↘	↘
BOD ₅	mg/L	↘	↘	↘	↘	↘	↘
COD	mg/L	↘	↘	↘	↘	↘	↘
Oil & grease	mg/L	↘	↘	–	–	–	–
<i>E. coli</i>	cfu/100mL	↘	↘	↘	↘	↘	↘
Faecal coliforms	cfu/100mL	↘	↘	↘	↘	↘	↘
Ammonia-nitrogen	mg/L	↘	–	↘	↘	↘	↘
Nitrate-nitrogen	mg/L	↗	–	↗	↗	↗	↗
Total Kjeldahl nitrogen	mg/L	↘	↘	↘	↘	↘	↘
Ortho-phosphate	mg/L	↘	↘	↘	↘	↘	↘
Total phosphorus	mg/L	↘	↘	↘	↘	↘	↘
Sulphide	mg/L	↘	–	–	–	–	–
Aluminium	µg/L	↘	↘	–	–	–	–
Cadmium	µg/L	↘	–	–	–	–	–
Chromium	µg/L	↘	–	↘	↘	↘	↘
Copper	µg/L	↘	↘	↘	↘	↘	↘
Lead	µg/L	↘	↘	↘	↘	–	–
Zinc	µg/L	↘	↘	–	–	–	–
Flow	L/s	–	↘	×	×	×	×

- Notes:
1. – indicates no significant trend was detected at $p < 0.05$.
 2. ↗ represents an increasing trend significant at $p < 0.05$.
 3. ↘ represents a decreasing trend significant at $p < 0.05$.
 4. × indicates no measurement was taken.
 5. * indicates the monitoring period for most of the parameters, a few commenced in different years during the period.

Water quality trends for Pai Min Kok Stream, Sam Dip Tam Stream and Kau Wa Keng Stream as analysed by the Seasonal Kendall Test

Watercourse		Pai Min Kok Stream		Sam Dip Tam Stream			Kau Wa Keng Stream
Monitoring station		AN1	AN2	TW1	TW2	TW3	KW3
Monitoring period*		88-15	88-15	86-15	86-15	86-15	87-15
Parameter	Unit	Results of the Seasonal Kendall Test					
Dissolved oxygen	mg/L	↗	↗	↗	↗	↗	↗
pH		↗	↗	↗	↗	↗	↗
Suspended solids	mg/L	↘	↘	↘	↘	↘	↘
BOD ₅	mg/L	↘	↘	↘	↘	↘	↘
COD	mg/L	↘	↘	↘	↘	↘	↘
Oil & grease	mg/L	–	–	–	–	–	–
<i>E. coli</i>	cfu/100mL	–	↘	↘	↘	↘	–
Faecal coliforms	cfu/100mL	–	–	↘	↘	↘	–
Ammonia-nitrogen	mg/L	↘	↘	↘	↘	↘	↘
Nitrate-nitrogen	mg/L	↘	↘	↗	–	↘	–
Total Kjeldahl nitrogen	mg/L	↘	↘	↘	↘	↘	↘
Ortho-phosphate	mg/L	↘	↘	↘	↘	↘	↘
Total phosphorus	mg/L	↘	↘	↘	↘	↘	↘
Sulphide	mg/L	–	–	–	–	–	–
Aluminium	µg/L	–	–	–	–	–	–
Cadmium	µg/L	–	–	–	–	–	–
Chromium	µg/L	–	–	–	–	–	–
Copper	µg/L	–	–	↘	↘	–	↗
Lead	µg/L	↘	–	↘	↘	–	↘
Zinc	µg/L	↘	↘	↘	–	–	↘
Flow	L/s	×	–	×	↗	×	–

- Notes:
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 2. ↗ represents an increasing trend significant at $p < 0.05$.
 3. ↘ represents a decreasing trend significant at $p < 0.05$.
 4. × indicates no measurement was taken.
 5. * indicates the monitoring period for most of the parameters, a few commenced in different years during the period.

Water quality trends for Kai Tak River as analysed by the Seasonal Kendall Test

Watercourse		Kai Tak River					
Monitoring station		KN1	KN2	KN3	KN4	KN5	KN7
Monitoring period*		86-15	86-15	86-15	86-15	86-15	86-15
Parameter	Unit	Results of the Seasonal Kendall Test					
Dissolved oxygen	mg/L	↗	↗	↗	↗	–	↗
pH		–	↗	↗	↗	↘	↘
Suspended solids	mg/L	↘	↘	↘	↘	↘	↘
BOD ₅	mg/L	↘	↘	↘	↘	↘	↘
COD	mg/L	↘	↘	↘	↘	↘	↘
Oil & grease	mg/L	–	–	–	–	–	–
<i>E. coli</i>	cfu/100mL	↘	–	–	↘	↘	↘
Faecal coliforms	cfu/100mL	↘	↘	–	↘	↘	↘
Ammonia-nitrogen	mg/L	↘	↘	↘	↘	↘	↘
Nitrate-nitrogen	mg/L	↗	↗	↗	↗	↗	↗
Total Kjeldahl nitrogen	mg/L	↘	↘	↘	↘	↘	↘
Ortho-phosphate	mg/L	↗	↗	↗	↗	↗	↗
Total phosphorus	mg/L	↘	–	–	↘	–	–
Sulphide	mg/L	↘	↘	↘	–	–	–
Aluminium	µg/L	–	–	–	↘	–	–
Cadmium	µg/L	–	↘	↘	–	–	–
Chromium	µg/L	↘	↘	↘	↘	↘	↘
Copper	µg/L	↘	↘	↘	↘	↘	–
Lead	µg/L	–	↘	↘	↘	↘	↘
Zinc	µg/L	–	–	–	↘	–	↘
Flow	L/s	×	×	×	–	–	–

- Notes:
1. – indicates no significant trend was detected at $p < 0.05$.
 2. ↗ represents an increasing trend significant at $p < 0.05$.
 3. ↘ represents a decreasing trend significant at $p < 0.05$.
 4. × indicates no measurement was taken.
 5. * indicates the monitoring period for most of the parameters, a few commenced in different years during the period.

Summary of river water quality improvements in 18 districts

District	Improvement	Remaining Problem
New Territories		
Kwai Tsing	All factory discharges are now controlled under the Water Pollution Control Ordinance. Domestic and commercial premises are connected to government sewers. Water quality in Kau Wa Keng Stream improved from “Bad” in 1991 to “Good” in 2015. Its <i>E. coli</i> level has been reduced by 40% compared with 1998. Sewerage works for villages are being planned.	Still affected by expedient connections and a small number of unsewered villages
Tsuen Wan	All factory discharges are now controlled under the Water Pollution Control Ordinance. Domestic and commercial premises are connected to government sewers. Livestock farm discharges eliminated under the Livestock Waste Control Scheme. Sam Dip Tam Stream improved from “Bad” in 1988 to “Excellent” in 2015. Its <i>E. coli</i> level has been reduced by 90% compared with 1999. Pai Min Kok Stream improved from “Bad” in 1988 to “Excellent” in 2015. Its <i>E. coli</i> level has been reduced by over 90% compared with 1989. Sewerage works at hinterland of affected beaches have been completed and progressive connection of unsewered properties to the new public sewers is made. Sewer provisions for other villages in the area are being planned.	Still affected by expedient connections and a small number of unsewered villages.
Tuen Mun	All factory discharges are now controlled under the Water Pollution Control Ordinance. Domestic and commercial premises are connected to government sewers. Livestock farm discharges are now controlled under the Livestock Waste Control Scheme. Tuen Mun River upstream improved from “Very Bad” in 1988 to “Bad” in 2015. Its <i>E. coli</i> level has been reduced by over 90% compared with 1988. Tuen Mun River midstream and downstream improved from “Bad” in 1988 to “Good” in 2015. The <i>E. coli</i> level has been reduced by over 90% compared with 1988. A new trunk sewer to the west of Tuen Mun River was completed in 2015. Sewerage works for villages are progressively being planned and implemented. The Pillar Point Sewage Treatment Works have been upgraded to chemically enhanced primary treatment level with UV disinfection since May 2014.	Still affected by expedient connections and unsewered villages especially in the upstream region.
Yuen Long	Under the Livestock Waste Control Scheme, most of the farms had ceased operation. Factory discharges are controlled under the Water Pollution Control Ordinance. Most of the domestic and commercial premises in the urbanized town center are connected to government sewers. Yuen Long Creek and Kam Tin River both changed from “Very Bad” in 1986 to “Bad” and “Fair” in 2015. The <i>E. coli</i> level of these rivers has reduced by over 70% compared with 1998. Tin Shui Wai Nullah improved from “Bad” in 1993 to “Fair” and “Good” in 2015. The <i>E. coli</i> level of this nullah has reduced by 94% compared with 1992. Trunk sewers, pumping stations, and village sewerage are being planned and constructed progressively. The sewage treatment works at San Wai and Yuen Long will be expanded in anticipation of projected population growth, and upgraded to produce higher quality effluent.	Still affected by polluted discharges from unsewered villages and expedient discharges from industrial enterprises and livestock farms.

District	Improvement	Remaining Problem
New Territories		
North	Under the Livestock Waste Control Scheme, most farms have ceased operation. Factory discharges are controlled under the Water Pollution Control Ordinance. Most of the domestic and commercial premises in the urbanized town centers are connected to government sewers. The downstream stations of River Indus, Beas and Ganges improved from "Very Bad" in 1986 to "Fair" in 2015. The <i>E. coli</i> level of these rivers has been reduced by over 90% compared with 1990. Sewerage works for villages are progressively being planned and implemented. The sewage treatment works at Sha Tau Kok and Shek Wu Hui Sewage Treatment Works will be expanded and upgraded to produce higher quality effluent, in anticipation of projected population growth and establishment of the New Development Areas.	Still affected by polluted discharges from unsewered villages and expedient discharges from industrial enterprises and livestock farms. Downstream of River Indus is affected by backflow of Shenzhen River water.
Tai Po	Factory discharges are controlled under the Water Pollution Control Ordinance. Most of the domestic and commercial premises in the urbanized town centers are connected to government sewers. Under the Livestock Waste Control Scheme, livestock farms have ceased operation. The water quality of downstream stations of rivers in the area improved from "Very Bad" in 1986 to "Good" or better in 2015. The <i>E. coli</i> level of these rivers has been reduced by over 70% compared with 1999. Sewerage works for villages are progressively being planned and implemented. The Tai Po Sewage Treatment Works acquired UV disinfection facilities in 2010 and its treatment capacity has been increased to 120,000 m ³ /d since 2015.	Still affected by polluted discharges from unsewered villages and expedient connections.
Sha Tin	Factory discharges are controlled under the Water Pollution Control Ordinance. Most of the domestic and commercial premises in the urbanized town center are connected to government sewers. Under the Livestock Waste Control Scheme, livestock farms have ceased operation. The water quality of the Shing Mun River main channel improved from "Fair" in 1986 to "Excellent" in 2015. The <i>E. coli</i> level of the river has been reduced by 85% compared with 1988. Village sewerage is being extended to most of the remaining villages in the area. The Sha Tin Sewage Treatment Works acquired UV disinfection facilities in 2010.	Still affected by polluted discharges from unsewered villages and expedient connections.
Sai Kung	Factory discharges are now controlled under the Water Pollution Control Ordinance. Most of the domestic and commercial premises in the urbanized town centers are connected to government sewers. The rural village houses are being connected to government sewers. Livestock farm discharges are now controlled under the Livestock Waste Control Scheme. The water quality of the rivers has improved from "Bad" in 1991 to "Good" or better in 2015. The <i>E. coli</i> level has been reduced by over 65% compared with 1997. Local town and village sewerage improvements are being planned and implemented. Upgrading of the Sai Kung Sewage Treatment Works is under planning.	Still affected by polluted discharges from unsewered villages and expedient connections.
Islands	Some of the domestic premises are connected to government sewers. Under the Livestock Waste Control Scheme, the small number of livestock farms has all been closed. Mui Wo River and Tung Chung River have stayed "Good" or better throughout the years. The <i>E. coli</i> level of Mui Wo River has been reduced by 97% compared with 1988. Additional village sewerage and sewage treatment facilities and upgrading of treatment facilities are being planned and provided to Southern Lantau, Mui Wo, Lamma Island, Cheung Chau and Peng Chau.	Still affected by discharges from unsewered villages and some expedient connections.

District	Improvement	Remaining Problem
Kowloon		
Yau Tsim Mong	The urban area of Kowloon contains few natural watercourses. Factory and restaurant discharges are now controlled under the Water Pollution Control Ordinance. Domestic and commercial premises are connected to government sewers. A large number of expedient connections have been rectified. The water quality of the Kai Tak River improved from "Very Bad" in 1986 to "Fair" or better in 2015. The <i>E. coli</i> level of the region has been reduced by over 75 % compared with 1999. Sewerage improvement works in the area are to be implemented. In 2010, the Civil Engineering and Development Department commissioned the Kai Tak Development - Environmental Improvement to Kai Tak Approach Channel. The Upgrading of Central & East Kowloon Sewerage, phase 1 was completed in August 2012. Phase II of the upgrading works will be completed in March 2016. Such works should further improve the water quality of Kai Tak River and the adjacent marine water.	Mainly affected by expedient connections, rear lane washing activities, urban run-offs and mis-connections to the storm drain system.
Sham Shui Po		
Kowloon City		
Wong Tai Sin		
Kwun Tong		
Hong Kong Island		
Central and Western	The urban area of Hong Kong Island contains few natural watercourses. Factory and restaurant discharges are now controlled under the Water Pollution Control Ordinance. Domestic and commercial premises are connected to government sewers. A large number of expedient connections have been rectified.	Mainly affected by expedient connections, rear lane washing activities, urban run-offs and mis-connections to the storm drain system. Some streams are affected by discharges from squatters built near the streams.
Wan Chai		
Eastern		
Southern		